

24MA403-PROBABILITY AND STATISTICS

(Common to Agri, BME, AD & IT)

UNIT-I PROBABILITY AND RANDOM VARIABLE

Introduction:

The word 'Probable' means chance or the likelihood of happening of an even.

Examples: It is quite probable that M.S.Naadash may go New Delhi next week.

But suppose a ball is thrown up in the air, it is sure that it may fall back to the ground.

In this action the outcome is sure or definite. If the outcome is certain it is deterministic or predictable. If the outcome is not unique, it is probabilistic or unpredictable.

But when we toss a coin, the fall of head or tail is not certain and it is probable. Similarly when a dice is thrown, we are not sure about the number of points that will show up on the upper most face. Similarly the life time of a chip, tube light, T.V etc are not definite. The situations are dependent on chance. The theory of probability tries to measure this element of chance in such situations.

The element of chance plays a vital role in our life. Many important business decisions are made on this basis.

The entire business of insurance and share market is based on the probability theory. Gamblers win or ruin fortunes, on actions the result of which depends on chance.

Important Definitions [Basic terms and Concepts]

[i] **Trial or attempt :**

Just tossing a coin or throwing a dice is a trial or an attempt. The collection of trials make an experiment.

[ii] **Random experiment :**

If an experiment is repeated any number of times under same conditions it does not give unique results, but may result in any one of the several possible outcome.

If an action or attempt produces any unpredictable outcome, it is called a random experiment or trail. The outcome are known as events or cases.

[iii] **Sample space :**

The collection of all possible outcome or sample points of a random experiment is called a sample space. It is denoted as S.

[or]

A set of all samples point connected with an experiment is called the sample space connected with that experiment

Note : The entire sample space is an even, since it is a subset of itself.

Illustrations

- [a] When a coin is tossed, there are two sample points Head (H) OR Tail (T) Then the sample space may be represented as $S = \{H, T\}, n(S) = 2$
- [b] When a dice is thrown, there are six sample points corresponding to the six faces which are in the uppermost face of the dice. The sample space may be represented by taking the number of points in the uppermost face as :

$$\{1, 2, 3, 4, 5, 6\} \text{ and } n(S) = 6$$

[iv] **Mutually exclusive events**
[or]

In compatible events

Two events of A and B in a sample space are said to be mutually exclusive events if they cannot occur together, i.e., if one occur the other cannot. On the other hand mutually exclusive means that the happening of an event excludes the possibility of happening of the other event in the same trial.

Note : In set notation if A and B represent mutually exclusive events and if they are disjointed , it means that

$$A \cap B \neq \phi, \phi \text{ is the nult set}$$

Illustrations :

- [i] If we toss a coin , either head or tail can be up, but both cannot be up at a time [or can happen simultaneously].
- [ii] If we throw a dice the outcome of getting 1,2,3.....6 are mutually exclusive events.

Note : Mutually exclusive event is applicable for a single trial only.

If a coin is tossed twice, the head appearing in the first trial will not affect the appearing of the tail in the next trail.

[v] **Equally likely events :**

The two events are said to be equally likely if either of them cannot be expected to occur in preference to other.

Illustration :

Cast a uniform die, then all the six sample points are equally likely events, since any one of them cannot be expected to turn up in preference to the other.

[vi] **Favorable events :**

The number of outcome favorable to an event in an experiment is the number of outcome which entail the happening of the event.

Illustration :

In tossing two coins the number of cases favorable to the event of getting a head is HT, TH, HH.

[Vii] **Exhaustive events :**

Outcome are said to be exhaustive when they include all possible outcome.

Illustration :

In drawing three cards from a pack of 52 cards, the exhaustive number of cases is $52 C_3$.

In case of throwing two dices , the exhaustive number of cases is 36, $[6^2]$.

Details about Probability : Mathematical definition of probability [Appiori definition of probability]

If there are ' n ' equally likely , mutually exclusive and exhaustive outcome or events or happenings , out of which ' m ' of them are favorable to an event ' A ' [say], then

probability of the happening to A is defined as the ratio $\frac{m}{n}$.

If P is the notation for probability of happening of the event, then

$$P(A) = p = \frac{\text{Number of favorable cases}}{\text{Total number of exhaustive cases}} = \frac{m}{n}$$

But total probability:

$$p + q = 1$$

$q = 1 - p$; is the probability of failure.

$$q = 1 - \frac{m}{n} = \frac{n - m}{n}$$

Statistical [or] empirical definition of Probability:

If in n trials an event E happens m times, then

$$p = \text{probability of } E = \lim_{n \rightarrow \infty} \left(\frac{m}{n} \right)$$

Note :

[i] If A is an event, then $0 \leq P(A) \leq 1$, otherwise the probability of an event lies between 0 and 1.

[ii] If A is an event say Head of the coin ; then $\bar{A}$, the complement of the event A , is Tail of the coin,

If $P(A)$ is the probability of $A = \frac{m}{n}$

$P(\bar{A})$ is the probability of $\bar{A} = \frac{1 - m}{n} = \frac{n - m}{n}$

$$P(A) + P(\bar{A}) = \frac{m}{n} + \frac{n - m}{n} = \frac{n}{n} = 1$$

$$P(A) + P(\bar{A}) = 1$$

$$P(\bar{A}) = 1 - P(A) \quad \text{i.e., } p + q = 1$$

[iii] If $P(A) = 1$, the event A is called as certain or sure event.

[iv] If $P(A) = 0$, the event A is called impossible event [or] $P(\phi) = 0$.

[v] The number of favorable events can either be counted or enumerated with the help of permutation and combination.

[vi] The above mathematical definition of probability fails to answer if :

[i] the events are not equally likely.

[ii] the exhaustive number of cases in a trial is infinity.

[vii] If $\{A_n\}$ is any finite or infinite sequence of disjoint events in 'S' then

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

Axioms of probability: (Axiomatic Treatment of probability theory)

Let S be the sample space, and σ be the class of events, and let P be a real valued function defined on σ . Then $P(A)$ is called the probability of the event A if P satisfies the following axioms :

[i] For every event $A \in \sigma, 0 \leq P(A) \leq 1$.

[ii] $P(S) = 1$.

[iii] For every finite or infinite sequence of disjoint events : $A_1, A_2, \dots$

$$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$$

$$\text{i.e., } P\left(\bigcup A_n\right) = \sum P(A_n)$$

(OR)

$$P\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n P(E_i)$$

Independent Events: [This is suitable for two or more events]

Two or more events are considered to be independent if the occurrence or non-occurrence of an event does not affect the occurrence or non-occurrence of the other.

For example:

If a coin is thrown twice, the result of the second throw is in no way affected by the result of the first throw. Thus the events are independent events.

Note : For independent events.

$$P(A \cap B) = P(A) \cdot P(B)$$

Theorems of Probability:

Theorem 1

Show that probability of an impossible event is zero. [or] $P(\phi) = 0$.

Proof:

Impossible event contains no sample point.

The sample space and the impossible event is zero. [or] $P(\phi) = 0$.

$$S \cup \phi = S$$

$$P(S \cup \phi) = P(S)$$

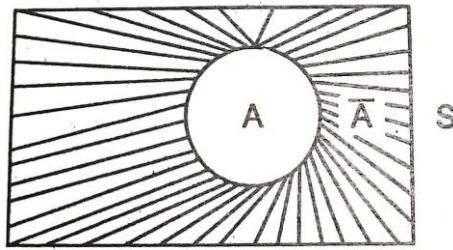
$$P(S) + P(\phi) = P(S)$$

$$P(\phi) = 0$$

Theorem 2

Show that the probability of the complement event $\bar{A}$ of A is $P(\bar{A}) = 1 - P(A)$.

Proof :



A and $\bar{A}$ are disjoint

$$P(A \cup \bar{A}) = P(S)$$

$$P(A) + P(\bar{A}) = 1$$

$$P(\bar{A}) = 1 - P(A)$$

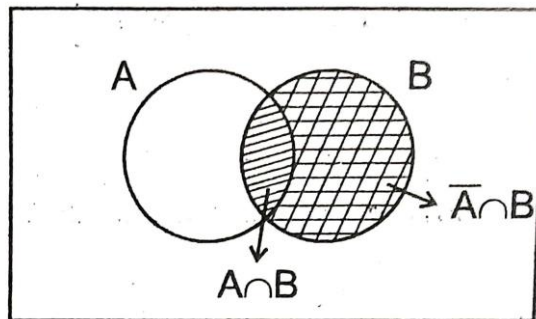
Theorem 3

Show that for any events A and B , is $P(\bar{A} \cap B) = P(B) - P(A \cap B)$.

Proof :

$$\bar{A} \cap B = \{p: p \in B \text{ and } p \in \bar{A} \text{ or } p \notin A\}$$

$\bar{A} \cap B$ and $A \cap B$ are disjoint sets.



$$P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

Note : Similarly we can prove,

$$P(A \cap \bar{B}) = P(A) - P(A \cap B)$$

Theorem 4

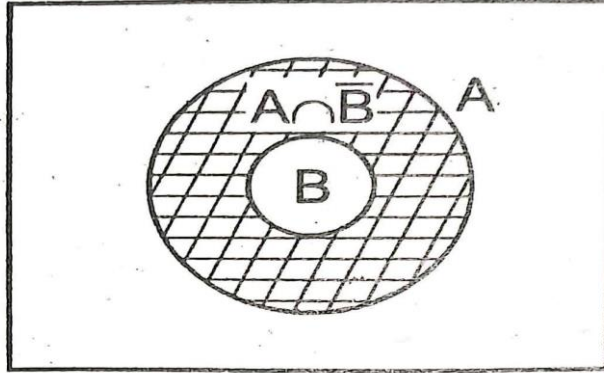
If $B \subset A$, then

$$(i) P(A \cap \bar{B}) = P(A) - P(B)$$

$$(ii) P(B) \leq P(A)$$

Proof :

[i] As $B \subset A$ B is the subset of A . B and $A \cap \bar{B}$ are disjoint and



$$P(B) + P(A \cap \bar{B}) = P(A)$$

$$P(A \cap \bar{B}) = P(A) - P(B)$$

[ii] If E is any element, then

$$0 \leq P(E) \leq 1, \quad P(E) \geq 0$$

$$0 \leq P(A \cap \bar{B}) \leq 1$$

$$P(A \cap B) \geq 0$$

$$P(A) - P(B) \geq 0$$

$$P(B) \leq P(A)$$

Theorem 5

Show that $P(A \cap B) \leq P(A)$ and $P(A \cap B) \leq P(B)$

Proof :

If $B \subset A$, then $P(B) \leq P(A)$ [proved]

As $A \cap B \subset A$ and $A \cap B \subset B$

$$P(A \cap B) \leq P(A), \quad P(A \cap B) \leq P(B)$$

Theorem 6 [Addition Theorem of Probabilities or Theorem of Total Probability]

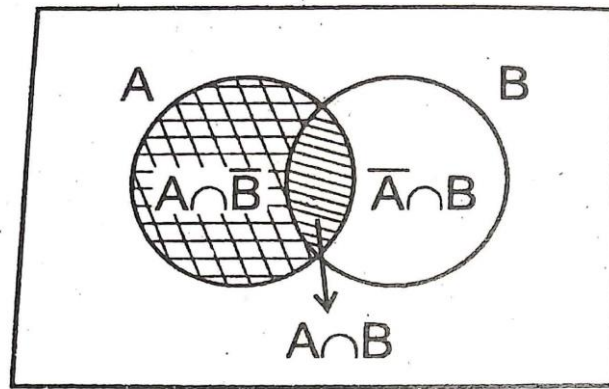
If A and B are any two events, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

(OR)

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Proof :



A and $\bar{A} \cap B$ are disjoint sets and their union is $A \cup B$

$$A \cup B = A \cup (\bar{A} \cap B)$$

$$P(A \cup B) = P(A) + P(\bar{A} \cap B) \quad (1)$$

But already proved: $P(\bar{A} \cap B) = P(B) - P(A \cap B)$

$$\text{Put in [1]} : P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Hence proved.

Note : [Extension or General Law of Addition of Probabilities]

For n events $A_1, A_2, \dots, A_n$;

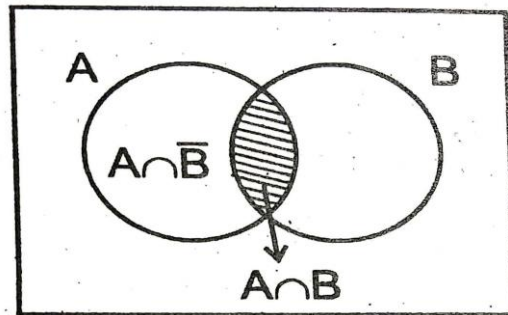
We have

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i \cap A_j) + \dots + (-1)^{n+1} P(A_1 \cap A_2 \cap \dots \cap A_n)$$

Theorem 7

If $A \cap B = \phi$, then show that

$$P(A) \leq P(B)$$



Proof :

$$A = (A \cap B) \cup (A \cap \bar{B})$$

But $A \cap B = \phi$, then

$$A = \phi \cup (A \cap \bar{B})$$

$$A = A \cap \bar{B}$$

$$A \subseteq \bar{B}$$

$$P(A) \leq P(\bar{B})$$

Theorem 8

If A, B and C are any three events, then prove that :

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

Proof :

Using Theorem 6, we know

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Consider $A \cup B$ as a single event, then

$$P((A \cup B) \cup C) = P(A \cup B) + P(C) - P((A \cup B) \cap C) \quad (1)$$

By distributive law

$$P((A \cup B) \cap C) = P[(A \cap C) \cup (B \cap C)]$$

Put in [1] :

$$\begin{aligned} &= (P(A) + P(B) - P(A \cap B) + P(C)) \\ &\quad [P(A \cap C) + P(B \cap C) - P((A \cap C) \cap (B \cap C))] \\ &= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) \\ &\quad - P(B \cap C) + P(A \cap B \cap C) \end{aligned}$$

Hence Proved.

Theorem 9

If $A_1, A_2, \dots, A_n$ are n mutually exclusive events, then the probability of the happening of any one of them is:

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3 \dots \cup A_n) &= P(A_1 + A_2 + \dots + A_n) \\ &= P(A_1) + P(A_2) + \dots + P(A_n) \end{aligned}$$

Proof :

Consider n -mutually exclusive and exhaustive and equally likely events, out of which m_1 are the event favorable to the event A_1 and m_2 are the events favorable to A_2 and so on.

$$P(A_1) = \frac{m_1}{n}, P(A_2) = \frac{m_2}{n} \dots P(A_n) = \frac{m_n}{n}$$

$$P(A_1 + A_2 + \dots A_n) = \frac{m_1 + m_2 + \dots + m_n}{n}$$

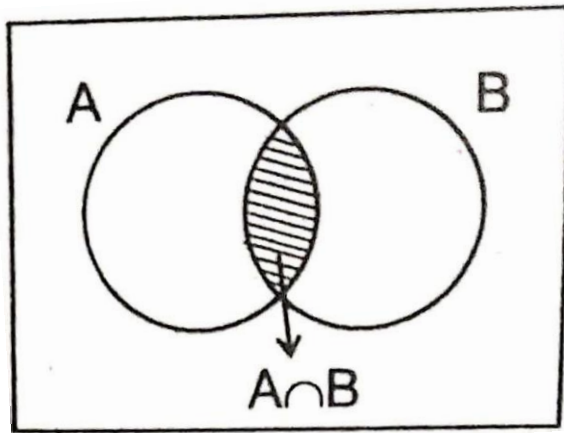
$$= \frac{m_1}{n} + \frac{m_2}{n} + \dots + \frac{m_n}{n}$$

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$$

Theorem 10

If A and B are two independent events, then show that :

- [i] $\bar{A}$ and $\bar{B}$ are also independent.
- [ii] $\bar{A}$ and B are also independent.
- [iii] A and B are also independent.



Proof :

- [i] $\bar{A}$ and $\bar{B}$ are also independent.

We know that

$$\overline{A \cap B} = \bar{A} \cap \bar{B}$$

$$P(\bar{A} \cap \bar{B}) = P(\overline{A \cap B})$$

$$= 1 - P(A \cap B)$$

$$= 1 - [P(A) + P(B) - P(A \cap B)] \quad (1)$$

[A and B are independent even, then $P(A \cap B) = P(A).P(B)$], put in [1] :

$$\begin{aligned}
 P(\bar{A} \cap \bar{B}) &= 1 - P(A) - P(B) + P(A).P(B) \\
 &= (1 - P(A))(1 - P(B)) \\
 P(\bar{A} \cap \bar{B}) &= P(\bar{A}).P(\bar{B})
 \end{aligned}$$

$\bar{A}$ and $\bar{B}$ are independent.

[ii] To show $\bar{A}$ and B are independent.

Proof :

The event $A \cap B$ and $\bar{A} \cap B$

$$\begin{aligned}
 (A \cap B) \cup (\bar{A} \cap B) &= B \\
 P[(A \cap B) \cup (\bar{A} \cap B)] &= P(B) \\
 P(A \cap B) + P(\bar{A} \cap B) &= P(B) \\
 P(\bar{A} \cap B) &= P(B) - P(A \cap B) \\
 &= P(B) - P(A).P(B),
 \end{aligned}$$

[A and B are independent $P(A \cap B) = P(A).P(B)$]

$$\begin{aligned}
 &= P(B) - [1 - P(A)] \\
 P(\bar{A} \cap B) &= P(\bar{A}).P(B)
 \end{aligned}$$

$\bar{A}$ and B are independent.

[iii] To show A and $\bar{B}$ are independent.

Proof :

$$\begin{aligned}
 A &= (A \cap B) \cup (A \cap \bar{B}) \\
 P(A) &= P(A \cap B) + P(A \cap \bar{B}) \\
 P(A \cap \bar{B}) &= P(A) - P(A \cap B) \\
 P(A \cap \bar{B}) &= P(A) - P(A).P(B) \\
 &= P(A)[1 - P(B)] \\
 P(A \cap \bar{B}) &= P(A).P(\bar{B})
 \end{aligned}$$

A and $\bar{B}$ are independent.

Theorem 11

If the even A is known to be independent of the events $B \cup C$ and $B \cap C$, then show that A is also independent of C .

Proof :

It is known that , by independence

$$P(A \cap B) = P(A).P(B) \quad (\text{or}) \quad P(AB) = P(A).P(B)$$

$$P(A \cap (B \cup C)) = P(A).P(B \cup C) \quad (1)$$

$$\begin{aligned} P((A \cap B) \cup (A \cap C)) &= P(A \cap B) + P(A \cap C) - P(A \cap (B \cup C)) \\ &= P(A).P(B) + P(A).P(C) - P(A).P(B \cap C) \quad (2) \end{aligned}$$

$$\begin{aligned} P(A \cap (B \cup C)) &= P(A).P(B \cup C) \\ &= P(A)[P(B) + P(C) - P(B \cap C)] \\ &= P(A)[P(B) + P(C) - P(B).P(C)] \quad (3) \end{aligned}$$

From [1] , [2] and [3]

$$P(A \cap C) = P(A).P(C)$$

A and C are independent.

Theorem 12

Find the probability of at least one of n independent events.

Proof :

Let $A_1, A_2, \dots, A_n$; are n mutually independent events, with

$$P(A_i) = p_i, \quad P(\overline{A_i}) = q_i, \quad (q_i = 1 - p_i)$$

Using Negation rule :

$$P(A_1 \cup A_2 \dots A_n) = 1 - P(\overline{A_1 \cup A_2 \dots A_n}) \quad (1)$$

Using De – Morgan's law

$$P(\overline{A_1 \cup A_2 \dots A_n}) = 1 - P(\overline{A_1} \cap \overline{A_2} \cap \dots \cap \overline{A_n})$$

Put in [1] : $\overline{A_1}, \overline{A_2}, \dots, \overline{A_n}$ are independent.

$$P(A_1 \cup A_2 \dots A_n) = 1 - P(\overline{A_1}).P(\overline{A_2}).\dots P(\overline{A_n})$$

$$P(A_1 \cup A_2 \dots A_n) = 1 - q_1 q_2 \dots q_n$$

Note : Thus if A and B are independent events, then

$$P(A \cup B) = 1 - P(\overline{A}).P(\overline{B})$$

Theorem 13

$$P.T \quad P(A \cap B) \leq P(A) \leq P(A \cup B) \leq P(A) + P(B)$$

Proof :

From the theorem 3,

$$P(A \cap \bar{B}) = P(A) - P(A \cap B)$$

$$\Rightarrow P(A) = P(A \cap \bar{B}) + P(A \cap B)$$

$$P(A) \geq P(A \cap B) \quad (1) \quad \because P(A \cap \bar{B}) \geq 0$$

We know that

But

$$P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

$$\Rightarrow P(B) = P(\bar{A} \cap B) + P(A \cap B)$$

$$P(B) \geq P(A \cap B) \quad (2) \quad \because P(\bar{A} \cap B) \geq 0$$

From the theorem 6,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) \geq P(A) + P(B) - P(B)$$

$$P(A \cup B) \geq P(A) \quad (3) \quad \because P(B) \geq P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) \leq P(A) + P(B) \quad (4) \quad \because P(A \cap B) \geq 0$$

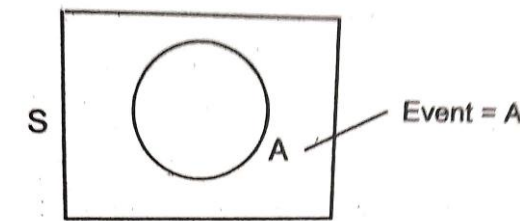
$$-P(A \cap B) \leq 0$$

Combining equations the [1], [3], [4], we get

$$P(A \cap B) \leq P(A) \leq P(A \cup B) \leq P(A) + P(B)$$

Hence $P(A \cap B) \leq P(A) \leq P(A \cup B) \leq P(A) + P(B)$

Note :



$n(A) = \text{Size of the event.}$

Solved Problem 1.1

From a bag containing 10 black and 12 white balls, a ball is drawn at random. What is the probability that it is a black?

Solution

Let A be the event of selecting a black ball.

Total number of balls = $10+12=22$

Total possible [Exhaustive] ways of choosing

$$\text{One ball} = {}^{22}C_1 = 22 \text{ ways}$$

$$n(S) = 22$$

Out of 10 black balls, the number of ways of choosing one black ball [favorable]

$$= {}^{10}C_1 = 10$$

$$n(A) = 10.$$

Probability of getting a black ball

$$= \frac{n(A)}{n(S)} = \frac{10}{22}$$

Solved Problem 1.2

In a class of 10 students, 4 are boys and the rest are girls. Find the probability that a student selected will be a girl.

Solution

Let A be the event of selecting a girl student

$$\text{Total number of students} = 10$$

$$\text{Given Boys} = 4$$

$$\text{Girls} = 6$$

$n(S)$ = Out of 10 students, one student can be selected in ${}^{10}C_1$ ways.

$$\frac{10}{1} = 10$$

$n(A)$ = Out of 6 girls students, one girl can be selected in 6C_1 ways.

$$= \frac{6}{1} = 6$$

The probability of selecting a girl student

$$P(A) = \frac{n(A)}{n(S)}$$

$$= \frac{6}{10} = \frac{3}{5}$$

Hence the probability of selecting a girl student $= \frac{3}{5}$.

Solved Problem 1.3

A box contains 4 white, 5 Red and 6 Black balls. Two balls are drawn at random. If both are black, what is the probability?

Solution

Let A be the event of drawing both balls are black.

$$\text{To find } P(A) = \frac{n(A)}{n(S)}$$

$$\text{Total number of balls} = 4 + 5 + 6 = 15$$

$n(S)$ = Out of 15 balls, 2 balls can be selected in ${}^{15}C_2$ ways.

$$\frac{15 \times 14}{1 \times 2} = 105$$

$n(A)$ = Out of 6 black balls, 2 balls can be selected in 6C_2 ways.

$$= \frac{6 \times 5}{1 \times 2} = 15$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{15}{105} = 0.1429$$

Hence the probability of drawing both the balls black is 0.1429.

Solved Problem 1.4

Two dice are rolled. Find the probability that the sum of the number thrown is 9.

W.K.T. the sample space S , for two dice are thrown is,

$$S = \left\{ \begin{array}{l} (1,1), (1,2), (1,3), (1,4), (1,5), (1,6) \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6) \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6) \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6) \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6) \end{array} \right.$$

$$n(S) = 36$$

Solution

Let A be the event, the sum of the numbers thrown is 9.

$$A = \{(3,6), (4,5), (5,4), (6,3)\}$$

$$n(A) = 4$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{4}{36} = \frac{1}{9}$$

Hence the probability of getting the sum of the number 9 is $\frac{1}{9}$.

Solved Problem 1.5

A Committee of 5 is formed by drawing lots from 8 boys and girls. Find the probability that the committee will consist of 2 boys and 3 girls.

Solution

Let A be the event of selecting 2 boys and 3 girls for the committee. Total number of persons in a committee = $8 + 6 = 14$.

Out of person, 5 can be drawn in ${}^{14}C_5$ ways

$$[\text{i.e.,}] \quad n(S) = {}^{14}C_5 = \frac{14 \times 13 \times 12 \times 11 \times 10}{1 \times 2 \times 3 \times 4 \times 5} = 2002.$$

Let A_1 be the event of selecting 2 boys. 2 boys out of 8 can be drawn in 8C_2 ways

$$n(A_1) = {}^8C_2 = \frac{8 \times 7}{1 \times 2} = 28.$$

Let A_2 be the event of selecting 3 girls. 3 girls out of 6 can be chosen in 6C_3 ways.

$$n(A_2) = {}^6C_3 = \frac{6 \times 5 \times 4}{1 \times 2 \times 3} = 20.$$

$$\text{Then } n(A) = n(A_1) \cdot n(A_2) = 28 \times 20 = 560$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{560}{2002} = 0.2797$$

Hence the probability that the committee will include 2 boys and 3 girls is = 0.2797 .

Solved Problem 1.6

Four cards are drawn from a pack of cards. Find the probability that there are two diamonds and two spades.

Solution

Let A be the event of selecting 4 cards.

Let A_1 be the event of selecting 2 diamonds.

Let A_2 be the event of selecting 2 spades.

Total number of cards = 52.

Then, 4 cards can be drawn from pack of 52 cards in ${}^{52}C_4$ ways.

$$n(S) = {}^{52}C_4 = \frac{52 \times 51 \times 50 \times 49}{1 \times 2 \times 3 \times 4} = 270725.$$

$n(A_1)$ = Two diamonds can be drawn from 13 cards in ${}^{13}C_2$ ways

$$n(A_1) = {}^{13}C_2 = \frac{13 \times 12}{1 \times 2} = 78$$

$n(A_2)$ = Two spades can be drawn from 13 cards in ${}^{13}C_2$ ways

$$n(A_2) = \frac{13 \times 12}{1 \times 2} = 78$$

$$\text{Then } n(A) = n(A_1) \cdot n(A_2) = 78 \times 78 = 6084$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{6084}{270725} = 0.02245$$

Hence the probability of getting 2 diamonds and 2 spades is = 0.0225

Conditional probability:

If A and B are dependent events. Conditional probability of A given B means the probability of occurrence of A when the event B has already happened. It is

denoted by $P\left(\frac{A}{B}\right)$ and is defined by $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$ if $P(B) \neq 0$

Conditional probability of B given A means the probability of occurrence of B where the event A has already happened. It is denoted by $P\left(\frac{B}{A}\right)$ is defined by

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} \text{ if } P(A) \neq 0$$

$$P(A \cap B) = P(A) \cdot P\left(\frac{B}{A}\right) = P(B) \cdot P\left(\frac{A}{B}\right)$$

Multiplication Law of Probability

If A and B are dependent events in a sample space S , Then,

$$\begin{aligned} P(A \cap B) &= P(A) \cdot P\left(\frac{B}{A}\right), P(A) \neq 0 \\ &= P(B) \cdot P\left(\frac{A}{B}\right), P(B) \neq 0 \end{aligned}$$

Where $P\left(\frac{B}{A}\right)$ represent the conditional probability of B given A and

$P\left(\frac{A}{B}\right)$ represent the conditional probability of A given B .

Proof:

Let N be the number of sample points in S and n_A, n_B are the number of cases favorable to the events A, B . Let n_{AB} be the number of cases favorable to the event $A \cap B$.

$$\text{Then } P(A) = \frac{n_A}{N}, P(B) = \frac{n_B}{N}, P(A \cap B) = \frac{n_{AB}}{N}$$

$$\text{Also } P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{n_{AB}}{n_B}, P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{n_{AB}}{n_A}$$

$$\text{Now } P(A \cap B) = \frac{n_{AB}}{N} = \frac{n_{AB}}{n_A} \cdot \frac{n_A}{N} = P\left(\frac{B}{A}\right) \cdot P(A)$$

$$\text{and } P(A \cap B) = \frac{n_{AB}}{N} = \frac{n_{AB}}{n_B} \cdot \frac{n_B}{N} = P\left(\frac{A}{B}\right) \cdot P(B)$$

$$\Rightarrow P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} \text{ if } P(A) \neq 0$$

$$\Rightarrow P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} \text{ if } P(B) \neq 0$$

Note :

1. Extension of multiplication law of probability for ' n ' events $A_1, A_2, \dots, A_n$ we have

$$P(A_1 \cap A_2, \dots, A_n) = P(A_1)P\left(\frac{A_2}{A_1}\right) \\ P\left(\frac{A_3}{A_1 \cap A_2}\right) \dots P\left(\frac{A_n}{A_1 \cap A_2 \dots A_{n-1}}\right)$$

2. If two events A and B are independent

$$P(A \cap B) = P(A) \cdot P(B)$$

EXAMPLE:1

If we draw a card from a well shuffled pack of 52 cards, what is the probability that the card is either an Ace or a king.

Solution

If A and B denote the events of drawing an ace and a king respectively, these events are mutually exclusive.

$$P(A) = \frac{4}{52}, \quad P(B) = \frac{4}{52} \\ P(A \cup B) = P(A) + P(B) = \frac{4}{52} + \frac{4}{52} = \frac{2}{13}$$

EXAMPLE:2

One card is drawn from a pack of 52 cards. What is the probability of the card being either a red or black king.

Solution

If A and B denote the events of drawing a red king respectively.

$$P(A) = \frac{26}{52}, \quad P(B) = \frac{4}{52}, \quad P(A \cap B) = \frac{2}{52} \\ P(A \cup B) = \frac{26}{52} + \frac{4}{52} - \frac{2}{52} = \frac{7}{13}$$

EXAMPLE:3

Let A and B be two events associated with an experiment. Given that

$$P(\bar{A}) = 0.2, \quad P(\bar{B}) = 0.15, \quad P(\bar{A}\bar{B}) = 0.15, \quad \text{find } P\left(\frac{\bar{A}}{\bar{B}}\right)$$

Solution

$$P\left(\frac{\bar{A}}{\bar{B}}\right) = \frac{P(\bar{A}\bar{B})}{P(\bar{B})} = \frac{0.15}{0.15} = 1$$

EXAMPLE:4

A Person is known to hit the target in 3 out of 4 shots whereas another person is known to hit the target in 2 out of 3 shots. Find the probability of the target being hit when they both try.

Solution

$$\text{Given } P(A) = \frac{3}{4}, \quad P(B) = \frac{2}{3}$$

Both A and B may hit the target. The events A and B are not mutually exclusive. Since A and B are independent

$$P(A \cap B) = P(A).P(B)$$

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= P(A) + P(B) - P(A).P(B) \\ &= \frac{3}{4} + \frac{2}{3} - \left(\frac{3}{4} \times \frac{2}{3} \right) \\ &= \frac{11}{12} \end{aligned}$$

EXAMPLE:5

A bag contains 5 white and 8 red balls. Two drawing of 3 balls are made that

[a] the balls are replaced before the second trial, and

[b] the balls are not replaced before the second trial.

Find the probability that the first draw will give 3 white and second 3 red balls in each case.

Solution

[a] When balls are replaced.

Total number of balls in the bag = $5 + 8 = 13$.

Among this three balls can be drawn 13 in $13C_3$ ways.

Three white balls can be drawn from 5 in $5C_3$ ways.

Three red balls can draw from 8 in $8C_3$ ways.

The probability of drawing 3 white balls in the first trial is $\frac{5C_3}{13C_3}$

The probability of 3 red balls at the second trial is $\frac{8C_3}{13C_3}$

The probability of 3 white and 3 red balls in first and second trial with replacement gives

$$\begin{aligned} P(A \cap B) &= P(A).P(B) \\ &= \frac{5C_3}{13C_3} \times \frac{8C_3}{13C_3} = 0.0068 \end{aligned}$$

[b] When balls are not replaced.

At the first draw, 3 white balls can be drawn in $5C_3$ ways, $P(A) = \frac{5C_3}{13C_3}$.

The balls are not replaced, in the second drawn 3 red balls can taken as $8C_3$ ways,

$$P\left(\frac{B}{A}\right) = \frac{8C_3}{10C_3}.$$

Probability of 3 white and 3 red balls in the first and second trial without replacement gives

$$\begin{aligned} P(A \cap B) &= P(A) \cdot P\left(\frac{B}{A}\right) \\ &= \frac{5C_3}{13C_3} \times \frac{8C_3}{10C_3} = 0.0163 \end{aligned}$$

Total probability-Bayes's Theorem

Statement:

Let $A_1, A_2, \dots, A_n$ be 'n' mutually exclusive and exhaustive events. Let B be independent event such that $B \subset \bigcup_{i=1}^n A_i$ is the conditional probability of B given that A_i has already occurred, then

$$P\left(\frac{A_i}{B}\right) = \frac{P(A_i) \cdot P\left(\frac{B}{A_i}\right)}{\sum_{i=1}^n P(A_i) \cdot P\left(\frac{B}{A_i}\right)}$$

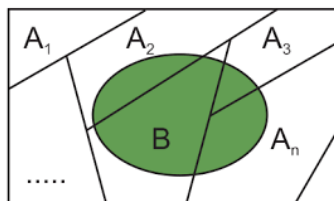
Proof:

Given $B \subset \bigcup_{i=1}^n A_i$

$$B = B \cap \left(\bigcup_{i=1}^n A_i \right) = \bigcup_{i=1}^n (B \cap A_i) \quad [\text{distributive law}]$$

$$\begin{aligned} P(B) &= P\left[\bigcup_{i=1}^n B \cap A_i \right] \\ &= \sum_{i=1}^n P(B \cap A_i) \\ &= \sum_{i=1}^n P(A_i) \cdot P\left(\frac{B}{A_i}\right) \end{aligned}$$

[$B \cap A_i$ one mutually disjoint events]



$$P(B \cap A_i) = P(B) \cdot P\left(\frac{A_i}{B}\right)$$

$$P\left(\frac{A_i}{B}\right) = \frac{P(B \cap A_i)}{P(B)}$$

$$P\left(\frac{A_i}{B}\right) = \frac{P(A_i) \cdot P\left(\frac{A_i}{B}\right)}{\sum_{i=1}^n P(A_i) \cdot P\left(\frac{B}{A_i}\right)}$$

Simple problems on Bayes' theorem:

EXAMPLE:1

Three urns A_1, A_2, A_3 denote the first, second and third urn chosen. Let B denote the event of with drawing red ball, then we have,

$$P(A_1) = \frac{1}{3}, \quad P(A_2) = \frac{1}{6}, \quad P(A_3) = \frac{1}{3}$$

$$P\left(\frac{B}{A_1}\right) = \frac{3}{8}, \quad P\left(\frac{B}{A_2}\right) = \frac{1}{6}, \quad P\left(\frac{B}{A_3}\right) = \frac{4}{9}$$

Now the probability drawing a red ball from urn A_2

$$P\left(\frac{A_2}{B}\right) = \frac{P(A_2) \cdot P\left(\frac{B}{A_2}\right)}{\sum_{i=1}^3 P(A_i) \cdot P\left(\frac{B}{A_i}\right)}$$

$$\sum_{i=1}^3 P(A_i) \cdot P\left(\frac{B}{A_i}\right) = P(A_1) \cdot P\left(\frac{B}{A_1}\right) + P(A_2) \cdot P\left(\frac{B}{A_2}\right) + P(A_3) \cdot P\left(\frac{B}{A_3}\right)$$

$$= \frac{1}{3} \times \frac{3}{8} + \frac{1}{6} \times \frac{1}{6} + \frac{1}{3} \times \frac{4}{9} = \frac{71}{216}$$

$$P\left(\frac{A_2}{B}\right) = \frac{\frac{1}{6} \times \frac{1}{6}}{\frac{71}{216}} = \frac{12}{71}$$

EXAMPLE:2

The contents of urns I, II and III are as follows 1 white, 2 black and 3 red balls. 2 white, 1 black and 1 red balls, and 4 white, 5 black and 3 red balls respectively. One urn is chosen at random and two balls are drawn from it. They happen to be white and red. What is the probability that they come from urns I, II or III.

Solution:

Let A_1, A_2, A_3 denote the events that the urn I, II, III is selected respectively and let B be the event that the two balls taken from the selected urn are white and red.

$$\text{Now } P(A_1) = \frac{1}{3}, P(A_2) = \frac{1}{3}, P(A_3) = \frac{1}{3}$$

$$P\left(\frac{B}{A_1}\right) = \frac{{}^1C_1 \times {}^3C_1}{{}^6C_2} = \frac{1}{5}$$

$$P\left(\frac{B}{A_2}\right) = \frac{{}^2C_1 \times {}^1C_1}{{}^4C_2} = \frac{1}{3}$$

$$P\left(\frac{B}{A_3}\right) = \frac{{}^4C_1 \times {}^3C_1}{{}^{12}C_2} = \frac{2}{11}$$

$$P\left(\frac{A_1}{B}\right) = \frac{P(A_1) \cdot P\left(\frac{B}{A_1}\right)}{\sum_{i=1}^3 P(A_i) \cdot P\left(\frac{B}{A_i}\right)}$$

Now

$$\begin{aligned} \sum_{i=1}^3 P(A_i) \cdot P\left(\frac{B}{A_i}\right) &= P(A_1) \cdot P\left(\frac{B}{A_1}\right) + P(A_2) \cdot P\left(\frac{B}{A_2}\right) + P(A_3) \cdot P\left(\frac{B}{A_3}\right) \\ &= \frac{1}{3} \times \frac{1}{5} + \frac{1}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{2}{11} = 0.238 \end{aligned}$$

$$P\left(\frac{A_1}{B}\right) = \frac{\frac{1}{3} \times \frac{1}{5}}{0.238} = 0.280$$

Similarly

$$P\left(\frac{A_2}{B}\right) = \frac{P(A_2) \cdot P\left(\frac{B}{A_2}\right)}{\sum_{i=1}^3 P(A_i) \cdot P\left(\frac{B}{A_i}\right)} = \frac{\frac{1}{3} \times \frac{1}{3}}{0.238} = 0.4668$$

$$P\left(\frac{A_3}{B}\right) = \frac{P(A_3) \cdot P\left(\frac{B}{A_3}\right)}{\sum_{i=1}^3 P(A_i) \cdot P\left(\frac{B}{A_i}\right)} = \frac{\frac{1}{3} \times \frac{2}{11}}{0.238} = 0.2546$$

EXAMPLE:3

Urn A contains 2 white and 2 black balls. Urn B contains 3 white and 2 black balls. One ball is transferred from A to B and then one ball is drawn out of B . Find the

chance the this ball is white. If this ball turns out to be white, find the probability that the transferred ball was white.

Solution

Let W_1 and W_2 be the events that the transferred ball is white and transferred ball is black respectively. Let W denote the event that the white ball is selected from B . Then The probability that the transferred ball is white is

$$P(W_1) = \frac{2}{4}; \quad P(W_2) = \frac{2}{4}$$

$$P\left(\frac{W}{W_1}\right) = \frac{4}{6}; \quad P\left(\frac{W}{W_2}\right) = \frac{3}{6}$$

The probability that the transferred ball white is :

$$\begin{aligned} P\left(\frac{W_1}{W_2}\right) &= \frac{P(W_1) \cdot P\left(\frac{W}{W_1}\right)}{\sum_{i=1}^2 P(W_i) \cdot P\left(\frac{W}{W_i}\right)} \\ &= \frac{\frac{2}{4} \times \frac{4}{6}}{\frac{2}{4} \times \frac{4}{6} + \frac{2}{4} \times \frac{3}{6}} = \frac{\frac{1}{3}}{\frac{7}{12}} = \frac{4}{7} \end{aligned}$$

EXAMPLE:4

In a bolt factory machines A, B, C manufactured 25%, 35%, 4%, respectively out of the total of their output 5,4,2 percents, are effective bolts. A bolt is drawn at random from the product and is found to be defective . what are the probabilities that it was manufactured by machines A, B and C ?

Solution

Let E_1, E_2, E_3 denote the respective events that a bolt selected at random is manufactured by the machines A, B, C . Let D denote the event that the bolt is defective. Then

$$P(E_1) = 0.25; \quad P(E_2) = 0.35; \quad P(E_3) = 0.40$$

$$P\left(\frac{D}{E_1}\right) = 0.05; \quad P\left(\frac{D}{E_2}\right) = 0.04; \quad P\left(\frac{D}{E_3}\right) = 0.02$$

The probability of the defective bolt manufactured by machine

$$A = P\left(\frac{E_1}{D}\right) = \frac{P(E_1) \cdot P\left(\frac{D}{E_1}\right)}{\sum_{i=1}^3 P(E_i) \cdot P\left(\frac{D}{E_i}\right)}$$

$$\begin{aligned}
 \text{Now } \sum_{i=1}^3 P(E_i) \cdot P\left(\frac{D}{E_i}\right) &= P(E_1) \cdot P\left(\frac{D}{E_1}\right) + P(E_2) \cdot P\left(\frac{D}{E_2}\right) + P(E_3) \cdot P\left(\frac{D}{E_3}\right) \\
 &= (0.25)(0.05) + (0.35)(0.04) + (0.40)(0.02) = 0.0345 \\
 P\left(\frac{E_1}{D}\right) &= \frac{(0.25)(0.05)}{0.0345} = 0.362
 \end{aligned}$$

Similarly

$$\begin{aligned}
 P\left(\frac{E_2}{D}\right) &= \frac{(0.04)(0.35)}{0.0345} = 0.406 \\
 P\left(\frac{E_3}{D}\right) &= \frac{(0.40)(0.02)}{0.0345} = 0.232
 \end{aligned}$$

EXAMPLE:5

In a certain college, the geographical distribution of men students is as follows: 50% come from the east, 30% come from the Midwest, 20% come from the far – west. The following proportions of the men students wear ties: 80% of the easterners, 60% of mid – westerners and 40% of the far – westerners. What is the probability that a student who wears a tie come from the east, mid and Far – west.

Solution

Let A_1, A_2, A_3 be the events that the student comes from east, mid – west and from far-west respectively.

$$P(A_1) = 0.50; \quad P(A_2) = 0.30; \quad P(A_3) = 0.20$$

Let D denote the men students wears ties

$$P\left(\frac{D}{A_1}\right) = 0.80; \quad P\left(\frac{D}{A_2}\right) = 0.60; \quad P\left(\frac{D}{A_3}\right) = 0.40$$

The probability that a student wears ties from east is given by

$$P\left(\frac{A_1}{D}\right) = \frac{P(A_1) \cdot P\left(\frac{D}{A_1}\right)}{\sum_{i=1}^3 P(A_i) \cdot P\left(\frac{D}{A_i}\right)}$$

$$\begin{aligned}
 \text{Now } \sum_{i=1}^3 P(A_i) \cdot P\left(\frac{D}{A_i}\right) &= P(A_1) \cdot P\left(\frac{D}{A_1}\right) + P(A_2) \cdot P\left(\frac{D}{A_2}\right) + P(A_3) \cdot P\left(\frac{D}{A_3}\right) \\
 &= 0.5 \times 0.8 + 0.3 \times 0.6 + 0.2 \times 0.4 = 0.66
 \end{aligned}$$

$$P\left(\frac{A_1}{D}\right) = \frac{0.8 \times 0.5}{0.66} = \frac{20}{33}$$

$$P(A_1) \cdot P\left(\frac{D}{A_1}\right) = P(A_1) \cdot P\left(\frac{D}{A_1}\right)$$

Similarly

$$P\left(\frac{A_2}{D}\right) = \frac{0.6 \times 0.3}{0.66} = \frac{3}{11}$$

$$P\left(\frac{A_3}{D}\right) = \frac{0.4 \times 0.2}{0.66} = \frac{4}{33}$$

EXAMPLE:6

The chances of three candidates A , B and C to become the manager of a company are in the ratio 3: 5: 4. The probability of introducing a special bonus scheme by them if selected are 0.6, 0.4 and 0.5 respectively. If the bonus scheme is introduced, what is the probability that B has become the manager.

Solution

Given

$$A:B:C = 3:5:4, \quad P(A) = \frac{3}{12}; \quad P(B) = \frac{5}{12}; \quad P(C) = \frac{4}{12}$$

Let D denote the event of introduction the bonus scheme

$$P\left(\frac{D}{A}\right) = 0.6; \quad P\left(\frac{D}{B}\right) = 0.4; \quad P\left(\frac{D}{C}\right) = 0.5$$

The probability of B becomes the manager is :

$$P\left(\frac{B}{D}\right) = \frac{P(B).P\left(\frac{D}{B}\right)}{P(A).P\left(\frac{D}{A}\right) + P(B).P\left(\frac{D}{B}\right) + P(C).P\left(\frac{D}{C}\right)}$$

$$= \frac{\frac{5}{12} \times 0.4}{\frac{3}{12} \times 0.6 + \frac{5}{12} \times 0.4 + \frac{4}{12} \times 0.5} = 0.345$$

Discrete and Continuous random variables:**Discrete Random Variable:**

Let X is a random variable. If the number of possible values 'x' is finite (or) countable infinite then 'x' is called discrete random variable.

Ex: - Consider the random experiment of tossing a dice $S = \{1,2,3,4,5,6\}$

Continuous Random Variable:

Let X is a random variable which takes all possible value is given interval is called a continuous random variable.

Ex: - The variable X = duration of telephonic conversation is a continuous random variable as X can have all values (integral as well as fraction)

Distribution function of random variables Properties:

Probability Mass Function:

If X is a discrete random variable which assume values $x_1, x_2, \dots, x_n$ with probabilities $p(x_1), p(x_2), \dots, p(x_n)$ respectively. Then p is called probability mass function (*pmf*) of X is provided the following conditions are satisfied.

$$(i). p(x_i) \geq 0 \quad (ii) \quad \sum_{i=1}^n p(x_i) = 1$$

Probability Density Function:

If X is a continuous random variable then the function f is said to be probability density function (*pdf*) of X , if it satisfies the following conditions

$$(i) \quad f(x) \geq 0, \forall x \quad (ii) \quad \int_{-\infty}^{\infty} f(x) dx = 1$$

Ex: - If a coin is tossed twice then the sample space is $S = \{HH, HT, TH, TT\}$. X is a number of heads obtained in the two tosses.

Hence $X(HH) = 2, X(HT) = 1 = X(TH), X(TT) = 0$

Cumulative Distribution Function (*cdf*):

Let X is a random variable the cumulative distribution function. Denoted by $F(x)$, is defined as $F(x) = P(X \leq x)$. probability that the random variable X assumes a value less than or equal to x . If $F(x) = P(X \leq x)$ and

$$P(X \leq x) = \sum_{i=0}^x P(x_i) \text{ if } X \text{ is discrete.}$$
$$= \int_{-\infty}^x f(x) dx \text{ if } X \text{ is continuous.}$$

Where $P(x_i)$ and $f(x)$ are probability mass function and probability density function.

Problems based on discrete random variables

Example: 1

A random variable X has the following probability function:

$X:$	0	1	2	3	4	5	6	7
$P(x):$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

- Find k
- Evaluate: $P(X < 6)$, $P(X \geq 6)$ and $P(0 < X < 5)$
- If $P(X \leq C) > 1/2$ find the minimum value of C
- Determine the distribution function of X

Solution:

(i) Since $\sum_{n=1}^7 p(x_i) = 1$, $\Rightarrow k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$

$$10k^2 + 9k - 1 = 0$$

$$\therefore k = -1 \text{ (Or) } k = 1/10$$

But since $P(x)$ cannot be negative, $k = -1$ is rejected.

Hence $k = 1/10$

$$(ii) P(X < 6) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5)$$

$$= \frac{1}{10} + \frac{2}{10} + \frac{2}{10} + \frac{3}{10} + \frac{1}{100} = \frac{81}{100}$$

$$P(X \geq 6) = 1 - P(X < 6) = 1 - \frac{81}{100} = \frac{19}{100}$$

$$P(0 < X < 5) = P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) = 8k = \frac{8}{10} = \frac{4}{5}$$

(iii) We form the table,

X	P(X)	$P(X \leq x)$	
0	0	0	0
1	k	k	1/10
2	2k	k+2k=3k	3/10
3	2k	3k+2k=5k	5/10=1/2
4	3k	5k+3k=8k	8/10

$P(X \leq 4) = \frac{8}{10} > \frac{1}{2}$, Hence 4 is the minimum value of a

X	0	1	2	3	4	5	6	7
$F(x) = P(X \leq x)$	0	k=1/10	3k=3/10	5k=5/10	8k=8/10	$8k+k^2$ =81/100	$8k+3k^2$ =83/100	$9k+10k^2$ =1

Example: 2

A random variable X has the following probability function:

x :	0	1	2	3	4	5	6	7	8
P(X=x):	a	3a	5a	7a	9a	11a	13a	15a	17a

(i) Find the value of a, (ii) Find $P(X < 3)$, $P(X \geq 3)$ & $P(0 < X < 3)$, (iii) the distribution function of X.

Solution :

$$(i) \text{ Since } \sum p(x_i) = 1, \Rightarrow a + 3a + 5a + 7a + 9a + 11a + 13a + 15a + 17a = 1$$

$$81a = 1 \quad \therefore a = \frac{1}{81}$$

$$(ii) P(X < 3) = P(X = 0) + P(X = 1) + P(X = 2) = a + 3a + 5a = 9a = \frac{9}{81} = \frac{1}{9}$$

$$P(X \geq 3) = 1 - P(X < 3) = 1 - \frac{1}{9} = \frac{8}{9}$$

$$P(0 < X < 3) = P(X = 1) + P(X = 2) = 3a + 5a + 7a = \frac{8}{81}$$

(iii) Distribution function $F(x)$ of X

X	0	1	2	3	4	5	6	7	8
$F(x) =$ $P(X \leq x)$	a 1/81	4a 4/81	9a 9/81	16a 16/81	25a 25/81	36a 36/81	49a 49/81	64a 64/81	81a 81/81

Example: 3

When a die is thrown, X denotes the number that turn up. Find $E(X)$, $E(X^2)$ and $\text{Var}(X)$.

Solution

Let X is a discrete random variable taking values 1,2,3,4,5,6 and with probability $1/6$ for each.

$$\begin{array}{lclclclcl} x : & 1 & 2 & 3 & 4 & 5 & 6 \\ P(x): & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 \end{array}$$

$$E(X) = \sum x_i P(x_i)$$

$$= 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6}$$

$$= \frac{1}{6} [1 + 2 + 3 + 4 + 5 + 6] = \frac{21}{6} = \frac{7}{2}$$

$$E(X^2) = \sum_i x_i^2 P(x_i) = 1 \times \frac{1}{6} + 4 \times \frac{1}{6} + 9 \times \frac{1}{6} + 16 \times \frac{1}{6} + 25 \times \frac{1}{6} + 36 \times \frac{1}{6}$$

$$= \frac{1}{6} [1 + 4 + 9 + 16 + 25 + 36] = \frac{91}{6}$$

$$\text{Var}(X) = E[X^2] - [E(X)]^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}$$

Example: 4

A random variable X has the following probability distribution

$$\begin{array}{lclclclcl} x : & -2 & -1 & 0 & 1 & 2 & 3 \\ P(x) : & 0.1 & k & 0.2 & 2k & 0.3 & 3k \end{array}$$

Find (i) The value of k (ii) Evaluate $P(X < 2)$ and $P(-2 < X < 2)$

(iii) The cumulative distribution of X (iv) Evaluate the mean of X .

Solution

(i) WKT $\sum p(x_i) = 1$ (ie) $0.1 + k + 0.2 + 2k + 0.3 + 3k = 1$; $6k + 0.6 = 1$

$$6k = 1 - 0.6$$

$$6k = 0.4$$

$$k = \frac{0.4}{6} = \frac{1}{15}$$

$$\begin{aligned}
 \text{(ii) } P(X < 2) &= P(X = -2) + P(X = -1) + P(X = 0) + P(X = 1) \\
 &= 0.1 + k + 0.2 + 2k = 0.3 + 3k = 0.3 + \frac{3}{15} = \frac{1}{2} = 0.5 \\
 P(-2 < X < 2) &= P(X = -1) + P(X = 0) + P(X = 1) \\
 &= k + 0.2 + 2k = 3k + 0.2 = \frac{3}{15} + 0.2 = \frac{2}{5} = 0.4
 \end{aligned}$$

(iii) Cumulative distribution of X:

x	P(x)	F(X)
-2	0.1	$F(-2) = P(X \leq -2) = P(X = -2) = 0.1$
-1	$k = \frac{1}{15}$	$F(-1) = 0.1 + \frac{1}{15} = 0.16 = \frac{1}{6}$
0	0.2	$F(0) = 0.16 + 0.2 = 0.36$
1	$2k = \frac{2}{15}$	$F(1) = P(X \leq 1) = 0.36 + 0.13 = 0.49$
2	0.3	$F(2) = P(X \leq 2) = 0.49 + 0.3 = 0.79$
3	$3k = \frac{3}{15}$	$F(3) = P(X \leq 3) = 0.79 + 0.2 = 0.99 \approx 1$

$$\text{(iv) Mean} = E(X) = \sum x_i P(x_i)$$

$$= -2 \times 0.1 + (-1) \times \left(\frac{1}{15}\right) + 0 \times 0.2 + 1 \times \left(\frac{2}{15}\right) + 2 \times 0.3 + 3 \times \left(\frac{3}{15}\right) = \frac{16}{15}$$

PROBLEMS BASED ON CONTINUOUS RANDOM VARIABLES

Exapmle: 1

A random variable X has the pdf f(x) is given by $f(x) = \begin{cases} cxe^{-x}; & x > 0 \\ 0; & x \leq 0 \end{cases}$ Find

the value of c?

Solution

$$\begin{aligned}
 \text{For a pdf } \int_{-\infty}^{\infty} f(x) dx &= 1; \int_0^{\infty} cxe^{-x} dx = 1; \quad c \left(x \frac{e^{-x}}{-1} - (1) \left(\frac{e^{-x}}{1} \right) \right) \Bigg|_0^{\infty} = 1 \\
 \Rightarrow c(-xe^{-x} - e^{-x}) \Bigg|_0^{\infty} &= 1; \Rightarrow c = 1
 \end{aligned}$$

Exapmle: 2

A continuous random variable X has the pdf f(x) given by $f(x) = \begin{cases} ce^{-|x|}; & -\infty < x < \infty \end{cases}$ Find the value of c

Solution

$$\text{For a pdf } \int_{-\infty}^{\infty} f(x)dx = 1; \quad \int_{-\infty}^{\infty} ce^{-|x|}dx = 1; \quad 2c \int_0^{\infty} e^{-x}dx = 1;$$

$$2c \left(\frac{e^{-x}}{-1} \right)_0^{\infty} = 1; \quad 2c = 1 \quad c = \frac{1}{2}$$

Exapmle: 3

Given that the pdf of random variable X is $f(x) = kx, 0 < x < 1$ Find k and $P(X > 0.5)$

Solution

$$\text{For a pdf } \int_{-\infty}^{\infty} f(x)dx = 1; \quad \int_0^1 kx dx = 1; \quad k \left(\frac{x^2}{2} \right)_0^1 = 1; \quad k = 2$$

$$P(X > 0.5) = \int_{0.5}^{\infty} kx dx = \int_{0.5}^1 2x dx = 2 \left(\frac{x^2}{2} \right)_{0.5}^1 = 1 - \frac{1}{4} = \frac{3}{4}$$

Exapmle: 4

A continuous random variable X follows the probability law $f(x) = Ax^2, 0 \leq x \leq 1$, Determine A and find the probability that X lies between 0.2 and 0.5.

Solution

$$\text{For a pdf } \int_{-\infty}^{\infty} f(x)dx = 1; \quad \int_0^1 Ax^2 dx = 1; \quad A \left(\frac{x^3}{3} \right)_0^1 = 1; \quad A = 3$$

$$P(0.2 \leq X \leq 0.5) = \int_{0.2}^{0.5} Ax^2 dx = \int_{0.2}^{0.5} 3x^2 dx = 3 \left(\frac{x^3}{3} \right)_{0.2}^{0.5} = 0.117$$

Exapmle: 5

If the random variable X has the following cdf $F(x) = 1 - (1+x)e^{-x}$ for $x \geq 0$, then find the probability that it assume the value less than?

Solution

Given $F(x)$ is the cdf of the continuous random variable X. Let f be the pdf of X.

$$\therefore f(x) = \frac{d}{dx} F(x); \quad F(x) = 1 - (1+x)e^{-x}$$

$$\begin{aligned}
&= 0 - \{(1+x)e^{-x}(-1) + e^{-x} \cdot 1\} \\
&= (1+x)e^{-x} - e^{-x} \\
&= e^{-x} + xe^{-x} - e^{-x} \\
&= xe^{-x}
\end{aligned}$$

$$\therefore f(x) = \begin{cases} xe^{-x}; 0 < x < \infty \\ 0; elsewhere \end{cases}$$

$$\begin{aligned}
P(X < 1) &= \int_{-\infty}^1 f(x)dx = \int_0^1 xe^{-x}dx \\
&= \left(x \frac{e^{-x}}{-1} - 1 \cdot \frac{e^{-x}}{1} \right)_0^1 \\
&= -e^{-1} - e^{-1} + 1 \\
&= 1 - 2e^{-1} \\
&= 1 - \frac{2}{e} = \frac{e-2}{e}
\end{aligned}$$

Moments:

If $g(x) = X^r$ for $r = 0, 1, 2, 3, \dots$ then the expectation of X^r is called the r^{th} moment about the origin of the random variable X . It is denoted by μ_r'

$$\mu_r' = E(X^r) = \begin{cases} \sum_x x^r p(x); X - \text{discrete} \\ \int_{-\infty}^{\infty} x^r f(x)dx; X - \text{continuous} \end{cases}$$

Mean:

Let X be a random variable with probability distribution $f(x)$ the mean (or) expected value of X is

$$E(X) = \sum_x xp(x), \text{ if } X \text{ is Discrete.}$$

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx, \text{ if } X \text{ is Continuous}$$

Relations:

$$\begin{aligned}
\mu_1 &= 0 = \mu_1' \text{ (always) } \& \mu_2 = \mu_2' - \mu_1'^2 \\
\mu_3 &= \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3 \quad \& \mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4
\end{aligned}$$

Moment Generating Functions and Properties:

Moment Generating Functions (MGF):

The *mgf* of a random variable X about the origin whose probability distribution is $f(x)$ is given by $E(e^{tx})$ and is denoted by $M_X(t)$.

$$\text{Hence } M_X(t) = E(e^{tx}) = \begin{cases} \sum_x e^{tx} p(x); & X - \text{discrete} \\ \int_{-\infty}^{\infty} e^{tx} f(x) dx; & X - \text{continuous} \end{cases}$$

Properties of (MGF):

Let $Y = aX + b$, where X is a random variable with *mgf*, $M_X(t)$ then

1. $M_Y(t) = E(e^{tY}) = E(e^{t(aX+b)}) = E(e^{taX} \cdot e^{bt}) = e^{bt} E(e^{Xat}) = e^{bt} M_X(at)$
2. $M_{cX}(t) = E(e^{ctX}) = E(e^{X(ct)}) = M_X(ct)$ Where c is constant.
3. If X and Y are two independent random variables then

$$\begin{aligned} M_{X+Y}(t) &= M_X(t) \cdot M_Y(t) \\ M_{X+Y}(t) &= E(e^{t(X+Y)}) = E(e^{tX+tY}) \\ &= E(e^{tX} e^{tY}) \\ &= E(e^{tX}) E(e^{tY}) \\ &= M_X(t) \cdot M_Y(t) \end{aligned}$$

Properties of Mathematical Expectation

If X and Y are random variable's and a and b are constant then

- (i) $E(a) = a$
- (ii) $E(aX) = aE(X)$
- (iii) $E(aX + b) = aE(X) + b$
- (iv) $E(X + Y) = E(X) + E(Y)$
- (v) $E(XY) = E(X)E(Y)$
- (vi) $E(X - \bar{X}) = 0$
- (vii) $E(ag(x)) = aE(g(x))$
- (viii) $E(g(x) + a) = E(g(x)) + a$
- (ix) $E(g(x)) = gE(X)$ if $g(x)$ is linear.

Exapmle: 1

$$\text{For the triangular distribution } f(x) = \begin{cases} x; & 0 < x \leq 1 \\ 2-x; & 1 \leq x < 2 \\ 0; & \text{otherwise} \end{cases} \text{ find the mean,}$$

Variance and the Moment generating function.

Solution

$$\text{Given } f(x) = \begin{cases} x; 0 < x \leq 1 \\ 2-x; 1 \leq x < 2 \\ 0; \text{otherwise} \end{cases}$$

$$\begin{aligned} \text{Mean} = E(X) &= \int_{-\infty}^{\infty} xf(x)dx \\ &= \int_0^1 x^2 dx + \int_1^2 x(2-x)dx \\ &= \int_0^1 x^2 dx + \int_1^2 (2x - x^2)dx \\ &= \left(\frac{x^3}{3}\right)_0^1 + \left(\frac{2x^2}{2} - \frac{x^3}{3}\right)_1^2 \\ &= \frac{1}{3} + \left[\left(4 - \frac{8}{3}\right) - \left(1 - \frac{1}{3}\right)\right] \\ &= \frac{1}{3} + \left[\frac{4}{3} - \frac{2}{3}\right] = \frac{1}{3} + \frac{2}{3} = 1 \end{aligned}$$

$$\text{Variance} = E(X^2) - (E(X))^2$$

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x)dx \\ &= \int_0^1 x^3 dx + \int_1^2 (2x^2 - x^3)dx \\ &= \left(\frac{x^4}{4}\right)_0^1 + \left(\frac{2x^3}{3} - \frac{x^4}{4}\right)_1^2 \\ &= \frac{1}{4} + \left(\frac{16}{3} - \frac{16}{4} - \frac{2}{3} + \frac{1}{4}\right) = \frac{7}{6} \end{aligned}$$

$$\text{Var}(X) = E(X^2) - (E(X))^2 = \frac{7}{6} - 1 = \frac{1}{6}$$

The Moment generating function of the random variable X is $M_X(t) = E(e^{tx})$

$$= \int_{-\infty}^{\infty} e^{tx} f(x)dx = \int_0^1 xe^{tx} dx + \int_1^2 (2-x)e^{tx} dx$$

$$\begin{aligned}
&= \left(\frac{x e^{tx}}{t} - (1) \frac{e^{tx}}{t^2} \right)_0^1 + \left((2-x) \frac{e^{tx}}{t} - (-1) \frac{e^{tx}}{t^2} \right)_1^2 \\
&= \frac{e^t}{t} - \frac{e^t}{t^2} + \frac{1}{t^2} + \frac{e^{2t}}{t^2} - \frac{e^t}{t} - \frac{e^t}{t^2} \\
&= \frac{1}{t^2} [e^{2t} - 2e^t + 1] \\
&= \frac{1}{t^2} [e^t - 1]^2
\end{aligned}$$

Exapmle: 2

The density function of a random variable X is given by

$f(x) = kx(2-x), 0 \leq x \leq 2$. Find k , mean, variance, r^{th} moment.

Solution

Given $f(x) = kx(2-x), 0 \leq x \leq 2$ is a pdf then $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\Rightarrow \int_0^2 k.x(2-x) dx = 1$$

$$\Rightarrow k \int_0^2 (2x - x^2) dx = 1$$

$$\Rightarrow k \left(\frac{2x^2}{2} - \frac{x^3}{3} \right)_0^2 = 1$$

$$\Rightarrow k \left(4 - \frac{8}{3} \right) = 1$$

$$\Rightarrow k \left(\frac{4}{3} \right) = 1$$

$$k = \frac{3}{4}$$

$$Mean = E(X) = \int_{-\infty}^{\infty} x f(x) dx = \frac{3}{4} \int_0^2 x^2 (2-x) dx$$

$$= \frac{3}{4} \int_0^2 (2x^2 - x^3) dx$$

$$\begin{aligned}
&= \frac{3}{4} \left(\frac{2x^3}{3} - \frac{x^4}{4} \right)_0^2 \\
&= \frac{3}{4} \left(\frac{16}{3} - \frac{16}{4} \right) \\
&= \frac{3}{4} \left(\frac{4}{3} \right) = 1
\end{aligned}$$

$$\begin{aligned}
E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx = \frac{3}{4} \int_0^2 x^3 (2-x) dx = \frac{3}{4} \int_0^2 (2x^3 - x^4) dx \\
&= \frac{3}{4} \left(\frac{2x^4}{4} - \frac{x^5}{5} \right)_0^2 \\
&= \frac{3}{4} \left(\frac{32}{4} - \frac{32}{5} \right) \\
&= \frac{3}{4} \left(\frac{40-32}{5} \right) \\
&= \frac{3}{4} \left(\frac{8}{5} \right) = \frac{6}{5}
\end{aligned}$$

$$Var(X) = E(X^2) - (E(X))^2 = \frac{6}{5} - 1 = \frac{1}{5}$$

$$\begin{aligned}
\mu_r' &= E(X^r) = \int_{-\infty}^{\infty} x^r f(x) dx \Rightarrow \int_0^2 x^r k \cdot x(2-x) dx = \frac{3}{4} \int_0^2 (2x^{r+1} - x^{r+2}) dx \\
&= \frac{3}{4} \left(\frac{2x^{r+2}}{r+2} - \frac{x^{r+3}}{r+3} \right)_0^2 \\
&= \frac{3}{4} \left(2 \frac{2^{r+2}}{r+2} - \frac{2^{r+3}}{r+3} \right) = \frac{3}{4} \left(\frac{2^{r+3}}{r+2} - \frac{2^{r+3}}{r+3} \right) \\
&= \frac{3}{4} 2^{r+3} \left(\frac{1}{r+2} - \frac{1}{r+3} \right) \\
&= \frac{3}{4} 2^{r+3} \left(\frac{r+3-r-2}{(r+2)(r+3)} \right) \\
\mu_r' &= \frac{3}{4} 2^{r+3} \left(\frac{1}{(r+2)(r+3)} \right)
\end{aligned}$$

Exapmle: 3

Find the moment generating function of the random variable X whose probability function $P(X = x) = \frac{1}{2^x}$, $x = 1, 2, 3, \dots$, and hence find its Mean.

Solution

$$\begin{aligned}
 M_X(t) &= E(e^{tx}) = \sum_{x=1}^{\infty} e^{tx} f(x) \\
 &= \sum_{x=1}^{\infty} e^{tx} \frac{1}{2^x} = \sum_{x=1}^{\infty} \frac{e^{tx}}{2^x} = \sum_{x=1}^{\infty} \left(\frac{e^t}{2}\right)^x \\
 &= \frac{e^t}{2} + \frac{(e^t)^2}{2^2} + \frac{(e^t)^3}{2^3} + \dots \\
 &= \frac{e^t}{2} \left(1 + \frac{e^t}{2} + \left(\frac{e^t}{2}\right)^2 + \left(\frac{e^t}{2}\right)^3 + \dots\right) \\
 &= \frac{e^t}{2} \left(1 - \frac{e^t}{2}\right)^{-1} \\
 &= \frac{e^t}{2} \left(\frac{2 - e^t}{2}\right)^{-1} = \frac{e^t}{2} \left(\frac{2}{2 - e^t}\right) \\
 M_X(t) &= \left(\frac{e^t}{2 - e^t}\right) \quad \because (1 - x)^{-1} = 1 + x + x^2 + x^3 + \dots
 \end{aligned}$$

$$\begin{aligned}
 \text{Mean } \bar{X} &= M_X'(t) \text{ at } t = 0 \\
 &= \left(\frac{(2 - e^t)e^t - e^t(-e^t)}{(2 - e^t)^2}\right)_{t=0} \\
 &= \left(\frac{(2 - e^t)e^t + e^t \cdot e^t}{(2 - e^t)^2}\right)_{t=0} \\
 &= \left(\frac{(2 - 1) + 1}{(2 - 1)^2}\right)
 \end{aligned}$$

$$\bar{X} = M_X'(t) = 2$$

Exapmle: 4

The density function of a random variable X is given by $f(x) = \begin{cases} kx(2-x)^2; 0 < x < 2 \\ 0; \text{otherwise} \end{cases}$ Find k , Mean and Variance of the distribution.

Solution

Given $f(x) = kx(2-x)^2$, $0 < x < 2$ is a pdf then $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\Rightarrow \int_0^2 k \cdot x(2-x)^2 dx = 1$$

$$\Rightarrow k \int_0^2 x(4+x^2-4x) dx = 1$$

$$\Rightarrow k \int_0^2 (4x+x^3-4x^2) dx = 1$$

$$\Rightarrow k \left(\frac{4x^2}{2} + \frac{x^4}{4} - 4 \frac{x^3}{3} \right)_0^2 = 1$$

$$\Rightarrow k \left(8 - \frac{16}{4} - \frac{32}{3} \right) = 1$$

$$\Rightarrow k \left(12 - \frac{32}{3} \right) = 1$$

$$\Rightarrow k \left(\frac{36-32}{3} \right) = 1$$

$$\Rightarrow k \left(\frac{4}{3} \right) = 1$$

$$k = \frac{3}{4}$$

$$\text{Mean} = E(X) = \int_{-\infty}^{\infty} x f(x) dx = \frac{3}{4} \int_0^2 x^2 (2-x)^2 dx$$

$$= \frac{3}{4} \int_0^2 x^2 (4+x^2-4x) dx$$

$$= \frac{3}{4} \int_0^2 (4x^2+x^4-4x^3) dx$$

$$= \frac{3}{4} \left(\frac{4x^3}{3} + \frac{x^5}{5} - 4 \frac{x^4}{4} \right)_0^2$$

$$= \frac{3}{4} \left(\frac{32}{3} + \frac{32}{5} - 16 \right) = \frac{3}{4} \left(\frac{16}{15} \right)$$

$$\text{Mean} = \bar{X} = \frac{4}{5}$$

$$\text{Variance} = \mu_2 = \mu_2' - \mu_1'^2$$

$$= \int_0^2 x^2 f(x) dx - \left(\int_0^2 x f(x) dx \right)^2$$

$$= \frac{3}{4} \int_0^2 x^2 \cdot x(2-x)^2 dx - \left(\frac{4}{5}\right)^2$$

$$= \frac{3}{4} \int_0^2 x^3 (4 + x^2 - 4x) dx - \left(\frac{4}{5}\right)^2$$

$$= \frac{3}{4} \int_0^2 (4x^3 + x^5 - 4x^4) dx - \left(\frac{4}{5}\right)^2$$

$$= \frac{3}{4} \left(\frac{4x^4}{4} + \frac{x^6}{6} - \frac{4x^5}{5} \right)_0^2 - \left(\frac{4}{5}\right)^2$$

$$= \frac{3}{4} \left(16 + \frac{64}{6} - \frac{128}{5} \right) - \left(\frac{4}{5}\right)^2$$

$$= \frac{3}{4} \left(\frac{16}{15} \right) - \left(\frac{4}{5}\right)^2$$

$$= \left(\frac{4}{5}\right) - \left(\frac{4}{5}\right)^2$$

$$= \left(\frac{4}{5}\right) \left(1 - \frac{4}{5}\right) = \left(\frac{4}{5}\right) \left(\frac{1}{5}\right)$$

$$\text{Variance} = \left(\frac{4}{25}\right)$$

UNIT-2

STANDARD DISTRIBUTIONS

Discrete Distributions:

Binomial Distribution

A random variable X is said to follow binomial distribution with parameter n and p if its probability mass function is given by

$$P(X = x) = \begin{cases} nC_x p^x q^{n-x}; & x = 0, 1, 2, 3, \dots, n \\ 0; & \text{otherwise} \end{cases} \quad X \text{ is called binomial distribution}$$

random variable.

Moment Generating function (MGF) of Binomial Distribution

Binomial distribution is $p(x) = P(X = x) = nC_x p^x q^{n-x}; x = 0, 1, 2, 3, \dots, n$

$$\begin{aligned} M_X(t) &= E(e^{tx}) = \sum_{x=0}^{\infty} e^{tx} p(x) \\ &= \sum_{x=0}^n e^{tx} nC_x p^x q^{n-x} \\ &= \sum_{x=0}^n nC_x (pe^t)^x q^{n-x} \\ &= q^n + nC_1 q^{n-1} pe^t + nC_2 q^{n-2} (pe^t)^2 + \dots \end{aligned}$$

$$M_X(t) = (q + pe^t)^n$$

$$[\because (x+a)^n = nC_0 x^n + nC_1 x^{n-1} a^1 + nC_2 x^{n-2} a^2 + \dots + nC_r x^{n-r} a^r + \dots + nC_n a^n]$$

Mean and Variance of Binomial Distribution

We know that $M_X(t) = (q + pe^t)^n$

$$M_X'(t) = n(q + pe^t)^{n-1} \cdot pe^t$$

$$M_X'(t) = npe^t (q + pe^t)^{n-1}; (t = 0) \dots \dots \dots (1)$$

$$\begin{aligned} M_X'(0) &= np(q + p)^{n-1} \\ &= np \end{aligned}$$

$$\text{Mean} = \bar{X} = M_X'(0) = np = E(x) = \mu_1'$$

$$\text{From (1), } M_X''(t) = np[e^t(n-1)(q + pe^t)^{n-2} pe^t + e^t(q + pe^t)^{n-1}] \Big|_{t=0}$$

$$\begin{aligned}
&= np[(n-1)(q+p)^{n-2}p + (q+p)^{n-1}] \\
&= np[(n-1)p + 1] \\
&= np[1 - p + np] \\
&= np[q + np] \\
&= npq + n^2 p^2 \\
M_x''(0) = \mu_2' &= npq + n^2 p^2 \\
\text{Variance} = \mu_2 = \mu_2' - \mu_1'^2 \\
&= npq + n^2 p^2 - n^2 p^2 \\
&= npq
\end{aligned}$$

$\therefore \text{Mean} = np$ and $\text{Variance} = npq$

Examples on Binomial Distribution:

Example:1

The mean of a binomial distribution is 20 and standard deviation is 4. Find the parameters of the distribution.

Solution

$$\text{Mean} = np = 20 \text{ and } \text{Variance} = npq = 16$$

$$npq = 16 \quad 20q = 16 \quad q = \frac{4}{5}$$

$$p + q = 1 \quad p = 1 - q \quad p = 1 - \frac{4}{5} \quad p = \frac{1}{5}$$

$$np = 20 \quad n\left(\frac{1}{5}\right) = 20 \quad n = 100$$

Example: 2

The probability of a bomb hitting a target is $\frac{1}{5}$. Two bombs are enough to destroy a bridge. If six bombs are aimed at the bridge; find the probability that the bridge is destroyed?

Solution

Let P denote the probability of a bomb hitting the target

$$\text{Given } p = \frac{1}{5}; p + q = 1 \Rightarrow q = 1 - p \Rightarrow q = 1 - \frac{1}{5} \Rightarrow q = \frac{4}{5}$$

Let X denote the number of bombs that destroy a bridge. The random variable X follows a binomial distribution with parameters $n = 6$; $p = \frac{1}{5}$ and is denoted by

$$X \sim B(n, p), \quad X \sim B\left(6, \frac{1}{5}\right)$$

To find $P(X = 2)$

The probability distribution is given by $P(X = x) = {}^nC_x p^x q^{n-x}$

$$\begin{aligned}
 P(X = 2) &= 6C_2 \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^4 \\
 P(X = 2) &= 15 \left(\frac{1}{25}\right) \left(\frac{256}{625}\right) \\
 &= \frac{768}{3125} \\
 &= 0.2458
 \end{aligned}$$

Example: 3

For a binomial distribution of mean 4 and variance 2, find the probability of getting (i) at least 2 success (ii) at most 2 success (iii) $p(5 \leq X \leq 7)$

Solution

The binomial distribution is $P(X = x) = nC_x p^x q^{n-x}$

$$\text{Mean} = np = 4 \text{ and } \text{Variance} = npq = 2; \quad 4q = 2 \quad q = \frac{1}{2}$$

$$p + q = 1; \quad p = 1 - q; \quad p = 1 - \frac{1}{2}; \quad p = \frac{1}{2}; \quad np = 4; \quad n\left(\frac{1}{2}\right) = 4; \quad n = 8$$

$$\begin{aligned}
 \text{(i) } P(\text{At least 2 success}) &= P(X \geq 2) \\
 &= 1 - P(X < 2) \\
 &= 1 - [P(X = 0) + P(X = 1)] \\
 &= 1 - \left[8C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^8 + 8C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^7 \right] \\
 &= 1 - \left[\left(\frac{1}{2}\right)^8 + 8 \left(\frac{1}{2}\right)^8 \right] \\
 &= 1 - 9 \left(\frac{1}{2}\right)^8 \\
 &= 1 - \left(\frac{9}{256}\right) \\
 &= \frac{247}{256}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } P(\text{at most 2 success}) &= P(X \leq 2) \\
 &= [P(X = 0) + P(X = 1) + P(X = 2)] \\
 &= \left[8C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^8 + 8C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^7 + 8C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^6 \right]
 \end{aligned}$$

$$= \left[\left(\frac{1}{2} \right)^8 + 8 \left(\frac{1}{2} \right)^8 + 28 \left(\frac{1}{2} \right)^8 \right] = 37 \left(\frac{1}{2} \right)^8 = \frac{37}{256}$$

$$\begin{aligned} \text{(iii) } p(5 \leq X \leq 7) &= [P(X=5) + P(X=6) + P(X=7)] \\ &= \left[8C_5 \left(\frac{1}{2} \right)^5 \left(\frac{1}{2} \right)^3 + 8C_6 \left(\frac{1}{2} \right)^6 \left(\frac{1}{2} \right)^2 + 8C_7 \left(\frac{1}{2} \right)^1 \left(\frac{1}{2} \right)^7 \right] \\ &= \left[56 \left(\frac{1}{2} \right)^8 + 28 \left(\frac{1}{2} \right)^8 + 8 \left(\frac{1}{2} \right)^8 \right] \\ &= 92 \left(\frac{1}{2} \right)^8 = \frac{92}{256} \end{aligned}$$

Poisson Distribution:

Definitions

A random variable X is said to follow Poisson distribution with parameter λ if its pdf is given by $P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}; x=0,1,2,3,\dots$

Moment Generating function (MGF) of Poisson distribution

Poisson distribution is given by $P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}; x=0,1,2,3,\dots$ By the definition of Moment Generating function (MGF) is

$$\begin{aligned} M_X(t) &= E(e^{tx}) = \sum_{x=0}^{\infty} e^{tx} p(x) \\ &= \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!} \\ &= e^{-\lambda} \left(1 + \frac{\lambda e^t}{1!} + \frac{(\lambda e^t)^2}{2!} + \dots \right) \\ &= e^{-\lambda} e^{\lambda e^t} \\ M_X(t) &= e^{\lambda(e^t - 1)} \end{aligned}$$

Mean and Variance of Poisson distribution

$$\begin{aligned} \text{Mean} = \mu_1' &= \frac{d}{dt} (M_X(t))_{t=0} \\ &= \frac{d}{dt} (e^{\lambda(e^t - 1)})_{t=0} \\ &= (e^{\lambda(e^t - 1)} \cdot \lambda e^t)_{t=0} \\ &= \lambda (e^{\lambda(e^t - 1) + t})_{t=0} \end{aligned}$$

$$\text{Mean} = \mu_1' = \lambda$$

$$\begin{aligned}\mu_2' &= \frac{d^2}{dt^2} (M_X(t))_{t=0} \\ &= \frac{d}{dt} \left(\lambda e^{\lambda(e^t - 1) + t} \right)_{t=0} \\ &= \lambda \left(\lambda e^{\lambda(e^t - 1) + t} (\lambda e^t + 1) \right)_{t=0} \\ &= \lambda(\lambda + 1) \\ \mu_2' &= \lambda^2 + \lambda\end{aligned}$$

$$\begin{aligned}\text{Variance} = \mu_2 - \mu_1'^2 \\ = \lambda^2 + \lambda - \lambda^2 = \lambda\end{aligned}$$

The Poisson distribution with parameter $\text{Mean} = \lambda; \text{Variance} = \lambda$

Example: 1

In a Poisson variate X is such that $P(X = 1) = 2P(X = 2)$ Find $P(X = 0)$ and $\text{Var}(X)$.

Solution

If X is a Poisson random variable with parameter λ then

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, 2, 3, \dots$$

$$\text{Given } P(X = 1) = 2P(X = 2)$$

$$\frac{e^{-\lambda} \lambda^1}{1!} = 2 \frac{e^{-\lambda} \lambda^2}{2!} \quad \lambda = 1$$

$$\text{To find } P(X = 0) = \frac{e^{-\lambda} \lambda^0}{0!} = e^{-\lambda}$$

$$\text{Put } \lambda = 1; e^{-\lambda} = e^{-1}$$

$$\text{Var}(X) = \lambda = 1$$

Example: 2

In a company the monthly break down of a machine is a random variable with Poisson distribution with an average 1.8, find the probability that the machine will function for a month

- (i) With out break down
- (ii) With exactly one break down
- (iii) With at least one break down

Solution

Let X denotes the number of break down in a month.

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

With mean $= 1.8 = \lambda$

$$P(X = x) = \frac{e^{-1.8}(1.8)^x}{x!}$$

$$(i) P(\text{With out break down}) = P(X = 0) = \frac{e^{-1.8}(1.8)^0}{0!} = e^{-1.8} = 0.1653$$

$$(ii) P(\text{With exactly one break down}) = P(X = 1) = \frac{e^{-1.8}(1.8)^1}{1!} = e^{-1.8}(1.8)^1 = 0.2975$$

$$\begin{aligned} (iii) P(\text{With at least one break down}) &= P(X \geq 1) = 1 - P(X < 1) \\ &= 1 - P(X = 0) \\ &= 1 - 0.1653 \\ &= 0.8347 \end{aligned}$$

Example:3

It is known that the probability of an item produced by a certain machine will defective is 0.05. If the produced item was sent to the market in packets of 20, find the number of packets containing at least, exactly and at most 2 defective items in a consignment of 1000 packets using Poisson approximation.

Solution

Given $n = 20$; $P = 0.05$ and $\lambda = np = 1$ Let X is a Poisson variate.

$$\begin{aligned} P(X = x) &= \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, 2, \dots \\ &= \frac{e^{-1} 1^x}{x!} = \frac{e^{-1}}{x!} \end{aligned}$$

$$P(X = 2) = \frac{e^{-1}}{2!} = 0.1839$$

The total number of packets containing **exactly two defective** items in a consignment of 1000 packets is

$$\begin{aligned} &= N \times P(\text{at least two defective}) \\ &= 1000 \times 0.1839 \\ &= 184 \text{ packets.} \\ P(X \geq 2) &= 1 - P(X < 2) \\ &= 1 - [P(X = 0) + P(X = 1)] \\ &= 1 - \frac{e^{-1}}{0!} - \frac{e^{-1}}{1!} \\ &= 0.2642 \end{aligned}$$

The total number of packets containing **atleast two defective** items in a consignment of 1000 packets.

$$\begin{aligned} &= N \times P(\text{at least two defective}) \\ &= 1000 \times 0.2642 \\ &= 264 \text{ packets.} \\ P(X \leq 2) &= P(X = 0) + P(X = 1) + P(X = 2) \end{aligned}$$

$$\begin{aligned}
&= \frac{e^{-1}}{0!} + \frac{e^{-1}}{1!} + \frac{e^{-1}}{2!} \\
&= e^{-1} \left[1 + 1 + \frac{1}{2} \right] \\
&= 0.9197
\end{aligned}$$

The total number of packets containing **atmost two defective** items in a consignment of 1000 packets.

$$= N \times P(\text{at least two defective})$$

$$= 1000 \times 0.9197 = 920 \text{ packets.}$$

Example 4:

In a data entry job, the variance of the number of mistakes per page is 0.3. what percentage of pages will have i) No mistakes, ii) One mistake and iii) Two mistakes. iv) In book containing 500 pages, how many pages are expected to have two or more mistakes?

Solution:

Here 'X number of mistakes per page' is a Poisson variate with variance = 0.3

$$\therefore \text{Variance} = \lambda = 0.3$$

$\therefore$ By Poisson probability law,

$$P(x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

$$\begin{aligned}
\therefore \text{i.e. } P(x) &= \frac{e^{-0.3} (0.3)^x}{x!} \\
&= 0.7408 \frac{(0.3)^x}{x!}
\end{aligned}$$

$$\begin{aligned}
\text{i) } \therefore P(\text{No mistake}) &= P(x = 0) \\
&= 0.7408 \frac{(0.3)^0}{0!} \\
&= 0.7408
\end{aligned}$$

$$\begin{aligned}
\therefore \text{Percentage of page with no mistakes} \\
&= 100 \times (0.7408) = 74.08\%
\end{aligned}$$

$$\begin{aligned}
\text{ii) } P(\text{one mistake}) &= P(x = 1) \\
&= 0.7408 \frac{(0.3)^1}{1!} \\
&= 0.222
\end{aligned}$$

$$\begin{aligned}
\therefore \text{Percentage of page with one mistakes} \\
&= 100 \times (0.222) = 22.2\%
\end{aligned}$$

$$\begin{aligned}
\text{iii) } P(\text{two mistake}) &= P(x = 2) \\
&= 0.7408 \frac{(0.3)^2}{2!} \\
&= 0.7408 \frac{0.09}{2} = 0.0333
\end{aligned}$$

$\therefore$ Percentage of page with two mistakes

$$\begin{aligned}
 &= 100 \times (0.0333) = 3.33 \% \\
 \text{iv) } P(\text{two or more mistakes}) &= P(x \geq 2) \\
 &= 1 - P(x < 2) \\
 &= 1 - P(x = 0 \text{ or } 1) \\
 &= 1 - [P(0) + P(1)] \\
 &= 1 - [0.7408 + 0.222] \\
 &= 0.0372
 \end{aligned}$$

$\therefore$ Expected number of pages in a book having (N = 500) pages with two mistakes is

$$\begin{aligned}
 f(x \geq 2) &= N \times P(x \geq 2) \\
 &= 500 \times 0.0372 \\
 &= 18.6 \\
 &= 19
 \end{aligned}$$

Geometric Distribution:

Definition

The discrete random variable X, which assumes non-negative values such that its pmf is given by $P(X = x) = q^x p; x = 0, 1, 2, 3, \dots, \infty$

MGF, MEAN VARIANCE AND STANDARD DEVIATION OF GEOMETRIC DISTRIBUTION

As the Geometric distribution is a discrete distribution by definition of mgf .we have

$$\begin{aligned}
 M_X(t) &= E(e^{tx}) = \sum_{x=0}^{\infty} e^{tx} p(x) \\
 &= \sum_{x=0}^{\infty} e^{tx} q^x p = p \sum_{x=0}^{\infty} (qe^t)^x \\
 &= p \left[(qe^t)^0 + (qe^t)^1 + (qe^t)^2 + \dots \right] \\
 &= p \left[1 + (qe^t) + (qe^t)^2 + \dots \right] = p [1 - qe^t]^{-1} \\
 &= \frac{p}{1 - qe^t}
 \end{aligned}$$

$$\begin{aligned}
 \text{Mean} = \mu'_1 &= \frac{d}{dt} [M_X(t)] \\
 &= \frac{d}{dt} \left[\frac{p}{1 - qe^t} \right] \\
 &= \left[\frac{(1 - qe^t)0 - p(-qe^t)}{(1 - qe^t)^2} \right]_{t=0} \\
 \mu'_1 &= \left[\frac{pqe^t}{(1 - qe^t)^2} \right]_{t=0} = \frac{pq}{p^2} = \frac{q}{p}
 \end{aligned}$$

$$\begin{aligned}
\mu_2' &= \frac{d}{dt} \left[\frac{pqe^t}{(1-qe^t)^2} \right]_{t=0} \\
&= pq \left[\frac{(1-qe^t)^2 e^t - e^t 2(1-qe^t)(-qe^t)}{(1-qe^t)^4} \right]_{t=0} \\
&= pq \left[\frac{(1-q)^2 + 2q(1-q)}{(1-q)^4} \right] \\
&= pq \left[\frac{1-q+2q}{(1-q)^3} \right] \\
&= pq \left[\frac{1+q}{p^3} \right] = \frac{q}{p^2} + \frac{q^2}{p^2} \\
\text{var}(x) &= \mu_2' - \mu_1'^2 \\
&= \frac{q}{p^2} + \frac{q^2}{p^2} - \frac{q^2}{p^2} = \frac{q}{p^2} \\
S.D &= \sqrt{\text{var}(x)} = \frac{\sqrt{q}}{p}
\end{aligned}$$

Example: 1

State and prove memoryless property of geometric distribution.

Statement

If X is a discrete random variable following Geometric Distribution then $P(X \geq m+n | X \geq m) = P(X \geq n)$ where m & n are any two non-negative integers.

Proof

Since X follows Geometric Distribution

$$P(X = x) = q^x p; \quad x = 0, 1, 2, 3, \dots, \infty; \quad p + q = 1$$

$$\begin{aligned}
P(X \geq k) &= \sum_{x=k}^{\infty} q^x p = q^k p + q^{k+1} p + q^{k+2} p + q^{k+3} p + \dots \\
&= q^k p (1 + q + q^2 + \dots) \\
&= q^k p (1 - q)^{-1} \\
&= \frac{q^k p}{1 - q} = \frac{q^k p}{p} = q^k
\end{aligned}$$

$$P(X \geq k) = q^k$$

Now consider the conditional probability of

$$P(X \geq m+n | X \geq m) = \frac{P(X \geq m+n \text{ and } X \geq m)}{P(X \geq m)}$$

$$\begin{aligned}
 &= \frac{P(X \geq m+n)}{P(X \geq m)} \\
 &= \frac{q^{m+n}}{q^m} = \frac{q^m q^n}{q^m} = q^n = P(X \geq n)
 \end{aligned}$$

Example: 2

Let X and Y be independent random variables such that $P(X = x) = P(Y = y) = q^x p$; $x = 0, 1, 2, \dots$ where $p + q = 1$. i. Find the distribution of $X+Y$ ii. The conditional distribution of X given $X + Y = n$

Solution:-

$$\begin{aligned}
 (i). P(X + Y = n) &= \sum_{x=0}^n P[X = x \cap Y = n - x] \\
 &= \sum_{x=0}^n P(X = x) \cdot P(Y = n - x) \\
 &= \sum_{x=0}^n q^x p q^{n-x} p \\
 &= p^2 \sum_{x=0}^n q^n = p^2 q^n \sum_{x=0}^n 1 \\
 &= p^2 q^n \left(\underbrace{1+1+1+\dots+1}_{n+1} \right) \\
 &= p^2 q^n (n+1) \\
 (ii). P\left(X = x / X + Y = n\right) &= \frac{P(X = x \cap X + Y = n)}{P(X + Y = n)} \\
 &= \frac{P(X = x \cap Y = n - x)}{P(X + Y = n)} \\
 &= \frac{P(X = x) \cdot P(Y = n - x)}{P(X + Y = n)} \\
 &= \frac{q^x p \cdot q^{n-x} p}{p^2 q^n (n+1)} = \frac{1}{n+1}
 \end{aligned}$$

Example: 3

If X is a Geometric Distribution taking the values $1, 2, 3, \dots, \infty$ Find $P(x \text{ is odd})$

Solution:

pmf Of Geometric Distribution is

$$P(X = x) = q^{x-1} p; x = 1, 2, 3, \dots, \infty$$

$$P(x \text{ is odd}) = P(x = 1, 3, 5, \dots)$$

$$\begin{aligned}
&= P(X=1) + P(X=3) + P(X=5) + \dots \\
&= p + q^2 p + q^4 p + \dots \\
&= p(1 + q^2 + q^4 + \dots) \\
&= p(1 - q^2)^{-1} \\
&= \frac{p}{1 - q^2} = \frac{1 - q}{(1 - q)(1 + q)} \\
&= \frac{1}{1 + q}
\end{aligned}$$

CONTINUOUS DISTRIBUTIONS:

Uniform Distribution:

The continuous random variable in the interval (a, b) is said to follow uniform distribution if its probability density function is given by

$$f(x) = \begin{cases} \frac{1}{b-a}; & a \leq x \leq b \\ 0; & \text{otherwise} \end{cases}$$

Mean and Variance of Uniform Distribution

The probability of uniform distribution (a, b) is given by $f(x) = \begin{cases} \frac{1}{b-a}; & a \leq x \leq b \\ 0; & \text{otherwise} \end{cases}$

$$\begin{aligned}
\text{Mean} = E(X) &= \int_a^b x f(x) dx \\
&= \int_a^b x \frac{1}{b-a} dx \\
&= \frac{1}{b-a} \int_a^b x dx \\
&= \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b \\
&= \frac{b^2 - a^2}{2(b-a)} = \frac{(b-a)(b+a)}{2(b-a)} = \frac{(b+a)}{2}
\end{aligned}$$

$$E(X) = \frac{b+a}{2}$$

$$E(X^2) = \int_a^b x^2 f(x) dx$$

$$\begin{aligned}
&= \int_a^b x^2 \frac{1}{b-a} dx \\
&= \frac{1}{b-a} \int_a^b x^2 dx \\
&= \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b \\
&= \frac{b^3 - a^3}{3(b-a)} = \frac{(b-a)(b^2 + ba + a^2)}{3(b-a)} = \frac{b^2 + ba + a^2}{3} \\
E(X^2) &= \frac{b^2 + ba + a^2}{3}
\end{aligned}$$

$$\begin{aligned}
Var(X) &= E(X^2) - (E(X))^2 = \frac{b^2 + ba + a^2}{3} - \frac{(b+a)^2}{4} \\
&= \frac{4(b^2 + ba + a^2) - 3(b^2 + 2ba + a^2)}{12} \\
&= \frac{4b^2 + 4ba + 4a^2 - 3b^2 - 6ba - 3a^2}{12} \\
&= \frac{1}{12} [b^2 - 2ba + a^2] = \frac{(b-a)^2}{12} \\
Var(X) &= \frac{(b-a)^2}{12}
\end{aligned}$$

Moment Generating function (MGF) of Uniform Distribution

$$\begin{aligned}
M_X(t) &= E(e^{tx}) = \int_a^b e^{tx} f(x) dx \\
&= \int_a^b e^{tx} \frac{1}{b-a} dx \\
&= \frac{1}{b-a} \int_a^b e^{tx} dx \\
&= \frac{1}{b-a} \left[\frac{e^{tx}}{t} \right]_a^b \\
&= \frac{e^{bx} - e^{ax}}{t(b-a)}
\end{aligned}$$

Example: 1

Show that the uniform distribution: $f(x) = \begin{cases} \frac{1}{2a}; -a < x < a \\ 0; \text{elsewhere} \end{cases}$ the moment generating

function about the origin is $\frac{1}{at} \sinh at$, Also moments of even order are given by

$$\mu_{2n} = \frac{a^{2n}}{2n+1}$$

Solution

Moment generating function about the origin is given by

$$\begin{aligned} M_X(t) &= E(e^{tx}) = \int_a^b e^{tx} f(x) dx \\ &= \int_{-a}^a e^{tx} \frac{1}{2a} dx = \frac{1}{2a} \int_{-a}^a e^{tx} dx \\ &= \frac{1}{2a} \left[\frac{e^{tx}}{t} \right]_{-a}^a = \frac{1}{at} \left[\frac{e^{at} - e^{-at}}{2} \right] \\ M_X(t) &= \frac{\sinh at}{at} \\ &= \frac{1}{at} \left[at + \frac{(at)^3}{3!} + \dots \right] \\ &= \frac{at}{at} \left[1 + \frac{a^2 t^2}{3!} + \frac{a^4 t^4}{5!} + \dots + \frac{a^{2n} t^{2n}}{(2n+1)!} \right] \\ &= 1 + \frac{a^2 t^2}{3!} + \frac{a^4 t^4}{5!} + \dots + \frac{a^{2n} t^{2n}}{(2n+1)!} \\ &= 1 + \frac{a^2 t^2}{2! \cdot 3} + \frac{a^4 t^4}{4! \cdot 5} + \dots + \frac{a^{2n} t^{2n}}{2n! (2n+1)} \end{aligned}$$

There are no add power of t is $M_X(t)$. All moments of add order vanish.

$$(ie) \mu_1' = \mu_3' = 0 \dots \mu_{2n+1}' = 0$$

The moment of even order is given by

$$\mu_{2n} = \text{coefficient of } \frac{t^{2n}}{2n!} \text{ in } M_X(t) = \frac{a^{2n}}{2n+1}$$

Example: 2

Let X be a uniform distributed r.v's over (0, 1) Determine the moment generating function.

Solution

Given X be a uniform distributed random variable over (0, 1) then

$$f(x) = \frac{1}{b-a}; a \leq x \leq b; \quad f(x) = \frac{1}{1-0}; 0 < x < 1$$

The moment generating function $f(x)$ is given by

$$\begin{aligned} M_X(t) &= E(e^{tx}) = \int_a^b e^{tx} f(x) dx \\ &= \int_0^1 e^{tx} dx \\ &= \left[\frac{e^{tx}}{t} \right]_0^1 = \frac{e^t - 1}{t} \\ M_X(t) &= \frac{e^t - 1}{t} \\ &= \frac{1}{t} (e^t - 1) = \frac{1}{t} \left(\left[1 + \frac{t}{1!} + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots \right] - 1 \right) \\ &= \frac{1}{1!} + \frac{t}{2!} + \frac{t^2}{3!} + \dots \end{aligned}$$

$$\text{Mean} = \mu_1' = \text{coefficient of } t = \frac{1}{2}$$

$$\mu_2' = \text{coefficient of } \frac{t^2}{2!} = \frac{1}{3}$$

$$\text{Var}(X) = \mu_2' - \mu_1'^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

Example: 3

Electric trains in a particular route run every half an hour between 12.am and 6.am. Find the probability that a passenger entering the station at any time during this period will have to wait at least twenty minutes?

Solution

Let X is a random variable denote the waiting time (in minutes) of a passenger for the next train. Given that a man arrives at the station at random.

$\Rightarrow$ X is distributed uniformly on (0, 30) with density.

$$f(x) = \frac{1}{b-a}; a \leq x \leq b; \quad f(x) = \frac{1}{30-0}; 0 < x < 30$$

The probability that he has to wait for at least 20 minutes is

$$P(X \geq 20) = \int_{20}^{30} f(x) dx$$

$$\begin{aligned}
&= \frac{1}{30} \int_{20}^{30} dx \\
&= \frac{1}{30} [x]_{20}^{30} \\
&= \frac{1}{30} [30 - 20] \\
&= \frac{10}{30} = \frac{1}{3}
\end{aligned}$$

Exponential Distribution:

Definition

A continuous random variable X defined in $(0, \infty)$ is said to follow an exponential distribution if the probability density function is given by $f(x) = \lambda e^{-\lambda x}; \lambda > 0; 0 < x < \infty$

MGF, Mean & Variance of Exponential Distribution

The *pdf* of an exponential distribution is given by $f(x) = \begin{cases} \lambda e^{-\lambda x}; x \geq 0 \\ 0; \text{otherwise} \end{cases}$ and the

moment generating function is given by $M_X(t) = E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$

$$M_X(t) = \frac{\lambda}{\lambda - t}$$

Mean and Variance:

$$M_X'(t) = \frac{(\lambda - t) \cdot 0 - \lambda(-1)}{(\lambda - t)^2} = \frac{\lambda}{(\lambda - t)^2}$$

$$M_X'(0) = \frac{\lambda}{(\lambda)^2} = \frac{1}{\lambda}$$

$$\mu_1' = \text{Mean} = M_X'(0) = \frac{1}{\lambda}$$

$$M_X''(0) = \frac{2\lambda}{(\lambda)^3} = \frac{2}{\lambda^2}$$

$$\mu_2' = M_X''(0) = \frac{2}{\lambda^2}$$

$$\text{Variance} = \mu_2 = \mu_2' - \mu_1'^2$$

$$= \frac{2}{\lambda^2} - \frac{1}{\lambda}$$

$$= \frac{1}{\lambda^2}$$

Example: 1

State and prove Memory less property of Exponential Distribution

Statement

If X is exponential distributed, then $P(X > s+t/X > s) = P(X > t), \forall s, t > 0$

Proof

$$P(X > K) = \int_k^{\infty} \lambda e^{-\lambda x} dx$$

$$= \lambda \int_k^{\infty} e^{-\lambda x} dx = \lambda \left[\frac{e^{-\lambda x}}{-\lambda} \right]_k^{\infty}$$

$$= e^{-\lambda k} \dots \dots \dots (1)$$

$$P(X > s+t/X > s) = \frac{P(X > s+t \text{ and } X > s)}{P(X > s)}$$

$$= \frac{P(X > s+t)}{P(X > s)} \quad \text{by (1)}$$

$$= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}}$$

$$= e^{-\lambda t}$$

$$= P(X > t)$$

Hence $P(X > s+t/X > s) = P(X > t)$. The converse of this result is also true.

(ie) $P(X > s+t/X > s) = P(X > t)$ Then X follows an exponential distribution.

Example: 2

Find the moment generating function of the exponential distribution

$$f(x) = \frac{1}{c} e^{-\frac{x}{c}}; \quad 0 \leq x < \infty, c > 0.$$

Solution

The moment generating function is defined as $E(e^{tx})$ and is denoted by

$$M_X(t) = E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$$

$$= \int_{-\infty}^{\infty} e^{tx} \frac{1}{c} e^{-\frac{x}{c}} dx = \frac{1}{c} \int_0^{\infty} e^{tx} e^{-\frac{x}{c}} dx$$

$$\begin{aligned}
&= \frac{1}{c} \int_0^{\infty} e^{\frac{-x}{c}(1-ct)} dx \\
&= -\frac{1}{c} \left[\frac{e^{\frac{-x}{c}(1-ct)}}{\frac{1-ct}{c}} \right]_0^{\infty} \\
&= \frac{1}{c} \cdot \frac{c}{1-ct} = \frac{1}{1-ct}
\end{aligned}$$

The *mgf* of an exponential distribution is given by $\frac{1}{1-ct}$, $c > 0$

To find Mean (or) the first moment

$$M_X'(t) = \frac{(1-ct) \cdot 0 - 1(-c)}{(1-ct)^2} = \frac{c}{(1-ct)^2}$$

$$M_X'(0) = c$$

$$\mu_1' = \text{Mean} = M_X'(0) = c$$

$$M_X''(t) = \frac{(1-ct)^2 \cdot 0 - c \cdot 2(1-ct)(-c)}{(1-ct)^4}$$

$$= \frac{2c^2(1-ct)}{(1-ct)^4} = \frac{2c^2}{(1-ct)^3}$$

$$\mu_2' = M_X''(0) = 2c^2$$

$$\begin{aligned}
\text{Variance} = \mu_2 &= \mu_2' - \mu_1'^2 \\
&= 2c^2 - c^2 = c^2
\end{aligned}$$

$$\text{Standard deviation} = \sqrt{\text{Variance}} = c$$

Example: 3

The length of time a person speaks over phone follows exponential distribution with mean $\frac{1}{6}$. What is the probability that the person will talk for (i) More than 8 minutes (ii) Between 4 and 8 minutes?

Solution: - Given $f(x) = \frac{1}{6} e^{\frac{-x}{6}}$

$$\begin{aligned}
P(X > 8) &= \int_8^{\infty} f(x) dx = \int_8^{\infty} \frac{1}{6} e^{\frac{-x}{6}} dx \\
&= \frac{1}{6} \int_8^{\infty} e^{\frac{-x}{6}} dx
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{6} \left[\frac{e^{-\frac{x}{6}}}{-\frac{1}{6}} \right]_8^\infty = - \left[e^{-\frac{x}{6}} \right]_8^\infty \\
&= - \left[e^{-\infty} - e^{-\frac{8}{6}} \right] = e^{-\frac{8}{6}} = 0.2635 \\
P(4 \leq X \leq 8) &= \int_4^8 \frac{1}{6} e^{-\frac{x}{6}} dx \\
&= \frac{1}{6} \int_4^8 e^{-\frac{x}{6}} dx \\
&= \frac{1}{6} \left[\frac{e^{-\frac{x}{6}}}{-\frac{1}{6}} \right]_4^8 \\
&= - \left[e^{-\frac{x}{6}} \right]_4^8 = - \left[e^{-\frac{8}{6}} - e^{-\frac{4}{6}} \right] \\
&= -[0.2636 - 0.5134] = 0.2498
\end{aligned}$$

Gamma Distribution:

Definition

A continuous random variable is said to follow the gamma distribution with parameter λ

If the probability density function is given by $f(x) = \begin{cases} \frac{e^{-x} x^{\lambda-1}}{\Gamma \lambda}; \lambda > 0; 0 < x < \infty \\ 0; \text{otherwise} \end{cases}$

Note:

1. $\Gamma \lambda = \int e^{-x} x^{\lambda-1} dx; \lambda > 0$
2. $\Gamma \lambda + 1 = \lambda \Gamma \lambda = \lambda!$
3. $\Gamma 1/2 = \sqrt{\pi}$
4. $\Gamma 1 = 1$
5. $\Gamma 0 = \infty$

Generalized Gamma Distribution

A continuous random variable X having the following density function is said to follow generalized gamma distribution with parameter α and λ

$$f(x) = \begin{cases} \frac{e^{-\lambda x} \lambda^\alpha x^{\alpha-1}}{\Gamma \alpha}; \alpha, \lambda > 0; 0 < x < \infty \\ 0; \text{otherwise} \end{cases}$$

Moment Generating Functions (MGF) of a random variable

The *mgf* of a random variable X about the origin whose probability function $f(x)$ is given by $E(e^{tx})$ and is denoted by $M_X(t)$.

$$\text{Hence } M_X(t) = E(e^{tx}) = \begin{cases} \sum_x e^{tx} f(x); X - \text{discrete} \\ \int_{-\infty}^{\infty} e^{tx} f(x) dx; X - \text{continuous} \end{cases}$$

Let X follows gamma distribution with density as bellow.

$$f(x) = \frac{a \lambda e^{-ax} x^{\lambda-1}}{\Gamma \alpha}; a, \lambda > 0; 0 < x < \infty$$

Moment Generating Functions (MGF) of a Gamma Distribution

$$\begin{aligned} M_X(t) &= E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} f(x) dx \\ &= \int_0^{\infty} e^{tx} \frac{e^{-ax} x^{\lambda-1}}{\Gamma \lambda} dx \\ &= \frac{1}{\Gamma \lambda} \int_0^{\infty} e^{tx} e^{-ax} x^{\lambda-1} dx \\ &= \frac{1}{\Gamma \lambda} \int_0^{\infty} e^{-(1-t)x} x^{\lambda-1} dx \end{aligned}$$

$$\begin{array}{lll} \text{Put} & (1-t)x = u & \text{when } x=0; u=0 \\ & (1-t)dx = du & x=\infty; u=\infty \end{array} \quad x = \frac{u}{1-t}$$

$$\begin{aligned} M_X(t) &= \frac{1}{\Gamma \lambda} \int_0^{\infty} e^{-u} \left(\frac{u}{1-t} \right)^{\lambda-1} \left(\frac{du}{1-t} \right) \\ &= \frac{1}{\Gamma \lambda} \int_0^{\infty} e^{-u} \frac{u^{\lambda-1}}{(1-t)^{\lambda-1+1}} du \\ &= \frac{1}{\Gamma \lambda} \cdot \frac{1}{(1-t)^\lambda} \int_0^{\infty} e^{-u} u^{\lambda-1} du \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\Gamma\lambda} \cdot \frac{1}{(1-t)^\lambda} \Gamma\lambda \\
&= \frac{1}{(1-t)^\lambda}; |t| < 1 \quad \left(\Gamma\lambda = \int e^{-x} x^{\lambda-1} dx \right) \\
&= (1-t)^{-\lambda}
\end{aligned}$$

$$M_X(t) = (1-t)^{-\lambda}$$

Mean and Variance of a Gamma Distribution and first four moments

We know that $M_X(t) = (1-t)^{-\lambda}$

$$M_X'(t) = -\lambda(1-t)^{-\lambda-1}(-1)$$

$$M_X'(t) = \lambda(1-t)^{-\lambda-1}$$

$$M_X'(0) = \lambda = \mu_1'$$

$$M_X''(t) = \lambda(-\lambda-1)(1-t)^{-\lambda-2}(-1)$$

$$M_X''(t) = \lambda(\lambda+1)(1-t)^{-\lambda-2}$$

$$M_X''(0) = \lambda(\lambda+1) = \mu_2'$$

$$\text{Mean} = \mu_1' = \lambda$$

$$\text{Variance} = \mu_2 = \mu_2' - (\mu_1')^2$$

$$= \lambda^2 + \lambda - \lambda^2$$

$$= \lambda$$

$$M_X'''(t) = \lambda(\lambda+1)(-\lambda-2)(1-t)^{-\lambda-3}(-1)$$

$$M_X'''(0) = \lambda(\lambda+1)(\lambda+2)(1-t)^{-\lambda-3}$$

$$\mu_3' = \lambda(\lambda+1)(\lambda+2)$$

$$M_X^{iv}(t) = \lambda(\lambda+1)(\lambda+2)(-\lambda-3)(1-t)^{-\lambda-4}(-1)$$

$$M_X^{iv}(0) = \lambda(\lambda+1)(\lambda+2)(\lambda+3)(1-t)^{-\lambda-4}$$

$$\mu_4' = \lambda(\lambda+1)(\lambda+2)(\lambda+3)$$

Additive Property of Gamma Distribution

Statement: If X and Y are independent random variable following gamma distribution with following parameters λ_1 and λ_2 then $X + Y$ also follow gamma distribution with

parameter $-(\lambda_1 + \lambda_2)$

Proof

Let X follows gamma distribution with parameter $\lambda_1 \quad \therefore M_X(t) = (1-t)^{-\lambda_1}$

Let Y follows gamma distribution with parameter $\lambda_2 \quad \therefore M_Y(t) = (1-t)^{-\lambda_2}$

Consider $X + Y$ Where X and Y are independent.

$$\begin{aligned}\therefore M_{X+Y}(t) &= M_X(t) \cdot M_Y(t) \\ &= (1-t)^{-\lambda_1} \cdot (1-t)^{-\lambda_2} \\ &= (1-t)^{-\lambda_1 - \lambda_2} \\ &= (1-t)^{-(\lambda_1 + \lambda_2)}\end{aligned}$$

Hence $X + Y$ also follow gamma distribution with parameter $-(\lambda_1 + \lambda_2)$

Example: 1

The daily consumption of milk in excess of 20,000 liters is approximately distributed as gamma variable with parameter $k = 2$ and $\lambda = \frac{1}{10,000}$ if the city has a daily stock of 30,000 liters on a given day, find the probability that the stock is insufficient.

Solution

Let X is the random variable for daily consumption of milk in the city. Let Y is a gamma variate ($Y = X - 20,000$). The pdf of gamma distribution with parameters

$$k \text{ and } \lambda. \quad k = 2 \text{ and } \lambda = \frac{1}{10,000}$$

$$f(x) = \frac{\lambda^k x^{k-1} e^{-\lambda x}}{\Gamma k}; x \geq 0 \quad \text{Here } x = y$$

$$f(y) = \frac{\left(\frac{1}{10,000}\right)^2 y^{2-1} e^{-\left(\frac{1}{10,000}\right)y}}{\Gamma 2}; y \geq 0$$

$$f(y) = \frac{1}{(10,000)^2} e^{-\left(\frac{1}{10,000}\right)y} y \quad \text{By note } (\Gamma 2 = \Gamma 1 + 1 = 1 \Gamma 1 = 1! = 1)$$

Stock of milk is 30,000 liters,

$$\begin{aligned}P(\text{for the stock is insufficient}) &= P(X > 30,000) \\ &= P(Y > 10,000) \quad (Y = X - 20,000)\end{aligned}$$

$$\begin{aligned}
&= \int_{10000}^{\infty} \frac{1}{(10000)^2} e^{-\frac{y}{10000}} y dy \\
\text{Put } \frac{y}{10000} &= t; \frac{dy}{10000} = dt \\
&= \frac{1}{10,000} \int_1^{\infty} t e^{-t} (10,000) dt \\
&= \int_1^{\infty} t e^{-t} dt \\
&= \left[t \frac{e^{-t}}{-1} - 1 \cdot \frac{e^{-t}}{1} \right]_1^{\infty} \\
&= \left[-t e^{-t} - e^{-t} \right]_1^{\infty} \\
&= 0 - \left[-e^{-1} - e^{-1} \right] \\
&= 2e^{-1} \\
&= 0.73575
\end{aligned}$$

Example 2 :

In a certain locality the daily consumption of water in millions of litres, can be treated as a r.v. having a gamma distribution with an average of 3 million litres if the pumping station of the locality has daily supply capacity of 4 million litres what is the probability that this water supply will be inadequate on any given day?

Solution: We have

X – Daily water consumption of the locality in million litres and a X follows gamma distribution with average of 3 million litres.

∴ Mean = λ = 3 million litres

∴ The density function of X is

$$f(x) = \frac{e^{-x} x^{\lambda-1}}{\Gamma(\lambda)}, x \geq 0$$

$$\text{i.e. } f(x) = \frac{e^{-x} x^2}{\Gamma 3}, x \geq 0 \quad \dots \because \lambda = 3$$

Now, P (water supply of 4 million litres is inadequate on a given day)

i.e. P(water consumption (demand) is more than 4 million litres)

$$\text{i.e. } P(X > 4) = P(4 < X < \infty)$$

$$= \int_4^{\infty} f(x) dx$$

$$= \int_4^{\infty} \frac{e^{-x} x^2}{\Gamma 3} dx \quad \dots \text{ using } f(x)$$

$$= \frac{1}{3} \int_4^{\infty} x^2 e^{-x} dx$$

$$= \frac{1}{2!} [x^2(-e^{-x}) - (2x)(e^{-x}) + (2)(-e^{-x})]_4^{\infty}$$

... using intergreation by parts (short cut) (Bernoulli method)

$$\int uv dx = uv_1 - u'v_2 + u''v_3 \dots \text{ and } \therefore [n = (n-1)!]$$

$$= -\frac{1}{2} [e^{-x}(x^2 + 2x + 2)]_4^{\infty}$$

$$= -\frac{1}{2} [0 - e^{-4}(16 + 8 + 2)] \quad \dots$$

$$\therefore \lim_{x \rightarrow \infty} e^{-x}(x^2 + 2x + 2) = 0$$

By using L' Hospitals rule (indeterminate form)

$$= -\frac{e^{-4}}{2} \quad (26)$$

$$= 13 e^{-4} = 13 (0.0183)$$

$$= 0.2379$$

Example 3:

If a random variable X follows gamma distribution with variance 3, find i) $P(1 < X < 4)$ ii) $P(|X| < 1)$ iii) If another independent Gamma variate Y has a mean 1, find the mean and standard deviation of $(X + Y)$ and $P(X + Y > 1)$

Solution :

We know that for a gamma variate, mean = variance = λ , the parameter of distribution.

$\therefore$ Parameter of the gamma distribution of X

$$\lambda_1 = \text{Var}(X) = 3 \text{ (given)}$$

and $\therefore$ its P.D.F. is

$$f(x) = \begin{cases} \frac{e^{-x} x^{3-1}}{\sqrt{3}} & , \quad x \geq 0 \\ 0 & , \quad \text{otherwise} \end{cases}$$

$$(i.e) \quad f(x) = \begin{cases} \frac{e^{-x} x^2}{2} & , \quad x \geq 0 \\ 0 & , \quad \text{otherwise} \end{cases}$$

$$\dots \therefore f(x) = \frac{e^{-x} x^{\lambda-1}}{\sqrt{\lambda}}, \quad X \geq 0$$

$$\dots \therefore \sqrt{3} = 2! = 2$$

$$i) \quad \therefore P(1 < X < 4) = \int_1^4 f(x) dx$$

$$= \int_1^4 \frac{e^{-x} x^2}{2} dx$$

$$= \frac{1}{2} [x^2(-e^{-x}) - (2x)(e^{-x}) + (2)(-e^{-x})]_1^4$$

$$\dots \therefore P(a < X < b) = \int_a^b f(x) dx$$

$$\begin{aligned}
&= -\frac{1}{2} [e^{-x}(x^2 + 2x + 2)]_1^4 \\
&= -\frac{1}{2} [e^{-4}(16 + 8 + 2) - e^{-1}(1 + 2 + 2)] \\
&= -\frac{1}{2} [(0.0183)(26) - (0.3679)(5)] \\
&= -\frac{1}{2} [0.4758 - 1.8395] \\
&= -\frac{1}{2} (-1.3637) = 0.682
\end{aligned}$$

$$\begin{aligned}
\text{ii) } P(|X| < 1) &= P(-1 < X < 1) \\
&= \int_{-1}^1 f(x) dx \\
&= \int_{-1}^0 f(x) dx + \int_0^1 f(x) dx \\
&= 0 + \int_0^1 \frac{e^{-x} x^2}{2} dx \quad \dots \text{ using } f(x) \\
&= -\frac{1}{2} [e^{-x}(x^2 + 2x + 2)]_0^1 \\
&= -\frac{1}{2} [e^{-1}(5) - e^{-0}(2)] \\
&= -\frac{1}{2} [(0.3679)(5) - (1)(2)] \\
&= -\frac{1}{2} [1.8395 - 2] \\
&= -\frac{1}{2} (-0.1605) = 0.08
\end{aligned}$$

iii) Y is another independent Gamma variate with mean = 1

∴ Its parameter is $\lambda_2 = \text{Mean} = 1$

Hence, by additive propert of Gamma variates the variable (X=X+Y) will also be a Gamma variate with parameter, $\lambda = \lambda_1 + \lambda_2 = 3+1=4$

∴ For (X + Y) mean = Variance = $\lambda = 4$

$$\begin{aligned}
\therefore \text{ Standard deviation} &= \sqrt{\text{Variance}} \\
&= \sqrt{4} = 2
\end{aligned}$$

Also, its p.d.f. is

$$G(z) = \begin{cases} \frac{e^{-z} z^{4-1}}{\Gamma 4} = \frac{e^{-z} z^3}{6}, & z \geq 0 \\ 0, & \text{other wise} \end{cases}$$

$$\text{and} \therefore P(X + Y > 1) = P(Z < 1)$$

$$= P(1 < Z < \infty)$$

$$= \int_1^{\infty} \frac{e^{-z} z^3}{6} dz$$

$$= \frac{1}{6} [z^3(-e^{-z}) - (3z^2)(e^{-z}) + 6z(-e^{-z}) - (6)(e^{-z})]_1^{\infty}$$

... By bernoullis method

$$= -\frac{1}{6} [e^{-z}(z^3 + 3z^2 + 6z + 6)]_1^{\infty}$$

$$= -\frac{1}{6} [0 - e^{-1}(1 + 3 + 6 + 6)] \quad \dots$$

$$\therefore \lim_{z \rightarrow \infty} e^{-z}(z^2 + 3z + 6z + 6) = 0$$

$$= -\frac{1}{6} [-(0.3679)(16)]$$

$$= 0.981$$

Normal Distribution and their properties:

Definition

A continuous random variable X is said to follow a normal with parameter μ (mean) and σ^2 (variance) if its probability density function (pdf) is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2}; -\infty < x < \infty, \quad -\infty < \mu < \infty \text{ and } \sigma > 0.$$

Here μ & σ is the parameters. Hence it is also written as $N(\mu, \sigma^2)$

AREA PROPERTY

1. $P(a \leq X \leq b) = P(z_1 \leq Z \leq z_2)$ Where $z = \frac{x-\mu}{\sigma}$ is the Standard Normal variate.
2. $P(-\infty < Z < \infty) = 1$ 3. $P(-\infty < Z < 0) = 0.5$ 4. $P(0 < Z < \infty) = 0.5$
5. $P(-z_1 < Z < 0) = P(0 < Z < z_1)$
6. $P(-\infty \leq Z \leq -z_1) = 0.5 - P(0 \leq Z \leq z_1)$ & $P(z_1 \leq Z \leq \infty) = 0.5 - P(0 \leq Z \leq z_1)$

MGF, MEAN, VARIANCE AND S.D OF NORMAL DISTRIBUTION

$$\text{WKT, } M_x(t) = E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$$

$$\text{Where } f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{[x-\mu]^2}{2\sigma^2}}; -\infty < x < \infty$$

$$M_x(t) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-\frac{[x-\mu]^2}{2\sigma^2}} dx$$

$$\text{Put } z = \frac{x-\mu}{\sigma}, \quad x = \sigma z + \mu, \quad 2x = \sigma dz$$

$$\begin{aligned}
M_x(t) &= \frac{1}{\sqrt{2\Pi}} \int_{-\infty}^{\infty} e^{t(\sigma z + \mu)} e^{-\frac{z^2}{2}} dz \\
&= \frac{1}{\sqrt{2\Pi}} \int_{-\infty}^{\infty} e^{t\mu} e^{\sigma t z - \frac{z^2}{2}} dz \\
&= \frac{e^{t\mu}}{\sqrt{2\Pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2} + \sigma t z} dz \\
&= \frac{e^{t\mu}}{\sqrt{2\Pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}[(z^2 - 2z\sigma t + \sigma^2 t^2) - (\sigma^2 t^2)]} dz \\
&= \frac{e^{t\mu}}{\sqrt{2\Pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}[(z - \sigma t)^2 - (\sigma^2 t^2)]} dz \\
&= \frac{e^{t\mu}}{\sqrt{2\Pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z - \sigma t)^2} e^{\frac{\sigma^2 t^2}{2}} dz \\
&= \frac{e^{t\mu} e^{\frac{\sigma^2 t^2}{2}}}{\sqrt{2\Pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z - \sigma t)^2} dz
\end{aligned}$$

Put $z - t\sigma = \mu$; $dz = d\mu$

$$\begin{aligned}
M_x(t) &= \frac{e^{\frac{t\mu}{2} + \frac{\sigma^2 t^2}{2}}}{\sqrt{2\Pi}} \int_{-\infty}^{\infty} e^{-\frac{\mu^2}{2}} dz \\
&= \frac{e^{\frac{t\mu}{2} + \frac{\sigma^2 t^2}{2}}}{\sqrt{2\Pi}} \cdot \sqrt{2\Pi}
\end{aligned}$$

$$M_x(t) = e^{\frac{t\mu}{2} + \frac{\sigma^2 t^2}{2}}$$

Find Mean and Variance of a normal distribution.

$$\begin{aligned}
\text{Mean} = \mu_1' &= \left[\frac{d}{dt} M_x(t) \right]_{t=0} \\
&= \left[\frac{d}{dt} \left(e^{\frac{t\mu}{2} + \frac{\sigma^2 t^2}{2}} \right) \right]_{t=0} = \left[e^{\frac{t\mu}{2} + \frac{\sigma^2 t^2}{2}} \cdot (\mu + \sigma^2 t) \right]_{t=0} \\
&= e^0 (\mu + 0)
\end{aligned}$$

$$\text{Mean} = \mu$$

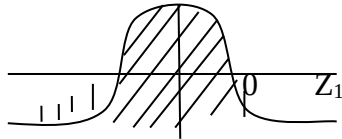
$$\mu_2' = \left[\frac{d^2}{dt^2} M_x(t) \right]_{t=0}$$

$$\begin{aligned}
&= \left[\frac{d}{dt} \left(e^{\mu t + \frac{\sigma^2 t^2}{2}} \cdot (\mu + \sigma^2 t) \right) \right]_{t=0} \\
&= \left[e^{\mu t + \frac{\sigma^2 t^2}{2}} \cdot (\sigma^2) + (\mu + \sigma^2 t) e^{\mu t + \frac{\sigma^2 t^2}{2}} \cdot (\mu + \sigma^2 t) \right]_{t=0} \\
&= e^0 (\sigma^2) + (\mu + 0) e^0 \cdot (\mu + 0) \\
&= \sigma^2 + \mu^2 \\
\text{Var}(X) &= \mu_2' - (\mu_1')^2 \\
&= \sigma^2 + \mu^2 - \mu^2 = \sigma^2
\end{aligned}$$

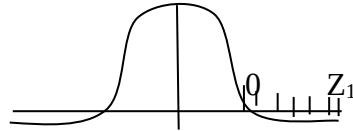
Standard Deviation of X . $\sigma_x = \sqrt{\text{Var}(X)} = \sigma$

Basic Properties of Standard Normal Curve

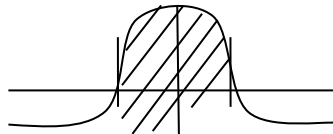
1. Total area under the Standard Normal Curve is equal to 1
2. Standard Normal Curve is symmetric about 0



$$\begin{aligned}
\text{Shaded Area} &= P(Z < Z_1) \\
&= 0.5 + P(0 < Z < Z_1)
\end{aligned}$$



$$\begin{aligned}
\text{Shaded Area} &= P(Z > Z_1) \\
&= 0.5 - P(0 < Z < Z_1)
\end{aligned}$$



$$\begin{aligned}
\text{Shaded Area} &= P(Z_1 < Z < Z_2) \\
&= P(0 < Z < Z_1) + P(0 < Z < Z_2)
\end{aligned}$$

Example: 1

The daily wages of 1000 workers are normally distributed with average wages of Rs.70 and standard deviation of Rs.5. Estimate the number of workers whose daily wages will be (i) between Rs.70 and 72 (ii) between Rs.69 and 72 (iii) more than Rs.75 (iv) less than Rs.63.

Solution

Let X denote daily wages in Rs. Thus, X is a normal variate with mean $\mu = 70$ & $\sigma = 5$ (ie) $\sigma^2 = 25$ (ie) $X \sim N(70, 25)$

The standard normal variate is $z = \frac{x - \mu}{\sigma} = \frac{x - 70}{5}$

i) Probability that a worker will have wages between Rs.70 and 72 = $P(70 < x < 72)$

Now when $x_1 = 70, z_1 = \frac{70-70}{5} = 0$ & $x_2 = 72, z_2 = \frac{72-70}{5} = 0.4$

$$P(70 < x < 72) = P(0 < z < 0.4) = 0.1554$$

Estimate the number of workers out of $N=1000$ having daily wages between Rs.70 and 72

$$\begin{aligned} \text{(ie) } P(70 < x < 72) &= N \cdot P(70 < x < 72) \\ &= 1000(0.1554) = 155.4 \approx 155 \end{aligned}$$

ii) we have to find $P(69 < x < 72)$

Now when $x_1 = 69, z_1 = \frac{69-70}{5} = -0.2$ & $x_2 = 72, z_2 = \frac{72-70}{5} = 0.4$

$$\begin{aligned} P(69 < x < 72) &= P(-0.2 < z < 0.4) \\ &= P(-0.2 < z < 0) + P(0 < z < 0.4) \\ &= P(0 < z < 0.2) + P(0 < z < 0.4) \\ &= 0.0793 + 0.1554 = 0.2347 \end{aligned}$$

Number of workers out of $N=1000$ having daily wages between Rs.69 and 72

$$\begin{aligned} P(69 < x < 72) &= N \cdot P(69 < x < 72) \\ &= 1000(0.2347) = 234.7 \approx 235 \end{aligned}$$

iii) we have to find $P(x > 75)$

When $x = 75, z_1 = \frac{75-70}{5} = 1$

$$\begin{aligned} P(x > 75) &= P(z > 1) = P(1 < z < \infty) \\ &= P(0 < z < \infty) - P(0 < z < 1) \\ &= 0.5 - 0.3413 = 0.1587 \end{aligned}$$

Number of workers out of $N=1000$ having daily wages more than Rs.75

$$= 1000(0.1587) = 158.7 \approx 159$$

iv) we have to find $P(x < 63)$

When $x = 63, z_1 = \frac{63-70}{5} = -1.4$

$$\begin{aligned} P(x < 63) &= P(z < -1.4) \\ &= P(Z > 1.4) \\ &= P(1.4 < z < \infty) \\ &= P(0 < z < \infty) - P(0 < z < 1.4) \\ &= 0.5 - 0.4192 = 0.0808 \end{aligned}$$

Number of workers out of $N=1000$ having daily wages less than Rs.63

$$= 1000(0.0808) = 80.8 \approx 81$$

Example: 2

An electrical firm manufactures light bulbs that have a life, before burn-out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find

1. The probability that a bulb burns more than 834 hours
2. The probability that a bulb burns between 778 and 834 hours.

Solution

Let X - life of a bulb (in hours), before it burn-out and thus X is normally distributed with mean $= \mu = 800$ and standard deviation $= \sigma = 40 \text{ hrs}$ (ie) variance $= \sigma^2 = 40^2$

Thus $X \sim N(800, 40^2)$

The standard normal variate is $z = \frac{x - \mu}{\sigma} = \frac{x - 800}{40}$

i) We have to find P(Bulb burns more than 834 hours) (ie) $P(X > 834)$

$$\text{When } x = 834, z_1 = \frac{834 - 800}{40} = 0.85$$

$$\begin{aligned} P(x > 834) &= P(z > 0.85) \\ &= P(0.85 < z < \infty) \\ &= P(0 < z < \infty) - P(0 < z < 0.85) \\ &= 0.5 - 0.3023 = 0.1977 \end{aligned}$$

ii) P(Bulb burns between 778 and 834 hours) (ie) $P(778 < X < 834)$

$$\text{When } x = 778, z_1 = \frac{778 - 800}{40} = -0.55$$

$$\begin{aligned} P(778 < X < 834) &= P(-0.55 < z < 0.85) \\ &= P(-0.55 < z < 0) + P(0 < z < 0.85) \\ &= P(0 < z < 0.55) + P(0 < z < 0.85) \\ &= 0.2088 + 0.3023 = 0.5111 \end{aligned}$$

UNIT-3

TESTING OF HYPOTHESIS

SAMPLING DISTRIBUTION:

If we draw a sample of size n from a given finite population of size N then the total number of possible samples is NC_n

$$NC_n = \frac{N!}{n!(N-n)!} = K$$

For each of these k samples we can compute some statistic say $t=t(x_1, x_2, \dots, x_n)$ in particular the mean $\bar{x}$ Variance (s^2), etc. as given below.

Sample No.	$\bar{x}$ Statistic	s^2
1	$\bar{x}_1$	s_1^2
2	$\bar{x}_2$	s_2^2
3	$\bar{x}_3$	s_3^2
...	$\bar{x}_4$	s_4^2
n	$\bar{x}_k$	s_k^2

The set of the values of the statistic so obtained, one for each sample, constitute the sampling distribution of the statistic.

Random sampling

A random sampling is one in which each number of population has an equal chance of being included in it. There are NC_n different samples of size n that can be picked up from a population size N

Error in Sampling

The main aim of the sampling theory is to draw a valid conclusion about the population parameters on the basis of the sample results. In doing this we may commit the following two types of error.

(i) Type-I error

When H_0 is true, we may reject it $P[\text{Reject } H_0 \text{ when it is true}] = \alpha$.

' α ' is called the size of the type I error, also referred to as producer's risk.

(ii) Type-II error

When H_0 is wrong, we may accept it. $P[\text{Accept } H_0 \text{ when it is wrong}] = \beta$.

' β ' is called the size of the type – II error, also referred to as consumer's risk.

	Accept H_0	Reject H_0
H_0 True	Correct decision	Type I error
H_0 False	Type II error	Correct decision

Procedure for Testing: (For two-tail test)

The null hypothesis is $H_0: \mu = \mu_0$

The alternative hypothesis is $H_1: \mu \neq \mu_0$

Since n is large the sampling distribution of $\bar{x}$ is approximately normal. (on the assumption that H_0 is true the statistic $z = \frac{\bar{x} - \mu}{S.E \text{ of } \bar{x}}$ is approximately $N(0,1)$. We take the level of significance as α).

Inference

For a significance level $\alpha = 0.05$ (5% level) if $|z| < 1.96$, H_0 is accepted at 5% level.

If $|z| > 1.96$, H_0 is rejected at 5% level.

For $\alpha = 0.01$ (1% level)

If $|z| < 2.58$, H_0 is accepted at 1% level.

If $|z| > 2.58$, H_0 is rejected at 1% level.

Procedure for one tail test (left tail)

(i) $H_0: \mu \geq \mu_0$

$H_1: \mu < \mu_0$ (left tail)

At $\alpha : 0.05$ the critical value of $|z| = 1.645$

if $z < -1.645$, H_0 is rejected

if $z > -1.645$, H_0 is accepted

At $\alpha : 0.01$, the critical value of z is 2.33

(ii) One tail test (right tail)

If $z < 1.645$, H_0 is accepted.

$Z > 1.645$, H_0 is rejected.

In all these cases the test we apply is called z -test.

Estimation of parameters:

A statistical measure (or) constant obtained from the population is known as a population parameter (or) parameter.

For example,

Population Mean (μ), Population Variance (σ^2), etc, are Parameters.

A statistical measure (or) constant obtained from the sample is known as a sample statistic (or) statistic.

For example, Sample mean ($\bar{x}$), sample variance (s^2), are Sample Statistic.

Notation:

	Size	Mean	Variance	S.D
Population	N	μ	σ^2	σ
Sample	n	$\bar{x}$	s^2	s

Statistical hypothesis - large sample test based on normal distribution for single mean and difference of means:

One Tailed and Two Tailed Tests

A test of any Statistical hypothesis where the alternative hypothesis is one tailed [right tailed or left tailed] is called one tailed test.

For example, in a test for testing the mean of a population in a single tailed we assume that the null hypothesis $H_0: \mu = \mu_0$ against

$$H_1: \mu > \mu_0 \text{ [Right Tailed]}$$

$$(\text{or}) H_1: \mu < \mu_0 \text{ [Left Tailed]}$$

is called one tailed test.

In a test of statistical hypothesis where the alternative hypothesis is two tailed is called two tailed test.

We assume that $H_0: \mu = \mu_0$

Against $H_1: \mu \neq \mu_0$ [$\mu > \mu_0$ & $\mu < \mu_0$] is called Two tailed Test.

Sampling

The process of selecting a sample from the population is called sampling.

Types of sampling

- (1) Purposive sampling
- (2) Random sampling
- (3) Stratified sampling
- (4) Systematic sampling

Standard Error: (S.E)

The standard deviation of the sampling distribution of a statistic is known as its standard error.

S.No	Statistic	Standard error
1.	$\bar{x}$	$\frac{\sigma}{\sqrt{n}}$
2.	Difference of two sample means $\bar{x}_1 - \bar{x}_2$	$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

Small Samples and Large Samples

When the size of the sample is greater than 30 [ie, $n > 30$], the sample is called large sample.

When the size of the sample is less than 30 [ie, $n < 30$], the sample is called small samples.

Test of Significance

An important aspect of the sampling theory is to study the test of significance which will enable us to decide, on the basis of the results of the sample, whether

- (i) The derivation between the observed sample statistic and the hypothetical parameter value (or)
- (ii) The derivation between two sample statistics is significant (or) might be attributed due to chance or the fluctuations of the sampling.

Testing a hypothesis

On the basis of sample informative, we make certain decisions about the population. In taking such decisions we make certain assumptions. These assumptions are known as statistical hypothesis. These hypothesis are tested.

Assuming the hypothesis correct we calculate the probability of getting the observed sample. If this probability is less than a certain assigned value, the hypothesis is to be rejected.

Null Hypothesis

For applying the test of significance, we first set up a hypothesis which is a definite statement about the Population Parameter. Usually such a hypothesis is a hypothesis of no difference and it is denoted by H_0 .

Alternative Hypothesis

Any hypothesis which is complementary to the null hypothesis H_0 is called an alternative hypothesis and it is denoted by H_1 .

Critical Value (or) Table Value

The Value of the test statistic which separates the sample space into rejection region and the acceptance region, is called critical value.

Critical Region

A region corresponding to a test statistic in the sample space which tends to rejection of H_0 is called critical region (or) region of rejection.

The region complementary to the critical region is called the region of acceptance.

Level of significance

The probability that the value of the statistic lies in the critical region is called the level of significance.

In general these levels are chosen as 0.01 or 0.05, called 1% level and 5% level of significance respectively.

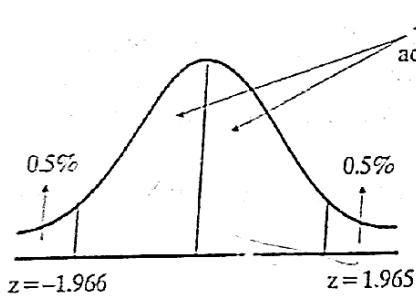


Fig. 1% level of significance

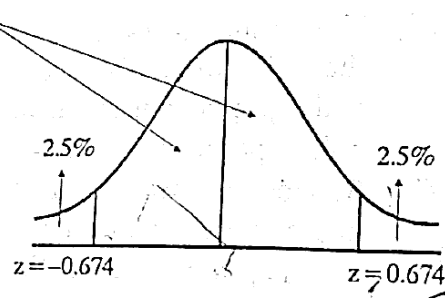


Fig. 5% level of significance

STATISTICAL HYPOTHESIS

- (i) Setup null and alternative hypothesis.
- (ii) Choose the appropriate level of significance. (either 5% or 1% level). This is to be decided before sample is drawn.
- (iii) Find the critical value (or) Table Value of Z.
- (iv) Compute the test statistic $Z = \frac{t - E(t)}{SE(t)}$.
- (v) Compare the computed value of Z with the significant value Z_α .
- (vi) If $|Z| < Z_\alpha$, we accept H_0
If $|Z| > Z_\alpha$, we reject H_0 .

Table for critical values on using normal probability

Critical Values	Level of Significance		
	1%	5%	10%
Two Tailed Test	$ Z_\alpha = 2.58$	$ Z_\alpha = 1.96$	$ Z_\alpha = 1.645$
Right Tailed Test	$Z_\alpha = 2.33$	$Z_\alpha = 1.645$	$Z_\alpha = 1.28$
Left Tailed Test	$Z_\alpha = -2.33$	$Z_\alpha = -1.645$	$Z_\alpha = -1.28$

Confidence limits

If a sample statistics lies in the interval

$\mu - 1.96 \sigma$, $\mu + 1.96 \sigma$, we call 95% confidence interval.

Similarly, Confidence limits as the area between $\mu - 2.58\sigma$, $\mu + 2.58\sigma$ is 99%.

The numbers 1.96, 2.58 are called confidence coefficients.

Random Sample: A Random sample of size n is a sample that is chosen in such a way that each subset of n elements of the population has the same probability of being selected.

S.No.	Statistic t	Standard error of t	Remark
1.	Sample mean $\bar{x}$	$\frac{\sigma}{\sqrt{n}}$	σ population S.D, n size of sample

2.	Sample standard deviation s	$\sqrt{\frac{\sigma^2}{2n}}$	σ population S.D, n size of sample
3.	Sample proportion p	$\sqrt{\frac{PQ}{n}}$	n size of sample P-population proportion $Q = 1-P$
4.	Difference of sample mean $(\bar{x}_1 - \bar{x}_2)$	$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$	n_1, n_2 - size of sample
5.	Difference of Sample standard deviation $S_1 - S_2$	$\sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}$	n_1, n_2 - size of sample, σ_1, σ_2 s.d's of populations
6.	Difference of proportion $p_1 - p_2$	$\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$	$P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$ $Q = 1-P$

ERRORS IN SAMPLING (TYPE – I AND II ERRORS)

- (i) Type I error: Reject H_0 when it is true.
- (ii) Type II error: Accept H_0 when it is wrong.

LARGE SAMPLE TESTS BASED ON NORMAL

DISTRIBUTION:

If the sample size $n > 30$, then the sample is called large sample.

Now we consider the following tests under large sample test based on normal distribution.

- (i) Test of significance for single mean.
- (ii) Test of significance for difference of means.

Test of Significance for single mean

Suppose we want to test whether the given sample of size 'n' has been drawn from a population with mean μ (or) to test whether the difference between sample mean and population mean is significant or not.

Working rule

- (i) We set up null hypothesis that there is no difference between μ and $\bar{x}$.

$$\text{ie, } H_0 : \mu = \bar{x}$$

$$\therefore H_1 : \mu \neq \bar{x} \quad [\text{Two tailed test}]$$

$$(\text{or}) \quad \mu < \bar{x} \quad [\text{left tailed test}]$$

$$(\text{or}) \quad \mu > \bar{x} \quad [\text{Right tailed test}]$$

(ii) Choose the appropriate level of significance.

(iii) Find the critical value.

(iv) Test Statistic $Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$ [If σ is given]

(or)

$$Z = \frac{\bar{x} - \mu}{s / \sqrt{n}} \text{ [If } s \text{ is given].}$$

(v) Compare the computed value of Z with significant value Z_α , write your conclusion.

Note:

1) 95% confidence (or) fiducial limits are $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$

2) 99% confidence (or) fiducial limits are $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$

EXAMPLES

1) A sample of 900 members has a mean 3.4cms. Can it be reasonably regarded as a sample from a large population of mean 3.2 and S.D 2.3cms?

Solution:

Given Sample size $n=900$

Sample mean $\bar{x}=3.4$

Population mean $\mu=3.2$

Population S.D $\sigma=2.3$ [σ is given]

(i) **Null hypothesis H_0 :** $\mu=3.2$

Alternative hypothesis H_1 : $\mu \neq 3.2$ [Two tailed test].

(ii) Level of significance $\alpha=5\%=0.05$

(iii) Critical value at 5% level is $Z_\alpha=1.96$

$$\begin{aligned} \text{(iv) Test Statistic } Z &= \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \\ &= \frac{3.4 - 3.2}{2.3 / \sqrt{900}} = 0.261 \end{aligned}$$

(v) Here $|Z| = 0.261 < Z_\alpha$

$\therefore H_0$ is accepted.

$\therefore$ The sample is drawn from the population with mean 3.2 and variance 2.3.

- 2) The average marks in Mathematics of a sample of 100 students was 51 with a S.D of 6 marks. Could this have been a random sample from a population with average marks of 50?

Solution:

Given $n=100$, $\bar{x}=51$, $s=6$, $\mu=50$ [s is given]

(i) $H_0: \mu=50$

$H_1: \mu \neq 50$ [Two tailed test]

(ii) Level of significance $\alpha=5\% = 0.05$

(iii) Critical value at $\alpha=5\%$ level is $Z_\alpha = 1.96$

(iv) Test statistic $Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{51 - 50}{\frac{6}{\sqrt{100}}} = 1.6667$

(v) $\therefore |Z| = 1.6667 < Z_\alpha$

$\therefore H_0$ is accepted.

$\therefore$ A random sample is drawn from a population with average marks of 50.

- 3) An insurance agent has claimed that the average age of policy holders who insure through him is less than the average for all agents which is 30.5 years. A random sample of 100 policy holders who had insured through him reveal that the mean and S.D are 28.8 years and 6.35 years respectively. Test his claim at 5% level of significance.

Solution:

Given $n=100$, $\mu=30.5$, $s=6.35$, $\bar{x}=28.8$, $\alpha=5\%$ [s is given]

(i) $H_0: \mu=30.5$

$H_1: \mu < 30.5$ [Left Tailed Test].

(ii) Level of significance $\alpha=5\%$

(iii) Critical value at $\alpha=5\%$ level is $Z_\alpha = -1.645$.

(iv) Test statistic $Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{28.8 - 30.5}{\frac{6.35}{\sqrt{100}}} = -2.68$

(v) Here $|Z| = 2.68 > Z_\alpha$

$\therefore H_0$ is rejected and H_1 is accepted.

- 4) The mean weight obtained from a random sample of size 100 is 64 gms. The S.D of the weight distribution of the population is 3gms. Test the statement that the mean weight of the population is 67gms at 5% level of significance. Also setup 99% confidence limits of the mean weight of the population.

Solution:

Given $n=100$, $\mu=67$, $\bar{x}=64$, $\sigma=3$.

(i) $H_0: \mu=67$

$H_1: \mu \neq 67$ [Two Tailed Test].

(ii) Level of significance $\alpha=5\%$

(iii) Critical value at $\alpha=5\%$ level is $Z_\alpha=1.96$

(iv) Test statistic
$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{64 - 67}{3 / \sqrt{100}} = -10$$

(v) Here $|Z| = 10 > Z_\alpha$

$\therefore H_0$ is rejected and H_1 is accepted.

The sample is not drawn from the population.

(vi) The 99% confidence limits are
$$\bar{x} \pm 2.58 \frac{\sigma}{\sqrt{n}}$$
$$= 64 \pm 2.58 \times \frac{3}{\sqrt{100}}$$
$$= 64.774, 63.226$$

$\therefore$ 99% confidence limits are 63.226 & 64.774.

- 5) The average income of persons was Rs. 210 with a S.D of Rs. 10 in a sample of 100 people of a city. For another sample of 150 persons, the average income was Rs. 220 with S.D of Rs. 12. The S.D of income of the people of the city was Rs.11. Test whether there is any significant difference between the average income of the localities.

Solution:

Given $n_1=100$, $\bar{x}_1=210$, $s_1=10$

$n_2=150$, $\bar{x}_2=220$, $s_2=12$

(i) $H_0: \mu_1=\mu_2$

$H_1: \mu_1 \neq \mu_2$ [Two tailed test]

(ii) Level of significance $\alpha=5\%$

(iii) Critical value at $\alpha=5\%$ level is $Z_\alpha=1.96$

(iv) Test statistic
$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$
$$= \frac{210 - 220}{\sqrt{\frac{10^2}{100} + \frac{12^2}{150}}} = -7.1428$$

(v) Here $|Z| = 7.1428$

$$|Z| > Z_\alpha$$

$\therefore H_0$ is rejected and H_1 is accepted.

$\therefore$ There is significant difference between the average income of the localities.

- 6) A simple sample of heights of 6,400 Englishmen has a mean of 170 inches and a S.D of 6.4 inches, while a simple sample of heights of 1600 Australians

has a mean of 172 inches and a S.D of 6.3 inches. Do the data indicate that Australians are on the average taller than Englishmen?

Solution:

Given $n_1=6400$, $\bar{x}_1 = 170$, $s_1=6.4$

$n_2=1600$, $\bar{x}_2 = 172$, $s_2=6.3$

(i) $H_0: \mu_1 = \mu_2$

$H_1: \mu_1 < \mu_2$ [Left tailed test]

(ii) Level of significance $\alpha=5\%$

(iii) Critical value $Z_\alpha = -1.645$

(iv) Test Statistic $Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{170 - 172}{\sqrt{\frac{(6.4)^2}{6400} + \frac{(6.3)^2}{1600}}} = -11.32$

(v) Here $|Z| = 11.32$

$\therefore |Z| > Z_\alpha$

$\therefore$ We reject H_0 and accept H_1 .

$\therefore$ The data indicate that Australian are the average taller than Englishmen.

TEST OF SIGNIFICANCE OF SMALL SAMPLES

When the size of the sample (n) is less than 30 then that sample is called a small sample.

The following are some important tests for small samples

(i) Student's 't' test

(ii) F- test

(iii) χ^2 - test

Small sample tests: t-test:

STUDENT'S 't'-TEST

The t- test is used when sample size $n < 30$ and the population S.D is unknown.

Applications of t- distribution

It is used to test

- (i) The significance of the difference of the mean of a random sample and the mean of the population.
- (ii) The significance of the difference between two sample means
- (iii) The significance of observed partial and multiple correlation coefficients.
- (iv) The significance of an observed sample correlation coefficients and sample regression coefficients.

t- test of significance for single mean

(i) Null hypothesis $H_0: \mu = \mu_0$

Alternative hypothesis $H_1: \mu \neq \mu_0$ [Two tailed test] (or)

$\mu > \mu_0$ [Right tailed test] (or)

$\mu < \mu_0$ [Left tailed test]

(ii) Write appropriate level of significance and d.f = n-1

(iii) Find critical values of 't' [ie, t_α] at the level of significance ' α ' with (n-1) d.f

(iv) Test Statistic $t = \frac{\bar{x} - \mu}{s / \sqrt{n-1}}$

(v) Compare the calculated value of 't' with critical value.

If $|t| < t_\alpha$ then H_0 is accepted.

If $|t| > t_\alpha$ then H_0 is rejected and H_1 is accepted.

Write the conclusion.

Confidence (or) fiducial limits

If t_α is the value of 't' at level of significance α at (n-1) degrees of freedom then

the confidence limits are $\bar{x} \pm t_\alpha \frac{s}{\sqrt{n}}$

EXAMPLE

1) A sample of 20 items has mean 42 units and S.D 5 units. Test the hypothesis that it is a random sample from a normal population with mean 45 units.

Solution:

Given $n=20$, $\bar{x}=42$, $s=5$, $\mu=45$

(i) $H_0: \mu=45$

$H_1: \mu \neq 45$ [Two tailed Test]

(ii) Level of significance $\alpha=0.05=5\%$, degrees of freedom d.f = $n-1=19$

(iii) Critical value at 5% level with 19 d.f is $t_\alpha = 2.093$

(iv) Test statistic $t = \frac{\bar{x} - \mu}{s / \sqrt{n-1}}$

$$= \frac{42 - 45}{5 / \sqrt{20-1}} = -2.615$$

(v) Here $|t| = 2.615$

$$\therefore |t| > t_\alpha$$

$\therefore H_0$ is rejected and H_1 is accepted. Therefore a random sample cannot be drawn from the normal population with mean 45 units.

2) A random sample of size 16 has 53 as mean. The sum of squares of the deviation from mean is 135. Can this sample be regarded as taken from the population having 56 as mean? Obtain 95% and 99% confidence limits of the mean of the population.

Solution:

Given $n=16$, $\bar{x}=53$, $\sum(x-\bar{x})^2 = 135$, $\mu=56$

(i) $H_0: \mu=56$

$H_1: \mu \neq 56$ [Two tailed test]

- (ii) Level of significance $\alpha=5\% = 0.05$ and d.f = 15
 (iii) Critical value at 5% level with 15 d.f is $t_{\alpha}=1.753$

(iv) Test statistic $t = \frac{\bar{x} - \mu}{s / \sqrt{n-1}}$

Where $s^2 = \frac{\sum(x - \bar{x})^2}{n} = \frac{135}{16} = 8.4375 \Rightarrow s = 2.9047$,

$$\therefore t = \frac{53-56}{2.9047 / \sqrt{15}} = -4$$

(v) Here $|t| = 4$

$$\therefore |t| > t_{\alpha}$$

$\therefore H_0$ is rejected and H_1 is accepted. $\therefore$ The sample cannot be regarded as taken from the population having 56 as mean

(vi) 95% confidence limits of the population mean $= \bar{x} \pm t_{0.05} \frac{s}{\sqrt{n}}$

$$= 53 \pm 1.753 \times \frac{2.9047}{\sqrt{16}}$$

$$= 53 \pm 1.2729$$

$$= (51.7271, 54.2729)$$

(vii) 99% confidence limits of the population mean $= \bar{x} \pm t_{0.01} \frac{s}{\sqrt{n}}$

$$= 53 \pm 2.602 \times \frac{2.9047}{\sqrt{16}}$$

$$= 53 \pm 1.8895$$

$$= (51.1105, 54.8895)$$

3) Test whether the sample having the values 63, 63,64,55,66,69,70,70 and 71 has been chosen from a population with mean of 65 at 5% level of significance.

Solution:

Here $n=9$, $\alpha=5\%$, $\mu=65$

(i) $H_0: \mu=65$

$H_1: \mu \neq 65$ [Two tailed test]

(ii) Level of significance $\alpha=5\% = 0.05$ and d.f = $n-1=8$

(iii) Critical value at 5% level with 8 d.f is $t_{\alpha}=2.31$

(iv) Test Statistic $t = \frac{\bar{x} - \mu}{s / \sqrt{n-1}}$

Where $\bar{x} = \frac{\sum x}{n}$, $s^2 = \frac{\sum (x - \bar{x})^2}{n}$

x	$x - \bar{x}$	$(x - \bar{x})^2$
63	-2.66	7.0756
63	-2.66	7.0756
64	-1.66	2.7556
55	-10.66	113.6356
66	0.34	0.1156
69	3.34	11.1556
70	4.34	18.8356
70	4.34	18.8356
71	5.34	28.5156
591		208.0004

$$\bar{x} = \frac{591}{9} = 65.66$$

$$s^2 = \frac{208.0004}{9} = 23.1111$$

$$\therefore s = 4.8074$$

$$\therefore t = \frac{65.66 - 65}{\frac{23.1111}{\sqrt{8}}} = 0.08077$$

(v) Here $|t| = 0.08077$

$$\therefore |t| < t_{\alpha}$$

$\therefore H_0$ is accepted.

$\therefore$ The sample has been drawn from a population with mean 65.

4) A random sample of 10 boys had the following I.Q's: 70,120,110,101, 88,83,95,98,107,100. Do those data support the assumption of a population mean I.Q of 100? Find a reasonable range in which most of the mean I.Q values of samples of 10 boys lie.

Solution:

Given $n=10$, $\mu=100$

(i) $H_0: \mu=100$

$H_1: \mu \neq 100$ [Two tailed test]

(ii) Level of significance $\alpha=5\%$, $=0.05$, d.f= $n-1=9$

(iii) Critical value at 5% level with 9 d.f is

$$t_{\alpha} = 2.262$$

(iv) Test Statistic $t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n-1}}}$

Where $\bar{x} = \frac{\Sigma x}{n}$, $s^2 = \frac{\Sigma(x - \bar{x})^2}{n}$ (or) $s^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2$

											Total
x	70	120	110	101	88	83	95	98	107	100	972
x^2	4900	14400	12100	10201	7744	6889	9025	9604	11449	10000	96312

$$\therefore \bar{x} = \frac{972}{10} = 97.2$$

$$s^2 = \frac{96312}{10} - (97.2)^2 = 183.36$$

$$\Rightarrow s = 13.5$$

$$\therefore t = \frac{97.2 - 100}{\frac{13.5}{\sqrt{9}}} = -0.62$$

(v) Here $|t| = 0.62$

$$\therefore |t| < t_{\alpha}$$

$\therefore H_0$ is accepted.

$\therefore$ The data support the assumption of a population I.Q. of 100.

(vi) 95% confidence limits are $\bar{x} \pm t_{0.05} \frac{s}{\sqrt{n}}$

$$= 97.2 \pm 2.262 \times \frac{13.5}{\sqrt{10}}$$

$$= 87.54, 106.86$$

$\therefore$ 95% confidence limits within which the mean I.Q. values of sample of 10 boys will lie is [86.99, 107.41]

STUDENT'S TEST (WHEN S.D. OF THE SAMPLE IS NOT GIVEN DIRECTLY)

- 1) A random sample of 10 boys had the following I.Q.'s 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these data support the assumption of a population mean I.Q. of 100? Find a reasonable range in which most of the mean I.Q. values of sample of 10 boys lie.

Solution:

Here S.D. and mean of the sample is not given directly. We have to determine these S.D. and mean as follows:

x	$x - \bar{x}$	$(x - \bar{x})^2$
70	-27.2	739.84

120	22.8	519.84
110	12.8	163.84
101	3.8	14.44
88	-9.2	84.64
83	-14.2	201.64
95	-2.2	4.84
98	0.8	0.64
107	9.8	96.04
100	2.8	7.84
972		1833.60

$$\text{Mean } \bar{x} = \frac{\Sigma x}{n} = \frac{972}{10} = 97.2$$

$$\text{We know that } s^2 = \frac{1}{n-1} (x - \bar{x})^2 = \frac{1833.60}{9}$$

$$\therefore \text{standard deviation} = \sqrt{203.73} = 14.27$$

Null Hypothesis H_0 : The data support the assumption of a population mean I.Q. of 100 in the population.

Alternative Hypothesis H_1 : $\mu \neq 100$

$$\text{The test statistic is 't'} = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{97.2 - 100}{\frac{14.27}{\sqrt{10}}} = \frac{-2.8}{4.514} = -0.62$$

$$|t| = 0.62$$

$\therefore$ Calculated value of $t = 0.62$.

Tabulated value of t for (10-1) d.f. i.e., 9 d.f. at 5% level of significance is 2.26 (two tailed test)

Since calculated $t <$ tabulated t , we accept the null hypothesis H_0 i.e., the data support the assumption of mean I.Q. of 100 in the population.

$$\begin{aligned} \text{95\% confidence limits are given by } & \bar{x} \pm t_{0.05} \frac{\sigma s}{\sqrt{n}} \\ & = 97.2 \pm 2.26 \times 4.514 \\ & = 97.2 \pm 10.20 \\ & = 107.40 \text{ and } 87.00 \end{aligned}$$

$\therefore$ The 95% confidence limits within which the mean I.Q. values of samples of 10 boys will lie is [87.00, 107.40].

2) The following table gives the lengths of 12 samples of Egyptian cotton taken from a large consignment. 48, 46, 49, 52, 45, 43, 47, 47, 46, 47, 50. Test if the mean length of the consignment be taken as 46.

Solution:

Calculation for sample mean and S.D.

x	$x - \bar{x}$	$(x - \bar{x})^2$
48	0.84	0.71
46	-1.16	1.35
49	1.84	3.39
46	-1.16	1.35
52	4.84	23.43
45	-2.16	4.67
43	-4.16	17.31
47	-0.16	0.03
47	-1.16	0.03
46	-0.16	1.35
47	2.84	0.03
		8.07
566		61.72

$$\text{Mean } \bar{x} = \frac{\Sigma x}{n} = \frac{566}{12} = 47.16$$

$$\text{We know that } s^2 = \frac{1}{n-1} \Sigma (x - \bar{x})^2 = \frac{1}{11} \times 61.72 = 5.61$$

$$\therefore s = 2.37$$

Null Hypothesis H_0 : The mean length is 46, i.e., $\mu = 46$.

Alternative Hypothesis H_1 : $\mu \neq 46$

$$\text{The test statistic is 't'} = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{47.16 - 46}{\frac{2.37}{\sqrt{12}}} = 1.69$$

$\therefore$ Calculated value of $t = 1.69$

Tabulated value of t for $12 - 1 = 11$ d.f. at 5% level of significance is 2.20 (two tailed test)

Since calculated value of $t >$ tabulated value of t , we accept the null hypothesis H_0

i.e., the mean length of the consignment be taken as 46

t-test of significance for difference of Means

The following assumptions are made in using this test.

- (i) Parent populations, from which the samples have been drawn are normally distributed.
- (ii) Population variances are equal and unknown
- (iii) The two samples are random and independent.

$$\text{Ttest statistic } t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{S^2 \left[\frac{1}{n_1} + \frac{1}{n_2} \right]}}$$

$$\text{Where } S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

s_1 and s_2 are sample standard deviations.

- (v) Compare the value of t with t_α . Write your conclusion

3) Two horses A and B were tested according to the time (in seconds) to run a particular race with the following results.

Horse A	28	30	32	33	33	29	34
Horse B	29	30	30	24	27	29	

Test whether the horse A is running faster than horse B at 5% level.

Solution:

Given $n_1=7$, $n_2=6$, $\alpha=5\%$

(i) $H_0: \mu_1 = \mu_2$

$H_1: \mu_1 > \mu_2$ [Right tailed test]

(ii) Level of significance $\alpha=5\%$, and degrees of freedom = $n_1+n_2-2= 11$

(iii) Critical value $t_\alpha = 1.796$.

x_1	x_2	x_1^2	x_2^2
28	29	784	841
30	30	900	900
32	30	1024	900
33	24	1089	576
33	27	1089	729
29	29	841	841
34		1156	
219	169	6883	4787

$$\therefore \bar{x}_1 = \frac{\Sigma x_1}{n_1} = \frac{219}{7} = 31.29$$

$$\bar{x}_2 = \frac{\Sigma x_2}{n_2} = \frac{169}{6} = 28.17$$

$$s_1^2 = \frac{\Sigma x_1^2}{n_1} - (\bar{x}_1)^2 = \frac{6883}{7} - (31.29)^2 = 4.23$$

$$s_2^2 = \frac{\Sigma x_2^2}{n_2} - (\bar{x}_2)^2 = \frac{4787}{6} - (28.17)^2 = 4.28$$

$$\therefore S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} = \frac{7 \times 4.23 + 6 \times 4.28}{7 + 6 - 2} = 5.03$$

$$\therefore \text{Test statistic } t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{S^2 \left[\frac{1}{n_1} + \frac{1}{n_2} \right]}} = \frac{31.29 - 28.17}{\sqrt{5.03 \left[\frac{1}{7} + \frac{1}{6} \right]}} = 2.498$$

(v) Here $|t| = 2.498$

$\therefore |t| > t_\alpha$

$\therefore H_0$ is rejected and H_1 is accepted.

$\therefore$ Horse A is running faster than Horse B.

4) To compare the prices of a certain product in two cities, ten shops were selected at random in each town. The prices were noted below.

City I:	61	63	56	63	56	63	59	56	44	61
City II:	54	54	47	59	51	61	57	55	64	58

Test whether the average prices can said to be the same in two cities.

Solution:

Given $n_1=10, n_2=10$

(i) $H_0: \mu_1 = \mu_2$

$H_1: \mu_1 \neq \mu_2$ [Two tailed test]

(ii) Level of significance $\alpha=5\%, =0.05, d.f = n_1+n_2-2 = 18$

(iii) Critical value at 5% significance level with 18 d.f is $t_\alpha = 2.10$.

$$(iv) \text{ Test statistic } t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{S^2 \left[\frac{1}{n_1} + \frac{1}{n_2} \right]}}$$

$$\text{Where } S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

x_1	61	63	56	63	56	63	59	56	44	61	582
x_1^2	3721	3969	3136	3969	3136	3969	3481	3136	1936	3721	34174
x_2	54	54	47	59	51	61	57	55	64	58	560
x_2^2	2916	2916	2209	3481	2601	3721	3249	3025	4096	3364	31578

$$\bar{x}_1 = \frac{\Sigma x_1}{n_1} = \frac{582}{10} = 58.2$$

$$\bar{x}_2 = \frac{\Sigma x_2}{n_2} = \frac{560}{10} = 56$$

$$s_1^2 = \frac{\Sigma x_1^2}{n_1} - (\bar{x}_1)^2 = \frac{34174}{10} - (58.2)^2 = 30.16$$

$$s_2^2 = \frac{\Sigma x_2^2}{n_2} - (\bar{x}_2)^2 = \frac{31578}{10} - (56)^2 = 21.8$$

$$\therefore S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} = \frac{10 \times 30.16 + 10 \times 21.8}{10 + 10 - 2} = 28.8667$$

$$\therefore t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{S^2 \left[\frac{1}{n_1} + \frac{1}{n_2} \right]}} = \frac{58.2 - 56}{\sqrt{28.8667 \left[\frac{1}{10} + \frac{1}{10} \right]}} = 0.9156$$

(v) Here $|t| = 0.9156$

$$\therefore |t| < t_\alpha$$

$\therefore H_0$ is accepted

$\therefore$ The average price is same in both cities.

- 5) A group of 10 rats fed on diet A and another group of 8 rats fed on diet B recorded the following increase in weight (gms).

Diet A:	5	6	8	1	12	4	3	9	6	10
Diet B:	2	3	6	8	10	1	2	8		

Does this show superiority of diet A over diet B.

Solution:

Given $n_1=10, n_2=8$.

(i) $H_0: \mu_1=\mu_2$

$H_1: \mu_1>\mu_2$ [Right tailed test]

(ii) Level of significance $\alpha=5\% = 0.05, d.f = n_1+n_2-2$

(iii) Critical value $t_\alpha=1.746$

(iv) Test statistic $t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{S^2 \left[\frac{1}{n_1} + \frac{1}{n_2} \right]}}$

$$\text{Where } S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

											Total
x_1	5	6	8	1	12	4	3	9	6	10	64
x_1^2	25	36	64	1	144	16	9	81	36	100	512
x_2	2	3	6	8	10	1	2	8	—	—	40
x_2^2	4	9	36	64	100	1	4	64	—	—	282

$$\therefore \bar{x}_1 = \frac{\Sigma x_1}{n_1} = \frac{64}{10} = 6.4$$

$$\bar{x}_2 = \frac{\Sigma x_2}{n_2} = \frac{40}{8} = 5$$

$$s_1^2 = \frac{\Sigma x_1^2}{n_1} - (\bar{x}_1)^2 = \frac{512}{10} - (6.4)^2 = 10.24$$

$$s_2^2 = \frac{\Sigma x_2^2}{n_2} - (\bar{x}_2)^2 = \frac{282}{8} - (5)^2 = 10.25$$

$$\therefore S^2 = \frac{10 \times 10.24 + 8 \times 10.25}{10 + 8 - 2} = 11.525$$

$$\therefore t = \frac{6.4 - 5}{\sqrt{11.525 \left[\frac{1}{10} + \frac{1}{8} \right]}} = 0.8694$$

(v) Here $|t| = 0.8694$

$|t| > t_\alpha$

$\therefore H_0$ is accepted.

$\therefore$ Diet A is superior to diet B.

F- TEST: (OR) SNEDECOR'S VARIANCE RATIO TEST

To test whether if there is any significant difference between two estimates of population variance.

$$\text{Test statistic } F = \frac{S_1^2}{S_2^2} \text{ if } S_1^2 > S_2^2$$

$$(\text{or}) \quad F = \frac{S_2^2}{S_1^2} \text{ if } S_2^2 > S_1^2$$

$$\text{Where } S_1^2 = \frac{n_1 s_1^2}{n_1 - 1} \quad \text{and} \quad S_2^2 = \frac{n_2 s_2^2}{n_2 - 1}$$

s_1^2, s_2^2 are sample variances

- (i) Choose the appropriate level of significance, degrees of freedom d.f. = $(n_1 - 1, n_2 - 1)$.
- (ii) Find the critical value F_α
- (iii) Compare the value of F with F_α
 If $F < F_\alpha$ then H_0 is accepted.
 If $F > F_\alpha$ then H_0 is rejected.

EXAMPLES

- 1) In the sample of 8 observations, the sum of the squares of deviations of the sample values from the sample mean was 94.5 and in another sample of 10

observations, it was found to be 101.7. Test whether the difference of variance is significant.

Solutions:

$$\text{Given } n_1=8, \Sigma(x_1 - \bar{x}_1)^2=94.5$$

$$n_2=10, \Sigma(x_2 - \bar{x}_2)^2=101.7$$

(i) $H_0: \sigma_1^2 = \sigma_2^2$ [The difference of variance is not Significant]

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

(ii) Test statistic $F = \frac{S_1^2}{S_2^2}$ if $S_1^2 > S_2^2$

$$\text{Where } S_1^2 = \frac{n_1 s_1^2}{n_1 - 1}, S_2^2 = \frac{n_2 s_2^2}{n_2 - 1}$$

$$s_1^2 = \frac{\Sigma(x_1 - \bar{x}_1)^2}{n_1}$$

$$= \frac{94.5}{8} = 11.8125$$

$$\therefore S_1^2 = \frac{8 \times 11.8125}{8 - 1} = 13.5$$

$$s_2^2 = \frac{\Sigma(x_2 - \bar{x}_2)^2}{n_2}$$

$$= \frac{101.7}{10} = 10.17$$

$$\therefore S_2^2 = \frac{10 \times 10.17}{10 - 1} = 11.3$$

$$\text{Here } S_1^2 > S_2^2$$

$$\therefore F = \frac{13.5}{11.3} = 1.195$$

(iii) Level of Significance $\alpha = 5\%$

$$D.f = (n_1 - 1, n_2 - 1) = (7, 9)$$

(iv) Critical Value $F_\alpha = 3.29$

(v) Here $F < F_\alpha$

$\therefore H_0$ is accepted.

$\therefore$ The difference of variances is not significant.

2) It is known that the mean diameters of rivets produced by two firms A and B are practically the same but the S.D may differ. For 22 rivets produced by firm A, the S.D is 2.9 mm, while for 16 rivets manufactured by firm B, the S.D is 3.8mm. Compute the statistic you would use to test whether the products of firm A have the same variability as those of firm B and test its significance.

Solution:

$$\text{Given } n_1=22, n_2=16, s_1=2.9, s_2=3.8.$$

(i) $H_0: \sigma_1^2 = \sigma_2^2$

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

$$(i) \quad \text{W.K.T } S_1^2 = \frac{n_1 s_1^2}{n_1 - 1}$$

$$S_2^2 = \frac{n_2 s_2^2}{n_2 - 1}$$

$$= \frac{22 \times (2.9)^2}{22 - 1} = \frac{16 \times (3.8)^2}{15}$$

$$= 8.8105 = 15.403$$

Here $S_2^2 > S_1^2$

$$\therefore \text{Test statistic } F = \frac{S_2^2}{S_1^2} = \frac{15.403}{8.8105} = 1.75$$

(iii) Level of significance $\alpha = 5\%$

$$\text{D.f.} = (n_2 - 1, n_1 - 1) = (15, 21)$$

(iv) Critical value of F with (15, 21) d.f at 5% level of significance is 2.20.
i.e, $F_\alpha = 2.20$.

(v) Here $F < F_\alpha$

$\therefore H_0$ is accepted.

$\therefore$ The products of both the firms A and B have the same variability.

3) A group of 10 rats fed on diet A and another group of 8 rats fed on diet B, recorded the following increase in weight.

Diet A	5	6	8	1	12	4	3	9	6	10
Diet B	2	3	6	8	10	1	2	8		

Find if the variances are significantly different.

Solution:

Given $n_1 = 10, n_2 = 8$

(i) $H_0: \sigma_1^2 = \sigma_2^2$

$H_1: \sigma_1^2 \neq \sigma_2^2$

$$(ii) \text{ W.K.T } S_1^2 = \frac{n_1 s_1^2}{n_1 - 1} \quad S_2^2 = \frac{n_2 s_2^2}{n_2 - 1}$$

$$\text{Where } s_1^2 = \frac{\sum x_1^2}{n_1} - (\bar{x}_1)^2, \quad s_2^2 = \frac{\sum x_2^2}{n_2} - (\bar{x}_2)^2$$

											Total
x_1 :	5	6	8	1	12	4	3	9	6	10	64
x_1^2 :	25	36	64	1	144	16	9	81	36	100	512
x_2 :	2	3	6	8	10	1	2	8			40
x_2^2 :	4	9	36	64	100	1	4	64			282

$$\bar{x}_1 = \frac{\sum x_1}{n_1} = \frac{64}{10} = 6.4$$

$$\bar{x}_2 = \frac{\sum x_2}{n_2} = \frac{40}{8} = 5$$

$$\therefore s_1^2 = \frac{512}{10} - (6.4)^2 = 10.24$$

$$\text{and } s_2^2 = \frac{282}{8} - (5)^2 = 10.25$$

$$\therefore S_1^2 = \frac{10 \times 10.24}{10 - 1} = 11.3777$$

$$S_2^2 = \frac{8 \times 10.25}{8 - 1} = 11.7143$$

Here $S_2^2 > S_1^2$

$$\therefore \text{Test statistic } F = \frac{S_2^2}{S_1^2} = \frac{11.7143}{11.3777} = 1.02958$$

(iii) Level of Significance $\alpha = 5\%$

D.f = $(n_2 - 1, n_1 - 1) = (7, 9)$

(iv) Critical value at 5% level of significance with (7, 9) d.f is $F_\alpha = 3.29$

(v) Here $F < F_\alpha$

$\therefore H_0$ is accepted.

$\therefore$ The difference of variances is not significant.

Chi-square test for Goodness of fit and Independence of attributes:

Test Based On X^2 -Test: [Chi-Square Test]

I-Application (or) uses of χ^2 distribution

- (i) To test the goodness of fit.
- (ii) To test the independence of attributes.
- (iii) To test the hypothetical value of the population variance.
- (iv) To test the homogeneity of independent estimates of the population variance.
- (v) To test the homogeneity of independent estimates of the population correlation coefficient.

II- Conditions for validity of χ^2 -test:

This is an approximate test for large values of 'n'. For the validity of χ^2 -test of goodness of fit between theory and experiment, the following conditions must be satisfied.

- (i) The sample observations should be independent.
- (ii) Constraints on the cell frequencies, if any, must be linear.
- (iii) N, the total frequency, should be reasonably large, say greater than 50.
- (iv) No theoretical cell frequency should be less than 5. If theoretical frequency is less than 5, then for the application of χ^2 -test, it is pooled with the succeeding (or) preceding so that the pooled frequency is more than 5.

III- Degrees of Freedom:

- (i) In the case of fitting a Binomial distribution, d.f = $n - 1$
- (ii) In the case of fitting a Poisson distribution d.f = $n - 2$
- (iii) In the case of fitting a Normal distribution d.f = $n - 3$

χ^2 -TESTS FOR GOODNESS OF FIT

χ^2 -test of goodness of fit is a test to find if the deviation of the experiment from theory is just by chance or it is due to the inadequacy of the theory to fit the observed data.

By this test, we test whether differences between observed and expected frequencies are significant or not.

$$\text{Test statistic } \chi^2 = \sum \frac{(O-E)^2}{E}$$

Where O – Observed frequency

E – Expected frequency

EXAMPLES

1) A coin were tossed 160 times and the following results were obtained.

No.of. Heads	0	1	2	3	4
Observed frequencies	17	52	54	31	6

Under the assumption that the coins are unbiased, find the expected frequencies of getting 0,1,2,3,4 heads and test goodness of fit.

Solution:

(i) H_0 : The coins are unbiased

H_1 : The coins are biased.

(ii) Level of significance $\alpha = 0.05$, d.f = $n-1 = 4$

(iii) Critical value of χ^2 is $\chi_{\alpha}^2 = 9.488$

(iv) Test Statistic $\chi^2 = \sum \frac{(O-E)^2}{E}$

Probability of getting head $p = \frac{1}{2}$

Probability of getting tail $q = \frac{1}{2}$

$\therefore$ Expected frequencies are

$$P(x) = nC_x p^x q^{n-x}, \quad x = 0, 1, 2, 3, 4$$

$$\therefore P(0) = 4C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^4 = 0.0625 \quad P(1) = 4C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^3 = 0.250$$

$$P(2) = 4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 = 0.375 \quad P(3) = 4C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^1 = 0.25$$

$$P(4) = 4C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^0 = 0.0625$$

No of heads	Observed frequency (O)	Expected frequency (E) [160×P(x)]	$\frac{(O-E)^2}{E}$
0	17	160×0.0625=10	4.9
1	52	160×0.25=40	3.6
2	54	160×0.375=60	0.6
3	31	160×0.25=40	2.025
4	6	160×0.0625=10	1.6
Total	160	160	12.725

$$\therefore \chi^2 = 12.725$$

(v) Here $\chi^2 > \chi_{\alpha}^2$

$\therefore$ We reject H_0 and accept H_1

$\therefore$ The coins are biased.

2) Records taken of the number of male and Female births in 800 families having four children are as follows.

No. of male births:	0	1	2	3	4
No. of female births:	4	3	2	1	0
No. of families:	32	178	290	206	94

Test whether the data are consistent with the hypothesis that the Binomial law holds and the chance of male birth is equal to that of female birth namely

$$p = q = \frac{1}{2}$$

Solution:

Here $n=5$

(i) H_0 : The data are consistent with the hypothesis that the Binomial law holds of equal probability for male and female births. ie. $p=q=1/2$

H_1 : The data are inconsistent.

(ii) Level of significance $\alpha=5\%$, d.f = $n-1=5-1=4$

(iii) Critical value $\chi_{\alpha}^2 = 9.488$

(iv) Test statistic $\chi^2 = \sum \frac{(O-E)^2}{E}$

We use binomial distribution to calculate theoretical frequency

$$P(X = x) = nC_x p^x q^{n-x}, \quad x = 0, 1, 2, 3, 4$$

$$P(x = 0) = 4C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^4 = 0.0625$$

$$P(x = 1) = 4C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^3 = 0.250$$

$$P(x = 2) = 4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 = 0.375$$

$$P(x=3) = 4C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^1 = 0.25$$

$$P(x=4) = 4C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^0 = 0.0625$$

No of male	Observed frequency (O)	Expected frequency (E) [N×P(x)]	$\frac{(O-E)^2}{E}$
0	32	800×0.0625=50	6.48
1	178	800×0.25=200	2.42
2	290	800×0.375=300	0.333
3	206	800×0.25=200	0.18
4	94	800×0.0625=50	38.72
Total	800	800	47.467

$$\therefore \chi^2 = 47.467$$

(v) Here $\chi^2 > \chi_{\alpha}^2$

$\therefore H_0$ is rejected and H_1 is accepted.

$\therefore$ The data are not consistent.

3) A die is thrown 270 times and the results of these throws are given below.

No. appeared on the die	1	2	3	4	5	6
Frequency	40	32	29	59	57	59

Test whether the die is biased or not.

Solution:

(i) H_0 : The die is unbiased

H_1 : The die is biased.

(ii) Level of significance $\alpha=5\%$, d.f = n-1 =5

(iii) Critical value $\chi_{\alpha}^2 = 11.070$

(iv) Test statistic $\chi^2 = \sum \frac{(O-E)^2}{E}$

	O	E	$\frac{(O-E)^2}{E}$
	40	46	0.7826
	32	46	4.2609
	29	46	6.2826
	59	46	3.6739
	57	46	2.6304
	59	46	3.6739
Total	276	276	21.30

$$\therefore \text{Expected frequency} = \frac{276}{6} = 46$$

$$\therefore \chi^2 = 21.30.$$

(v) Here $\chi^2 > \chi_{\alpha}^2$

$\therefore H_0$ is rejected and H_1 is accepted.

$\therefore$ The die is biased.

4) The table below gives the number of aircraft accidents that occurred during the various days of the week. Test whether the accidents are uniformly distributed over the week.

Days	Mon	Tue	Wed	Thu	Fri	Sat
No.of.Accidents	14	18	12	11	15	14

Solution:

(i) H_0 : The accidents are uniformly distributed over the week.

H_1 : The accidents are not uniformly distributed over the week.

(ii) Levels of significance $\alpha=5\%$, d.f = n-1

(iii) Critical value $\chi_{\alpha}^2 = 11.07$

(iv) Test statistic $\chi^2 = \sum \frac{(O-E)^2}{E}$

	O	E	O-E	$(O-E)^2$	$\frac{(O-E)^2}{E}$
	14	14	0	0	0
	18	14	4	16	1.143
	12	14	-2	4	0.286
	11	14	-3	9	0.643
	15	14	1	1	0.071
	14	14	0	0	0
Total	84	84			2.143

Here N=total frequency = 84

$\therefore$ Expected frequency = $84/6 = 14$

$$\therefore \chi^2 = 2.143$$

(v) Here $\chi^2 < \chi_{\alpha}^2$

$\therefore H_0$ is accepted

$\therefore$ The accidents are uniformly distributed over the week.

CHI-SQUARE TEST OF INDEPENDENCE OF ATTRIBUTES (CONTINGENCE TABLE)

Let us consider two attributes A and B. A is divided in 'r' classes $A_1, A_2, \dots, A_r$ and B is divided in 's' classes $B_1, B_2, \dots, B_s$. If this expressed as $r \times s$ matrix, the matrix is called $r \times s$ contingency table.

Suppose A is divided into two classes and B is divided in two classes. The various cell frequencies can be expressed in the following table known as 2×2 contingency table.

		B		Total
		B ₁	B ₂	
A	A ₁	a	b	a+b
	A ₂	c	d	c+d
Total		a+c	b+d	a+b+c+d=N

The expected frequencies are given by

$$\begin{array}{lll}
 E(a) = \frac{(a+b)(a+c)}{N} & E(b) = \frac{(a+b)(b+d)}{N} & a+b \\
 E(c) = \frac{(a+c)(c+d)}{N} & E(d) = \frac{(b+d)(c+d)}{N} & c+d \\
 a+c & b+d & N=a+b+c+d
 \end{array}$$

In this χ^2 test we test if two attributes A and B under consideration are independent or not.

Degrees of freedom d.f = $(r-1)(s-1)$

Where r – number of rows
 s – number of columns

EXAMPLES

1) What are the expected frequencies of 2×2 contingency tables given below

(i)	<table><tr><td>a</td><td>b</td></tr><tr><td>c</td><td>d</td></tr></table>	a	b	c	d	(ii)	<table><tr><td>2</td><td>10</td></tr><tr><td>6</td><td>6</td></tr></table>	2	10	6	6
a	b										
c	d										
2	10										
6	6										

Solution:

(i) Observed frequencies

a	b	a+b
c	d	c+d
a+c	b+d	a+b+c+d=N

Expected frequencies

$\frac{(a+b)(a+c)}{N}$	$\frac{(a+b)(b+d)}{N}$	a+b
$\frac{(a+c)(c+d)}{N}$	$\frac{(b+d)(c+d)}{N}$	c+d
a+c	b+d	N

(ii) Observed frequencies			Expected frequencies		
2	10	12	$\frac{8 \times 12}{24} = 4$	$\frac{16 \times 12}{24} = 8$	12
6	6	12	$\frac{8 \times 12}{24} = 4$	$\frac{16 \times 12}{24} = 8$	12
8	16	24	8	16	24

2) On the basis of information noted below, find out whether the new treatment is comparatively superior to the conventional one.

	Favourable	Non-Favourable	Total
Conventional	40	70	110
New	60	30	90
Total	100	100	200

Solution:

(i) H_0 : The two treatments are independent.

H_1 : The two treatments are dependent.

(ii) Level of significance $\alpha=5\%$

$$d.f = (r-1)(s-1) = (2-1)(2-1) = 1$$

(iii) Critical value $\chi_{\alpha}^2 = 3.841$

(iv) Test statistic $\chi^2 = \sum \frac{(O-E)^2}{E}$

The expected frequencies are

$\frac{110 \times 100}{200} = 55$	$\frac{110 \times 100}{200} = 55$
$\frac{90 \times 100}{200} = 45$	$\frac{90 \times 100}{200} = 45$

$$\therefore \chi_{\alpha}^2 = \frac{(40-55)^2}{55} + \frac{(70-55)^2}{55} + \frac{(60-45)^2}{45} + \frac{(30-45)^2}{45} = 18.1818$$

(v) Here $\chi^2 > \chi_{\alpha}^2$

$\therefore H_0$ is rejected and H_1 is accepted.

$\therefore$ The treatments are dependent.

3) From the following table regarding the colour of eyes of father and son. Test if the colour of son's eye is associated with that of the father.

		Eye Colour of son	
		Light	Not Light
Eye colour of father	Light	471	51
	Not light	148	230

Solution:

The observed frequencies are

		B		Total
		B₁	B₂	
A	A₁	471	51	522
	A₂	148	230	378
Total		619	281	900

The expected frequencies are

		B		Total
		B₁	B₂	
A	A₁	$\frac{619 \times 522}{900}$ = 359.02	$\frac{289 \times 522}{900}$ = 162.98	522
	A₂	$\frac{619 \times 378}{900}$ = 259.98	$\frac{289 \times 378}{900}$ = 118.02	378
Total		619	281	900

(i) H_0 : The colour of eyes of father and sons are independent.

H_1 : The colour of eyes of father and sons are dependent.

(ii) Level of significance $\alpha=5\%$

$$D.f=(r-1)(s-1) = (2-1)(2-1)=1.$$

(iii) Critical value $\chi_{\alpha}^2 = 3.841$

(iv). Test Statistic $\chi^2 = \sum \frac{(O-E)^2}{E}$

$$= \frac{(471-359.02)^2}{359.02} + \frac{(51-162.98)^2}{162.98} + \frac{(148-259.98)^2}{259.98} + \frac{(230-118.02)^2}{118.02}$$

$$= 266.366$$

(v) Here $\chi^2 > \chi_{\alpha}^2$

$\therefore H_0$ is rejected and H_1 is accepted.

$\therefore$ The colour of eyes of father and sons are dependent.

UNIT-4

DESIGN OF EXPERIMENTS

Basic Principles in the Design of Experiment:

There are three basic principles of an experimental design. They are

- 1) Randomization
- 2) Replication
- 3) Local Control

1. Randomization

Random assignment of treatment to the experimental units is the most effective way of eliminating any unknown bias in the experiment. For example, if we select plots for use of manure at random, the plots are experimental units and application of manure is treatment. The plots-selected for not using the manure are called control units or control group. The control group provide a standard for comparison.

2. Replication

Replication is the process of repeating the same treatment on more than one of the experimental units. Two identically treated plots (or units) will not give identical results. The differences are attributed to uncontrollable random causes. Such differences are called experimental errors. As the number of replication increases this error is reduced. So, replication is necessary to increase the accuracy of estimates of the treatment effects.

3. Local control

Another way of controlling the effects of extraneous variables is by employing the principle of local control. It consists of techniques of grouping, blocking and balancing of the experimental units.

Grouping means combining sets of homogeneous experimental unit into groups so that different groups may be subjected to different treatments. Each group can have different number of experimental units.

Blocking means assigning the same number of plots or experimental unit to different group, called blocks. The plots in the same block are relatively similar or homogeneous. We may use at random different manures to different plots in a block.

Balancing mean act of equalizing total effect of the extraneous variables on all the elements in the control group and in the experimental group.

Analysis of variance

Definition

In small samples, to test the single or equality of means of two populations, we use t-test. If there are more than two samples, t-test cannot be used. In such cases we

use a new technique called Analysis of Variance. So the main aim of Analysis of Variance is to test the equality of several means. It is shortly written as ANOVA.

Definition of ANOVA

Analysis of variance (ANOVA) is the separation of variance ascribable to one group of causes from the variance ascribable to other groups”.

It is nothing but an arithmetical procedure, used to express the total variation of data, as the sum of its non-negative components.

Assumptions

For the validity of the F-test in ANOVA, the following assumptions are made:

The observations are independent

- i. Parent population from which observations are taken is normal and
- ii. Various treatment and environmental effects are additive, in nature.

S.S.C - Sum of Squares between columns

T.S.S - Total Sum of Squares

S.S.T - S.S. due to treatments

M.S.S - Mean sum of squares

S.S.E - Error sum of squares (or) within sum of squares

R.S.S - Row sum of squares

C.F. - Correction factor

C.D - Critical Difference

S.S.R - Sum of squares between Rows

M.S.C - Mean sum of squares (between columns)

M.S.E - Mean sum of squares (within columns)

M.S.R - Mean sum of squares (between rows)

N - Number of Observations

N_1 - Number of elements in each column

N_2 - Number of elements in each row

ONE –WAY CLASSIFICATION – COMPLETELY RANDOMIZED DESIGN: (CRD)

Definition

It is a Design of Experiment in which treatments are allocated at random to the entire experimental units.

Working Procedure (One Way Classification CRD)

H_0 : There is no significant difference between the treatments.

H_1 : There is significant difference between the treatments.

Step 1: Find N i.e., The number of observations.

Step 2: Find T i.e., The total value of all observations.

Step 3: Find $\frac{T^2}{N}$ i.e., the correction factor.

Step 4: Calculate the total sum of squares.

$$TSS = \sum X_1^2 + \sum X_2^2 + \dots - \frac{T^2}{N}$$

Step 5: Calculate the column /row sum of squares

$$SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \dots - \frac{T^2}{N}$$

Here N_1 is the number of elements in each column.

(or)

$$SSR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} + \dots - \frac{T^2}{N}$$

Here N_2 is the number of elements in each row.

Step 6: Find $SSE = TSS - SSC$

Step 7: Prepare the ANOVA table to calculate F-ratio

Analysis of Variance Table

Source of Variation	Sum of squares	Degrees of freedom	Mean sum of squares	Variance ratio
Between columns	SSC	$c - 1$	$MSC = \frac{SSC}{c - 1}$	$F = \frac{MSC}{MSE}$
Error	SSE	$N - c$	$MSE = \frac{SSE}{N - c}$	(or)
Total	TSS	$N - 1$		$F = \frac{MSE}{MSC}$

The F- ratio should be calculated in such a way that $F > 1$

Step 8: Find the table value

Step 9: Conclusion

PROBLEMS ON COMPLETELY RANDOMIZED DESIGN: (CRD)

EXAMPLE

- 1) The following are the numbers of mistakes made in 5 successive days of 4 technicians working for a photographic laboratory :

Technician I (X_1)	Technician II (X_2)	Technician III (X_3)	Technician IV (X_4)
6	14	10	9
14	9	12	12
10	12	7	8
8	10	15	10
11	14	11	11

Test at the level of significance $\alpha = 0.01$, whether the difference among the 4 sample means, can be attributed to chance.

Solution:

H_0 : There is no significant difference between the technicians.

H_1 : There is significant difference between the technicians.

X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
6	14	10	9	39	36	196	100	81
14	9	12	12	47	196	81	144	144
10	12	7	8	37	100	144	49	64
8	10	15	10	43	64	100	225	100
11	14	11	11	47	121	196	121	121
49	59	55	50	213	517	717	639	510

Step 1: $N=20$

Step 2: $T=213$

Step 3: $\frac{T^2}{N} = \frac{(213)^2}{20} = 2268.45$

Step 4: $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N}$
 $= 517 + 717 + 639 + 510 - 2268.45 = 114.55$

Step 5 : $SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N}$

$N_1 \rightarrow$ number of elements in each column.

$$= \frac{(49)^2}{5} + \frac{(59)^2}{5} + \frac{(55)^2}{5} + \frac{(50)^2}{5} - 2268.45$$

$$= 480.2 + 696.2 + 605 + 500 - 2268.45$$

$$= 12.95$$

Step 6: $SSE = TSS - SSC$

$$= 114.55 - 12.95 = 101.6$$

Step 7: ANOVA Table

Source of variation	Sum of squares	Degrees of freedom	Mean sum of squares	Variance ratio	Table value at 1% level
Between columns	$SSC=12.95$	$c-1= 4-1$ $=3$	$MSC = \frac{SSC}{c-1}$ $MSC = \frac{12.95}{3}$ $MSC=4.317$	$F_c = \frac{MSE}{MSC}$ $= \frac{6.35}{4.317}$ $= 1.471 > 1$	$F_c(16, 3)$ $= 26.9$

Error	SSE=101.6	$N-c=20-4=16$	$MSE = \frac{SSE}{N-c}$ $MSE = \frac{101.6}{16}$ $MSE = 6.35$	Since $\frac{MSC}{MSE} < 1$	
Total	TSS=114.55				

Step 8 : Conclusion

Cal $F_c < \text{Tab } F_c$, So we accept H_0 .

It is concluded that there is no significant difference between the technicians at 1% level of significance.

- 2) There are three main brands of a certain powder. A set of 120 sample values is examined and found to be allocated among four groups (A,B,C and D) and three brands (I,II,III) as shown under.

Brands	Groups			
	A	B	C	D
I	0	4	8	15
II	5	8	13	6
III	8	19	11	13

Is there any significant difference in brands preference? Answer at 5% level.

Solution:

H_0 : There is no significant difference in brands.

H_1 : There is significant difference in brands.

Brands	Groups				Total	X_1^2	X_2^2	X_3^2	X_4^2
	A (X_1)	B (X_2)	C (X_3)	D (X_4)					
I (Y_1)	0	4	8	15	27	0	16	64	225
II (Y_2)	5	8	13	6	32	25	64	169	36
III (Y_3)	8	19	11	13	51	64	361	121	169
Total	13	31	32	34	110	89	441	354	430

Step 1: $N = 12$

Step 2: $T=110$

Step 3: $\frac{T^2}{N} = \frac{(110)^2}{12} = 1008.3$

Step 4: $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N}$
 $= 89 + 441 + 354 + 430 - 1008.3$

$$\text{Step 5: } SSR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} + \frac{(\sum Y_4)^2}{N_2} - \frac{T^2}{N}$$

$N_2 \rightarrow$ No. of elements in each row.

$$= \frac{(27)^2}{4} + \frac{(32)^2}{4} + \frac{(51)^2}{4} - 1008.3$$

$$= 182.25 + 256 + 650.25 - 1008.3$$

$$= 80.2$$

$$\text{Step 6: } SSE = TSS - SSR$$

$$= 305.7 - 80.2$$

$$SSE = 225.50$$

Step 7: ANOVA Table

Source of Variation	Sum of Squares	d.f	Mean sum of Squares	Variance ratio	Table value at 5% level
between rows	SSR = 80.2	$r-1=3-1=2$	$MSR = \frac{SSR}{r-1}$ $= \frac{80.2}{2}$ $MSR=40.1$	$F_R = \frac{MSR}{MSE}$ $= \frac{40.1}{20.06}$ $F_R=1.999$	$F_R(2,9)=4.26$
Error	SSE = 225.5	$N-r=12-3=9$	$MSE = \frac{SSE}{N-r}$ $= \frac{225.5}{9}$ $MSE=20.06$		
Total	TSS = 305.7				

Step 8: Conclusion

Cal $F_R < \text{Tab } F_R$,

so we accept H_0 .

3) The following table shows the lives in hours of four brands of electric lamps.

Brand A :	1610	1610	1650	1680	1700	1720	1800	
B :	1580	1640	1640	1700	1750			
C :	1460	1550	1600	1620	1640	1660	1740	1820
D :	1510	1520	1530	1570	1600	1680		

Perform an analysis of variance test the homogeneity of the mean lives of the four brands of Lamps.

Solution:

H_0 : There is no significant difference between the four brands.

H_1 : There is significant difference between the four brands.

Subtract 1600 and then divide by 10, we get

X_1 A	X_2 B	X_3 C	X_4 D	Total	X_1^2	X_2^2	X_3^2	X_4^2
1	-2	-14	-9	-24	1	4	196	81
1	4	-5	-8	-8	1	16	25	64
5	4	0	-7	2	25	16	0	49
8	10	2	-3	17	64	100	4	9
10	15	4	0	29	100	225	16	0
12	-	6	8	26	144	-	36	64
20	-	14	-	34	400	-	196	-
-	-	22	-	22	-	-	484	-
57	31	29	-19	98	735	361	957	267

Step 1: $N=26$

Step 2: $T = 98$

Step 3: $\frac{T^2}{N} = \frac{9604}{26} = 369.39$

Step 4: $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N}$
 $= 735 + 361 + 957 + 267 - 369.39$
 $= 1950.61$

Step 5: $SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N}$

$N_1 \rightarrow$ number of elements in corresponding column.

$$\begin{aligned}
 &= \frac{(57)^2}{7} + \frac{(31)^2}{5} + \frac{(29)^2}{8} + \frac{(-19)^2}{6} - 369.39 \\
 &= \frac{3249}{7} + \frac{961}{5} + \frac{841}{8} + \frac{361}{6} - 369.39 \\
 &= 464.14 + 192.2 + 105.13 + 60.17 - 369.39 \\
 &= 452.25
 \end{aligned}$$

Step 6: $SSE = TSS - SSC$

$$= 1950.61 - 452.25$$

$$= 1498.36$$

Step 7: ANOVA TABLE

Source of Variation	Sum of Squares	d.f	Mean sum of Square	Variance ratio	Table value at 5% level
Between columns	SSC = 452.25	$c-1$ $=4-1$ $=3$	$MSC=\frac{SSC}{c-1}$ $=\frac{452.25}{3}$ $MSC=150.75$	$F_c=\frac{MSC}{MSE}$ $=\frac{150.75}{68.11}$ $=2.21>1$	$F_c(3, 22)=3.05$
Error	SSE= 1498.36	$N-c$ $=26-4$ $=22$	$MSE=\frac{SSE}{N-c}$ $=\frac{1498.36}{22}$ $MSE=68.11$	Since $MSE< MSC$	
Total	TSS = 1950.61				

Step 8 : Conclusion.

Cal $F_c < \text{Tab } F_c$, so we accept H_0 .

TWO-WAY CLASSIFICATIONS – RANDOMIZED BLOCK

DESIGN (RBD):

Definition

If the given experimental area is not homogenous, it can be divided into many subgroups which are homogenous among themselves. Such groups are called Blocks. If the treatments are applied at random to these blocks, this design is known as Randomized Block Design (RBD).

Working Procedure: (Two – Way Classification, RBD)

H_0 : There is no significant difference

H_1 : There is significant difference

Step 1: Find N

Step 2: Find T

Step 3: Find $\frac{T^2}{N}$

Step 4: Find TSS

Step 5: Find SSC

Step 6: Find SSR

Step 7: $SSE = TSS - SSC - SSR$

Step 8: Prepare the ANOVA Table.

Source of Variation	Sum of Squares	Degrees of freedom	Mean sum of Squares	Variance ratio
Column treatments	SSC	(c-1)	$MSC = \frac{SSC}{c-1}$	$F_c = \frac{MSC}{MSE}$
Row treatments	SSR	(r-1)	$MSR = \frac{SSR}{r-1}$	$F_r = \frac{MSR}{MSE}$
Error	SSE	(N-c-r+1)	$MSE = \frac{SSE}{(N-c-r+1)}$	
Total	TSS	rc-1		

Note: The F- ratios F_c , F_r are calculated in a such way that they are each greater than one.

For example, if $\frac{MSC}{MSE} < 1$, then take $F_c = \frac{MSE}{MSC}$

Step 9: Find Table F_c and F_r

Step 10: Conclusion.

Problems On Randomized Block Design:

EXAMPLE

- 1) An experiment was designed to study the performance of 4 different detergents for cleaning fuel injectors. The following “Cleanliness” readings were obtained with specially designed equipment for 12 tanks of gas distributed over 3 different modes of engines.

	Engine 1	Engine 2	Engine 3	Total
Detergent A	45	43	51	139
Detergent B	47	46	52	145
Detergent C	48	50	55	153
Detergent D	42	37	49	128
Total	182	176	207	565

Perform the ANOVA and test at 0.01 level of significance, whether there are differences in the detergents or in the engines.

Solution:

The above data are classified according to criteria (i) Detergent (ii) Engine.

In order to simplify calculations, we code the data by subtracting 50 from each figure.

Detergent	Engine			Total	X_1^2	X_2^2	X_3^2
	(X ₁)	(X ₂)	(X ₃)				
A (Y ₁)	-5	-7	1	-11	25	49	1
B (Y ₂)	-3	-4	2	-5	9	16	4
C (Y ₃)	-2	0	5	3	4	0	25
D (Y ₄)	-8	-13	-1	-22	64	169	1
Total	-18	-24	7	-35	102	234	31

- 1) H_0 : There is no significant difference between Detergents and Engines
- 2) H_1 : There is significant difference between Detergents and Engines.

Step 1: $N = 12$ [Total Number of entries]

Step 2: $T = -35$

Step 3: $\frac{T^2}{N} = \frac{(-35)^2}{12} = 102.08$

Step 4: $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 - \frac{T^2}{N}$
 $= (102) + (234) + (31) - 102.08$
 $= 367 - 102.08$
 $= 264.92$

Step 5: $SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} - \frac{T^2}{N}$
 ($N_1 \rightarrow$ No of elements in each column)
 $= \frac{(-18)^2}{4} + \frac{(-24)^2}{4} + \frac{(7)^2}{4} - 102.08$
 $= 81 + 144 + 12.25 - 102.08$
 $= 135.17$

Step 6: $SSR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} + \frac{(\sum Y_4)^2}{N_2} - \frac{T^2}{N}$
 ($N_2 \rightarrow$ No of elements in each row)
 $= \frac{(-11)^2}{3} + \frac{(-5)^2}{3} + \frac{(3)^2}{3} + \frac{(-22)^2}{3} - 102.08$
 $= 40.33 + 8.33 + 3 + 161.33 - 102.08$
 $SSR = 110.91$

Step 7:

$$SSE = TSS - SSC - SSR$$

$$= 264.92 - 135.17 - 110.91$$

$$SSE = 18.84$$

Step 8: Table of Analysis of Variance

Source of Variation	Sum of Squares	Degree of freedom	Mean sum of Squares	Variance ratio	Table value at 1% level
Between Columns	SSC = 135.17	$c-1 = 3-1=2$	$MSC = \frac{SSC}{c-1} = \frac{135.17}{2}$ $MSC = 67.585$	$F_c = \frac{MSC}{MSE} = \frac{67.585}{3.14}$ $F_c = 21.52$	$F_c(2,6) = 10.92$
Between Rows	SSR = 110.91	$r-1 = 4-1 = 3$	$MSR = \frac{SSR}{r-1} = \frac{110.91}{3}$ $MSR = 36.97$	$F_R = \frac{MSR}{MSE} = \frac{36.97}{3.14}$ $F_R = 11.77$	$F_R(3,6) = 9.78$
Error	SSE = 18.84	$N-c-r+1 = 12-3-4+1 = 6$	$MSE = \frac{SSE}{N-c-r+1} = \frac{18.84}{6}$ $MSE = 3.14$		
Total	TSS = 264.92	$rc-1 = 3 \times 4 - 1 = 11$			

Step 8 : Conclusion

Cal $F_c > \text{Table } F_c$, so we reject H_0 .

Cal $F_R > \text{Table } F_R$, so we reject H_0 .

2) Analyses the following RBD and find your conclusion.

Blocks	Treatments				
		T ₁	T ₂	T ₃	T ₄
	B ₁	12	14	20	22
	B ₂	17	27	19	15
	B ₃	15	14	17	12
	B ₄	18	16	22	12
	B ₅	19	15	20	14

Solution:

H_0 : There is no significant difference between blocks and treatments.

H_1 : There is significant difference between blocks and treatments

Subtract 15 from each number we get

	X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
Y_1	-3	-1	5	7	8	9	1	25	49
Y_2	2	12	4	0	18	4	144	16	0
Y_3	0	-1	2	-3	-2	0	1	4	9
Y_4	3	1	7	-3	8	9	1	49	9
Y_5	4	0	5	-1	8	16	0	25	1
Total	6	11	23	0	40	38	147	119	68

Step 1: $N = 20$

Step 2: $T = 40$

Step 3: $\frac{T^2}{N} = \frac{(40)^2}{20} = 80$

Step 4: $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N}$
 $= 38 + 147 + 119 + 68 - 80$
 $TSS = 292$

Step 5: $SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N}$
 $(N_1 \rightarrow \text{No of elements in each column})$
 $= \frac{(6)^2}{5} + \frac{(11)^2}{5} + \frac{(23)^2}{5} + 0 - 80$
 $SSC = 57.2$

Step 6: $SSR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} + \frac{(\sum Y_4)^2}{N_2} + \frac{(\sum Y_5)^2}{N_2} - \frac{T^2}{N}$
 $(N_2 \rightarrow \text{No of elements in each row})$
 $= \frac{8^2}{4} + \frac{18^2}{4} + \frac{(-2)^2}{4} + \frac{8^2}{4} + \frac{8^2}{4} - 80$
 $SSR = 50$

Step 7: $SSE = TSS - SSC - SSR$
 $= 292 - 57.2 - 50$
 $SSE = 184.8$

Step 8: ANOVA Table

Source of Variation	Sum of Squares	Degree of freedom	Mean sum of Squares	Variance ratio	Table value at 1% level
Between Columns	SSC = 57.2	$c-1 = 4-1=3$	$MSC = \frac{SSC}{c-1} = \frac{57.2}{3}$ $MSC=19.1$	$F_c = \frac{MSC}{MSE} = \frac{19.1}{15.4}$ $F_c=1.24$	$F_c(3,12) = 3.49$
Between Rows	SSR = 50	$r-1 = 5-1 = 4$	$MSR = \frac{SSR}{r-1} = \frac{50}{4}$ $MSR=12.5$	$F_r = \frac{MSE}{MSR} = \frac{15.4}{12.5}$ $F_r=1.23$	$F_r(12,4) = 5.91$
Error	SSE = 184.8	$N-c-r+1 = 20-4-5+1 = 12$	$MSE = \frac{SSE}{N-c-r+1} = \frac{184.8}{12}$ $MSE=15.4$		
Total	TSS = 292				

Step 9: Conclusion.

Cal $F_c < \text{Table } F_c$, so we accept H_0 .

Cal $F_r < \text{Table } F_r$, so we accept H_0 .

LATIN SQUARE DESIGN (LSD):

Definition:

The design of experiment having equal number of replications and treatments is known as Latin Square Design. Here the data is classified in the form of a square matrix. It controls the experimental error in two perpendicular directions.

Working Rule:

Step 1: Find N

Step 2: Find T

Step 3: Find $\frac{T^2}{N}$

Step 4: Find TSS

Step 5: Find SSC

Step 6: Find SSR,

Step 7: Find SSK

Step 8: Find SSE

Step 9: ANOVA Table

Step 10: Conclusion

Preparation of ANOVA Table

Source of Variation	Sum of Squares	Degree of freedom	Mean Sum of Squares	Variance ratio
Between Columns	SSC	K-1	$MSC = \frac{SSC}{K-1}$	$F_C = \frac{MSC}{MSE}$
Between Rows	SSR	K-1	$MSR = \frac{SSR}{K-1}$	$F_R = \frac{MSR}{MSE}$
Between treatments	SSK	K-1	$MSK = \frac{SSK}{K-1}$	$F_T = \frac{MSK}{MSE}$
Error	SSE	(K-1)(K-2)	$MSE = \frac{SSE}{(K-1)(K-2)}$	
Total	TSS	K^2-1		

The F-ratios F_C , F_R , F_T are calculated in a such way that they are each greater than one.

For eg, if $\frac{MSC}{MSE} < 1$, then take $F_C = \frac{MSE}{MSC}$.

EXAMPLE

- The following is a Latin Square of a design, when 4 varieties of seeds are being tested. Set up the analysis of variance table and state your conclusion. You may carry out suitable change of origin and scale.

A	105	B	95	C	125	D	115
C	115	D	125	A	105	B	105
D	115	C	95	B	105	A	115
B	95	A	135	D	95	C	115

Solution:

Subtract 100 and then divide by 5, we get

A	1	B	-1	C	5	D	3
C	3	D	5	A	1	B	1
D	3	C	-1	B	1	A	3
B	-1	A	7	D	-1	C	3

H_0 : There is no significant difference

H_1 : There is significant difference

	X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
Y_1	1	-1	5	3	8	1	1	25	9
Y_2	3	5	1	1	10	9	25	1	1
Y_3	3	-1	1	3	6	9	1	1	9
Y_4	-1	7	-1	3	8	1	49	1	9
Total	6	10	6	10	T = 32	20	76	28	28

Step 1: $N = 16$

Step 2: $T = 32$

Step 3: $\frac{T^2}{N} = \frac{(32)^2}{16} = 64$

Step 4: $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N}$
 $= 20 + 76 + 28 + 28 - 64 = 88$

$TSS = 88$

Step 5: $SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N}$

($N_1 \rightarrow$ No of elements in each column)

$$= \frac{6^2}{4} + \frac{10^2}{4} + \frac{6^2}{4} + \frac{10^2}{4} - 64$$

$$= 9 + 25 + 9 + 25 - 64$$

$SSC = 4$

Step 6: $SSR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} + \frac{(\sum Y_4)^2}{N_2} - \frac{T^2}{N}$

($N_2 \rightarrow$ No of elements in each row)

$$= \frac{8^2}{4} + \frac{10^2}{4} + \frac{6^2}{4} + \frac{8^2}{4} - 64$$

$$= 16 + 25 + 9 + 16 - 64$$

$SSR = 2$

Step 7: To find SSK

Arrange the elements in the order of treatments.

	Total				
A	1	1	3	7	12
B	-1	1	1	-1	0
C	5	3	-1	3	10
D	3	5	3	-1	10

$$SSK = \frac{(12)^2}{4} + \frac{0^2}{4} + \frac{(10)^2}{4} + \frac{(10)^2}{4} - \frac{T^2}{N}$$

$$= 36 + 0 + 25 + 25 - 64$$

$SSK = 22$

Step 8: $SSE = TSS - SSC - SSR - SSK$
 $= 88 - 4 - 2 - 22$

$SSE = 60$

Step 9: ANOVA Table

Source of Variation	Sum of Squares	Degree of freedom	Mean sum of Squares	Variance ratio	Table value at 5% level
Between Rows	SSR = 2	K-1=3	$MSR = \frac{SSR}{K-1}$ $= 0.67$	$F_R = \frac{MSR}{MSE}$ $= \frac{10}{0.67}$ $F_R = 14.9$	$F_{R(6,3)} = 8.94$
Between Columns	SSC = 4	K-1=3	$MSC = \frac{SSC}{K-1}$ $= 1.33$	$F_C = \frac{MSE}{MSC}$ $= \frac{10}{1.33}$ $F_C = 7.52$	$F_{C(6,3)} = 8.94$
Between treatments	SSK=22	K-1=3	$MSK = \frac{SSK}{K-1}$ $= 7.33$	$F_T = \frac{MSE}{MSK}$ $= \frac{10}{7.33}$ $F_T = 1.36$	$F_{T(6,3)} = 8.94$
Error	SSE=60	$(K-1)(K-2) = 6$	$MSE = \frac{SSE}{(K-1)(K-2)}$ $= 10$		
Total	TSS=88	15			

Step 10: Conclusion.

Here Cal $F_C < \text{Table } F_C$, so we accept H_0 .

Cal $F_R > \text{Table } F_R$, so we reject H_0 .

Cal $F_T < \text{Table } F_T$, so we accept H_0 .

- A farmer wishes to test the effects of four different fertilizers A, B, C, and D on the yield of wheat. In order to eliminate sources of error due to variability in soil fertility he uses the fertilizers in a Latin Square arrangement as indicated in the following table, where the numbers indicate yields in bushels per unit area.

A	C	D	B
18	21	25	11
D	B	A	C
22	12	15	19
B	A	C	D
15	20	23	24
C	D	B	A
22	21	10	17

Perform an analysis of variance to determine, If there is a significant difference between the fertilizers at $\alpha = 0.05$ level of significance.

Solution:

Subtract 20 we get

A	-2	C	1	D	5	B	-9
D	2	B	-8	A	-5	C	-1
B	-5	A	0	C	3	D	4
C	2	D	1	B	-10	A	-3

H_0 : There is no significant difference

H_1 : There is significant difference

	X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
Y_1	-2	1	5	-9	-5	4	1	25	81
Y_2	2	-8	-5	-1	-12	4	64	25	1
Y_3	-5	0	3	4	2	25	0	9	16
Y_4	2	1	-10	-3	-10	4	1	100	9
Total	-3	-6	-7	-9	-25	37	66	159	107

Step 1: $N = 16$

Step 2: $T = -25$

Step 3: $\frac{T^2}{N} = \frac{(-25)^2}{16} = \frac{625}{16} = 39.06$

Step 4: $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N}$

$$= 37 + 66 + 159 + 107 - 39.06$$

$$= 369 - 39.06 = 329.94$$

$TSS = 329.94$

$$\text{Step 5: } SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N}$$

($N_1 \rightarrow$ No of elements in each column)

$$= \frac{9}{4} + \frac{36}{4} + \frac{49}{4} + \frac{81}{4} - 39.06$$

$$= \frac{175}{4} - 39.06 = 4.69$$

$$SSC = 4.69$$

$$\text{Step 6: } SSR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} + \frac{(\sum Y_4)^2}{N_2} - \frac{T^2}{N}$$

(N₂ → No of elements in each row)

$$= \frac{(-5)^2}{4} + \frac{(-12)^2}{4} + \frac{(2)^2}{4} + \frac{(-10)^2}{4} - 39.06$$

$$= \frac{273}{4} - 39.06 = 29.19$$

$$SSR = 29.19$$

Step 7: To find SSK

Arrange the elements in the order of treatments.

	Total				
A	-2	-5	0	-3	-10
B	-9	-8	-5	-10	-32
C	1	-1	3	2	5
D	5	2	4	1	12

$$SSK = \frac{(-10)^2}{4} + \frac{(-32)^2}{4} + \frac{(5)^2}{4} + \frac{(12)^2}{4} - \frac{T^2}{N}$$

$$= \frac{100}{4} + \frac{1024}{4} + \frac{25}{4} + \frac{144}{4} - 39.06$$

$$SSK = 284.19$$

Step 7: SSE = TSS – SSC – SSR - SSK

$$= 329.94 - 4.69 - 29.19 - 284.19$$

$$SSE = 11.87$$

Step 8: ANOVA Table

Source of Variation	Sum of Squares	Degree of freedom	Mean sum of Squares	Variance ratio	Table value at 5% level
Between Rows	SSR = 29.19	K-1=3	MSR = $\frac{SSR}{K-1}$ = 9.73	$F_R = \frac{MSR}{MSE}$ $F_R = 4.91$	$F_R (3,6)$ = 4.76
Between Columns	SSC = 4.69	K-1=3	MSC = $\frac{SSC}{K-1}$ = 1.56	$F_C = \frac{MSE}{MSC}$ $F_C = 1.26$	$F_C (6,3)$ = 8.94

Between treatments	SSK=284.1 9	K-1=3	$MSK = \frac{SSK}{K-1}$ = 94.73	$F_T = \frac{MSK}{MSE}$ $F_T = 47.8$	$F_T (3,6)$ = 4.76
Error	SSE=11.87	(K-1) (K-2)= 6	$MSE = \frac{SSE}{(K-1)(K-2)}$ = 1.98		

Step 10: Conclusion.

Here Cal $F_c < \text{Table } F_c$, so we accept H_0 .

Cal $F_R > \text{Table } F_R$, so we reject H_0 .

Cal $F_T > \text{Table } F_T$, so we reject H_0 .

3. In a Latin Square experiment given below are the yields in quintals per acre on the paddy crop carried out for testing the effect of five fertilizers A, B, C, D, E. Analyze the data for variations.

B	25	A	18	E	27	D	30	C	27
A	19	D	31	C	29	E	26	B	23
C	28	B	22	D	33	A	18	E	27
E	28	C	26	A	20	B	25	D	33
D	32	E	25	B	23	C	28	A	20

Solution:

It is a 5×5 Latin Square. Let us subtract 25 from all the items.

	X₁	X₂	X₃	X₄	X₅	Total	X₁²	X₂²	X₃²	X₄²	X₅²
Y₁	0	-7	2	5	2	2	0	49	4	25	4
Y₂	-6	6	4	1	-2	3	36	36	16	1	4
Y₃	3	-3	8	-7	2	3	9	9	64	49	4
Y₄	3	1	-5	0	8	7	9	1	25	0	64
Y₅	7	0	-2	3	-5	3	49	0	4	9	25
Total	7	-3	7	2	5	18	103	95	113	84	101

H_0 : There is no significant difference

H_1 : There is significant difference

Step 1: N = 25

Step 2: T = 18

Step 3: $\frac{T^2}{N} = \frac{(18)^2}{25} = 12.96$

Step 4: $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 + \sum X_5^2 - \frac{T^2}{N}$

$$= 103 + 95 + 113 + 84 + 101 - 12.96$$

$$TSS = 483.04$$

$$\text{Step 5: } SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} + \frac{(\sum X_5)^2}{N_1} - \frac{T^2}{N}$$

($N_1 \rightarrow$ No of elements in each column)

$$= \frac{49}{5} + \frac{9}{5} + \frac{49}{5} + \frac{4}{5} + \frac{25}{5} - 12.96$$

$$SSC = 14.24$$

$$\text{Step 6 : } SR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} + \frac{(\sum Y_4)^2}{N_2} + \frac{(\sum Y_5)^2}{N_1} - \frac{T^2}{N}$$

($N_2 \rightarrow$ No of elements in each row)

$$= \frac{4}{5} + \frac{9}{5} + \frac{9}{5} + \frac{49}{5} + \frac{9}{5} - 12.96$$

$$SSR = 3.04$$

Step 7: To find SSK

Arrange the elements in the order of treatments.

Treatment	1	2	3	4	5	Total
A	-6	-7	-5	-7	-5	-30
B	0	-3	-2	0	-2	-7
C	3	1	4	3	2	13
D	7	6	8	5	8	34
E	3	0	2	1	2	8

$$SSK = \frac{1}{5} [(-30)^2 + (-7)^2 + (13)^2 + (34)^2 + (8)^2] - \frac{T^2}{N}$$

$$= \frac{1}{5} [900 + 49 + 169 + 1156 + 64] - 12.96$$

$$= 454.64$$

$$\text{Step 7: } SSE = TSS - SSC - SSR - SSK$$

$$= 483.04 - 14.24 - 3.04 - 454.64$$

$$SSE = 11.12$$

Step 8: ANOVA Table

Source of Variation	Sum of Squares	Degree of freedom	Mean sum of Squares	Variance ratio	Table value at 5% level
Between Columns	SSC = 14.24	$K-1=5-1=4$	$MSC = \frac{SSC}{K-1}$ $= \frac{14.24}{4}$ $= 3.56$	$F_C = \frac{MSC}{MSE}$ $= \frac{3.56}{0.927}$ $F_C = 3.84$	$F_C(4,12)$ $= 3.26$

Between Rows	SSR = 3.04	$K-1=5-1=4$	$MSR = \frac{SSR}{K-1}$ $= \frac{3.04}{4}$ $= 0.76$	$F_R = \frac{MSE}{MSR}$ $= \frac{0.927}{0.76}$ $F_R = 1.22$	$F_R (12,4)$ $= 5.91$
Between treatments	SSK=454.64	$K-1=5-1=4$	$MSK = \frac{SSK}{K-1}$ $= \frac{454.64}{4}$ $= 113.66$	$F_T = \frac{MSK}{MSE}$ $= \frac{113.66}{0.927}$ $F_T = 122.6$	$F_T (4,12)$ $= 3.26$
Error	SSE=11.12	$(K-1)(K-2)$ $= 12$	$MSE = \frac{SSE}{(K-1)(K-2)}$ $= \frac{11.12}{12} = 0.927$		

Step 10: Conclusion.

Here Cal $F_c >$ Table F_c , (There is significant difference)

Cal $F_R <$ Table F_R , (There is no significant difference)

Cal $F_T >$ Table F_T , (There is significant difference)

2² FACTORIAL DESIGN:

Definition

The Design of experimental in which “K” factors are applied each at 2 levels (presence and absence) is called 2² Factorial Design.

Drawbacks of 2ⁿ factor design

1. Each factor has 2 levels and so it is impossible to judge whether the effects produced by variations in a factor are linear or parabolic or exponential.
2. If the number of factors is more, then the number of experimentation becomes very large. For example in a 2⁸ factorial design, 256 treatment combinations are required which is large.

Because of the large number of treatment combinations in 2ⁿ design quite often we use only a fraction of the experimental runs. This type of design is called fractional factorial design. The idea in this design is to reduce the size of the total experiment retaining the key information regarding the factors.

EXAMPLE

1. Find out the main effects and interactions in the following 2² factorial experiment and write down the Analysis of Variance table.

	(1)	a	b	ab
	00	10	01	11
Block I	64	25	30	6
II	75	14	50	33
III	76	12	41	17
IV	75	33	25	10

Solution:

Taking deviation from $y = 37$, we get

Treatment Combination	I X_1	II X_2	III X_3	IV X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
(Y ₁) (1)	27	38	39	38	142	729	1444	1521	1444
(Y ₂) a	-12	-23	-25	-4	-64	144	529	625	16
(Y ₃) b	-7	13	4	-12	-2	49	169	16	144
(Y ₄) ab	-31	-4	-20	-27	-82	961	16	400	729
Total	-23	24	-2	-5	-6	1883	2158	2562	2333

Step 1: $N = 16$

Step 2: $T = -6$

Step 3: Correction Factor $= \frac{T^2}{N} = \frac{36}{16} = 2.25$

Step 4: $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N}$
 $= 1883 + 2158 + 2562 + 2333 - 2.25$
 $TSS = 8933.75$

Step 5: $SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N}$
 $(N_1 \rightarrow \text{No of elements in each column})$
 $= \frac{(-23)^2}{4} + \frac{(24)^2}{4} + \frac{(-2)^2}{4} + \frac{(-82)^2}{4} - 2.25$
 $= \frac{529}{4} + \frac{576}{4} + \frac{4}{4} + \frac{25}{4} - 2.25$
 $SSC = 281.25$

Step 6: $SSR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} + \frac{(\sum Y_4)^2}{N_2} - \frac{T^2}{N}$
 $(N_2 \rightarrow \text{No of elements in each row})$
 $= \frac{(142)^2}{4} + \frac{(-64)^2}{4} + \frac{(-2)^2}{4} + \frac{(-82)^2}{4} - 2.25$
 $= \frac{20164}{4} + \frac{4096}{4} + \frac{4}{4} + \frac{6724}{4} - 2.25$
 $SSR = 7744.75$

Step 7: $SSE = TSS - SSC - SSR$

$$= 8933.75 - 281.25 - 7744.75$$

$$SSE = 907.75$$

For 2² experiment

$$\text{Contrast A} = [a + ab - b - (1)]$$

$$= -64 - 82 + 2 - 142 = -286$$

$$\text{Contrast B} = [b + ab - a - (1)]$$

$$= -2 - 82 + 64 - 142 = -162$$

$$\text{Contrast AB} = [ab + (1) - a - b]$$

$$= -82 + 142 + 64 + 2 = 126$$

$$\text{Main effect A} = \frac{1}{2} [a + ab - b - (1)]$$

$$= -143$$

$$B = \frac{1}{2} [b + ab - a - (1)]$$

$$= -81$$

$$AB = \frac{1}{2} [ab + (1) - a - b]$$

$$= 63$$

$$SS_A = \frac{[a + ab - b - (1)]^2}{16}$$

$$= \frac{(-286)^2}{16} = 5112.25$$

$$SS_B = \frac{[b + ab - a - (1)]^2}{16}$$

$$= \frac{(-162)^2}{16} = 1640.25$$

$$SS_{AB} = \frac{[ab + (1) - a - b]^2}{16}$$

$$= \frac{(126)^2}{16} = 992.25$$

Analysis of variance table for the 2² experiment

S.V	d.f	S.S	MSS	Variance ratio	Table value of F	
					5%	1%
Blocks	3	281.5	93.83	$F_c = \frac{100.86}{93.83} = 1.075$	$F_c(9,3) = 8.81$	27.35
Treatment	3	7744.75	2581.58	$F_R = \frac{2581.58}{100.86} = 25.60$	$F_R(3,9) = 3.86$	6.99
A	1	5112.25	5112.25	$\frac{5112.25}{100.86} = 50.69$	$F_A(1,9) = 5.12$	6.99

B	1	1640.2 5	1640.2 5	$\frac{1640.25}{100.86} = 16.26$	$F_B (1,9) = 5.12$	6.99
AB	1	992.25	992.25	$\frac{992.25}{100.86} = 9.84$	$F_{AB} (1,9) = 5.12$	6.99
Error	9	907.75	100.86			

$$\begin{aligned}
 \text{Error (d.f)} &= N - c - r + 1 \\
 &= 16 - 4 - 4 + 1 \\
 &= 9
 \end{aligned}$$

Conclusion:

Here Cal FA > Table FA
 Cal FB > Table FB
 Cal FAB > Table FAB

2. An experiment was planned to study the effect of sulphate of potash and super phosphate on the yield of potatoes. The (y_{70}) yields [lb. per plot = (y_{70})] acre obtained are given in the table. Analyse the data and give your conclusions.

Block	Yields (lbs per plot)			
I	(1) 23	a 25	b 22	ab 38
II	b 40	(1) 26	a 36	ab 38
III	(1) 29	a 20	ba 30	b 20
IV	ab 34	a 31	b 24	(1) 28

Solution:

Treatment combination	Blocks				Total	X_1^2	X_2^2	X_3^2	X_4^2
	X_1	X_2	X_3	X_4					
Y_1 (1)	-6	-3	0	1	-10	36	9	0	1
Y_2 a	-4	7	-9	2	-4	16	49	81	4
Y_3 b	-7	11	-9	-5	-10	49	121	81	25
Y_4 ab	9	9	1	5	24	81	81	1	25
Total	-8	24	-17	1	0	182	260	163	55

Step 1: $N=16$

Step 2: $T=0$

Step 3: $CF = \frac{T^2}{N} = 0$

Step 4: $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N}$

$$=182+260+163+55-0$$

$$=660$$

$$\text{Step 5: } SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N}$$

$$= \frac{64}{4} + \frac{576}{4} + \frac{289}{4} + \frac{1}{4} - 0$$

$$=232.5$$

$$\text{Step 6: } SSR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} + \frac{(\sum Y_4)^2}{N_2} - \frac{T^2}{N}$$

$$= \frac{100}{4} + \frac{16}{4} + \frac{100}{4} + \frac{576}{4} - 0$$

$$=198$$

$$\text{Step 7: } SSE = TSS - SSC - SSR$$

$$=660-232.5-198$$

$$=229.50$$

For 2² Experiment

$$\text{Contrast A} = [a + ab - b - (1)]$$

$$= [-4+24+10+10]$$

$$A = 40$$

$$\text{Contrast B} = [b + ab - a - (1)]$$

$$= [-10+24+4+10]$$

$$B = 28$$

$$\text{Contrast AB} = [ab + (1) - a - b]$$

$$= [24-10+4+10]$$

$$AB = 28$$

$$SS_A = \frac{[a + ab - b - (1)]^2}{16}$$

$$= \frac{40^2}{16} = 100$$

$$SS_B = \frac{[b + ab - a - (1)]^2}{16}$$

$$= \frac{(28)^2}{16} = 49$$

$$SS_{AB} = \frac{[ab + (1) - a - b]^2}{16}$$

$$= \frac{(28)^2}{16} = 49$$

Analysis of variance table for the 2² experiment.

S.V	d. t	S.S	MSS	Variance ratio	Table value of F	
					5%	1%
Blocks	3	232.5	$\frac{232.5}{3} = 77.5$	$F_C = \frac{77.5}{25.5} = 3.04$	$F_C(3,9)=3.86$	6.99
Treatments	3	198.00	$\frac{198}{3} = 66.00$	$F_R = \frac{66}{25.5} = 2.59$	$F_R(3,9)=3.86$	6.99
A	1	100	$\frac{100}{1} = 100$	$F_A = \frac{100}{25.5} = 3.92$	$F_A(1,9) = 5.12$	10.56
B	1	49	$\frac{49}{1} = 49$	$F_B = \frac{49}{25.5} = 1.92$	$F_B(1,9) = 5.12$	10.56
AB	1	49	$\frac{49}{1} = 49$	$F_{AB} = \frac{49}{25.5} = 1.92$	$F_{AB}(1,9) = 5.12$	10.56
Error	9	229.5	$\frac{229.5}{9} = 25.5$			

$$\begin{aligned}
 \text{Error (d.f.)} &= N - c - r + 1 \\
 &= 16 - 4 - 4 + 1 \\
 &= 17 - 8 \\
 &= 9
 \end{aligned}$$

As in each of the cases, the computed value of F is less than the corresponding tabulated value, there is no significant difference.

3) The following data represents the results of a 2^2 factorial design with 2 factors and 2 levels each with four replications. Analyses the data for response.

Treatment Combination	Replications			
	I	II	III	IV
(1)	12	12.3	11.8	11.6
a	12.8	12.6	13.7	14
b	11.5	11.9	12.6	11.8
ab	14.2	14.5	14.4	15

Solution:

Let A and B be the two factors.

Let high and low be the levels.

N = number of replications = 4

We code the data by subtracting 12 from each value. The coded data is

Treatment Combination	Replications				Total	X_1^2	X_2^2	X_3^2	X_4^2
	X_1	X_2	X_3	X_4					
(1)	0	0.3	-0.2	-0.4	-0.3	0	0.09	0.04	0.16
a	0.8	0.6	1.7	2	5.1	0.64	0.36	2.89	4
b	-0.5	-0.1	0.6	-0.2	-0.2	0.25	0.01	0.36	0.04
ab	2.2	2.5	2.4	3	10.1	4.84	6.25	5.76	9
Total					14.7	5.73	6.71	9.05	13.2

$$N = 4 \times 4 = 16,$$

$$T = 14.7$$

$$\text{Correction factor} = \frac{(14.7)^2}{16} = 13.5$$

$$\begin{aligned} \text{A Contrast} &= a + ab - b - (1) \\ &= 5.1 + 10.1 - (-0.2) - (-0.3) = 15.7 \end{aligned}$$

$$\begin{aligned} \text{B Contrast} &= b + ab - a - (-1) \\ &= -0.2 + 10.1 - 5.1 - (-0.3) = 5.1 \end{aligned}$$

$$\begin{aligned} \text{AB Contrast} &= (1) + ab - a - b \\ &= -0.3 + 10.1 - 5.1 - (-0.2) = 4.9 \end{aligned}$$

$$SSA = \frac{(\text{A Contrast})^2}{4n} = \frac{(15.7)^2}{4 \times 4} = 15.41$$

$$SSB = \frac{(\text{B Contrast})^2}{4n} = \frac{(5.1)^2}{4 \times 4} = 1.63$$

$$SSAB = \frac{(\text{AB Contrast})^2}{4n} = \frac{(4.9)^2}{4 \times 4} = 1.5$$

$$SST = \sum x_i^2 - \frac{T^2}{N} = 5.73 + 6.71 + 9.05 + 13.2 - 13.5$$

$$= 34.7 - 13.5 = 21.2$$

$$SSE = SST - SSA - SSAB$$

$$= 21.2 - 15.41 - 1.63 - 1.5 = 2.66$$

ANOVA TABLE

Source	S.S	d.f.	MS	Variation ratio F
Factor A	SSA = 15.41	1	MSA = SSA=15.41	$F_A = \frac{MSA}{MSE} = \frac{15.41}{0.222} = 69.41$
Factor B	SSB = 1.63	1	MSB = SSB = 1.63	$F_B = \frac{MSB}{MSE} = \frac{1.63}{0.222} = 7.34$
Interaction AB	SSAB = 1.5	1	MSAB=SSAB=1.5	$F_{AB} = \frac{MSAB}{MSE} = \frac{1.5}{0.222} = 6.76$
Error	SSE = 2.66	$4(n-1) = 12$	$MSE = \frac{SSE}{12} = \frac{2.66}{12} = 0.222$	
Total	SST = 21.2	$4n-1 = 15$		

Null hypothesis: H_0 : All the mean effects are equal.

Alternative hypothesis: H_1 : Not all equal.

Factor A: The calculated value of $F_A = 69.41$

At 5% level of significance, table value of

$$F_A(1, 12) = 4.75$$

$\therefore$ the calculated value of $F_A > |$ the table value of F_A

$\therefore H_0$ is rejected at |5%| level.

i.e., the effect of A is significant.

Factor B: At 5% level, the table value of $F_A(1, 12) = 4.75$

$\therefore$ the calculated value of $|F_A > |$ the table value of F_A

$\therefore H_0$ is rejected. i.e. the effect of B is significant.

Interaction AB: The calculated value of $F_{AB} = 6.76$

At 5% level, the table value of $F_{AB}(1, 12) = 4.75$

$\therefore$ the calculated value of $F_{AB} > |$ the table value of F_{AB}

$\therefore H_0$ is rejected. i.e. the effect of interaction AB is significant.

UNIT- 5

STATISTICAL QUALITY CONTROL

CONTROL CHARTS FOR MEASUREMENTS [X and R charts]:

- **Control Charts for attributes [p, c and np charts]**
- **Tolerance limits – Acceptance sampling**

Control Chart

A control chart provides a basis for deciding whether the variation in the output is due to random cause or due to assignable cause. It will assist us in making decisions whether to adjust the process or not.

A control chart is designed to display successive measurements of a process with a centre line and control limits.

The control limits are above and below the center line and are equidistant from the centre line and are known as Upper Control Limit (UCL) and Lower Control Limit (LCL).

The control chart helps us decide whether the process of production is in control or not.

Types of control charts

- (i) Control charts for variables
- (ii) Control charts for attributes

PROCEDURE TO DRAW THE $\bar{X}$ - CHART AND R – CHART

1. The sample values in each of the N samples each of size n will be given. Let $\bar{X}_1, \bar{X}_2, \dots, \bar{X}_N$ be the means of the N samples and $R_1, R_2, \dots, R_n$ be the ranges of the N samples. By range of a sample, we mean the maximum sample value minus the minimum sample value in that sample.
2. We can compute

$$\bar{\bar{x}} = \frac{1}{N} (\bar{x}_1 + \bar{x}_2 + \dots + \bar{x}_N) \text{ and}$$

$$\bar{R} = \frac{1}{N} (R_1 + R_2 + \dots + R_N)$$

3. The values of A_2 , D_3 , D_4 for the given sample size n are taken from the table of control chart constants.
4. Then the values of the control limits $\bar{X} \pm A_2 \bar{R}$ (for the mean chart) and the control limits $D_3 \bar{R}$ and $D_4 \bar{R}$ (for the range chart) are computed.
5. On the ordinary graph sheet, the sample numbers are represented on the x – axis and the sample means on the y – axis (for the mean chart) and the sample ranges on the y – axis (for the range chart).
6. For drawing the mean chart, we draw the three lines $y = \bar{X}$, $y = \bar{X} - A_2 \bar{R}$ and $y = \bar{X} + A_2 \bar{R}$ which represent respectively the central line, the L.C.L line and UCL line. Also we plot the points whose coordinates are $(1, \bar{X}_1)$, $(2, \bar{X}_2)$, $\dots (N, \bar{X}_N)$ join adjacent points by line segments. The graph thus obtained is the $\bar{X}$ – chart.
7. For drawing the range chart, we draw the three line $y = \bar{R}$, $y = D_3 \bar{R}$ and $y = D_4 \bar{R}$ which represent respectively the central line, the LCL line and UCL line. Also we plot the points whose co-ordinate are $(1, R_1)$, $(2, R_2)$, $\dots, (N, R_N)$ and join adjacent points by line segments. The graph thus obtained is the R – Chart.

CONSTRUCTION OF $\bar{X}$ – CHART:

Draw independent samples, each of size n from a large production process. Let $\bar{x}_1, \bar{x}_2, \bar{x}_3, \dots, \bar{x}_k$ be the means of these samples.

The mean of all these means is

$$\bar{X} = \frac{\bar{x}_1, \bar{x}_2, \bar{x}_3, \dots, \bar{x}_n}{K}$$

The control limits are given by,

$$UCL = \bar{X} + 3.SE(\bar{X})$$

$$LCL = \bar{X} - 3.SE(\bar{X})$$

$$CL = \bar{X}$$

where, $SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$, σ being the SD of the production

If σ is not available the SD of the Sampling Distribution of the mean can be taken as the best estimate of σ . In the case of small sample, the estimate of

$SE\text{of}\bar{X} = \text{is } \frac{\sigma}{\sqrt{n-1}}$. Alternatively in the case of small samples of size less than 20, the following results for control limits are available:

$$LCL = \bar{X} - A_2\bar{R}$$

$$CL = \bar{X}$$

Here $\bar{R}$ is the mean of the sample range $R_1, R_2, R_3, \dots, R_n$ obtained from the k samples. The factor A_2 has to be determined from statistical tables when the sample size n is known.

Range Chart (R – Chart)

For samples of size less than 20 the range provides a good estimates of σ . Hence to measure the variance in the variable, Range chart is based.

CONSTRUCTION OF R – CHART

Let $R_1, R_2, R_3, \dots, R_k$ be the values of the ranges in k samples. The mean of all these ranges is

$$\bar{R} = \frac{R_1 + R_2 + R_3 + \dots + R_k}{k}$$

The control limits are given by,

$$LCL = D_3\bar{R}$$

$$UCL = D_4\bar{R}$$

The factors D_3 and D_4 are determined from Statistical table for known sample size.

In the case of production process if the characteristics to be tested is non – measurable (if a unit is defective or not) control charts for attributes are used to find out whether a production process is under control or not. The most common control charts under this category are,

- (1) Control charts for number of defectives
- (2) Control charts for fraction defectives.

Problems Based on $\bar{X}$ and $\bar{R}$ chart

Example 1:

Given below are the values of sample mean $\bar{X}$ and sample range R for 10 samples, each of size 5. Draw the appropriate mean and range charts and comment on the state of control of the process.

Sample No.:	1	2	3	4	5	6	7	8	9	10
Mean $\bar{X}_i$:	43	49	37	44	45	37	51	46	43	47
Range R_i :	5	6	5	7	7	4	8	6	4	6

Solution:

$$\bar{\bar{X}} = \frac{1}{N} \sum \bar{x}_i = \frac{1}{10} (43 + 49 + 37 + \dots + 47) [\because N = 10]$$

$$= 44.2$$

$$\bar{R} = \frac{1}{N} \sum R_i$$

$$= \frac{1}{10} (5 + 6 + 5 + \dots + 5) = 5.8$$

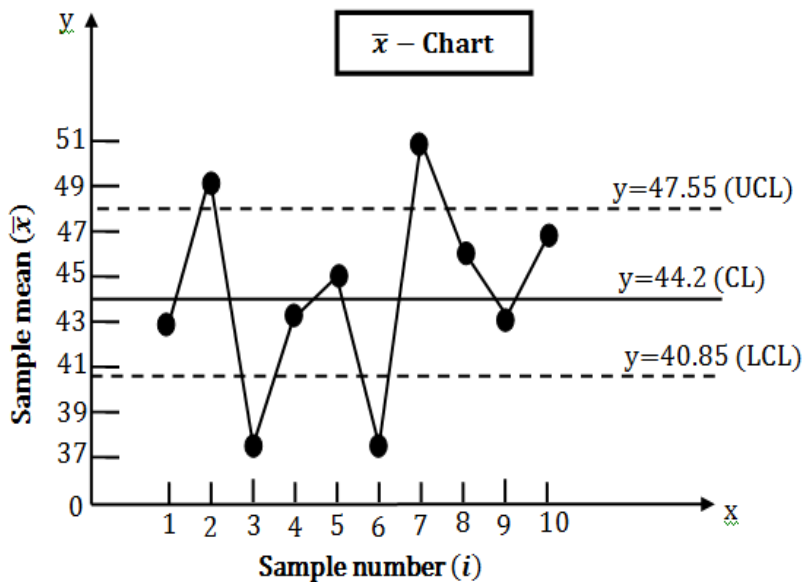
From the table of control chart, for sample size $n = 5$, we have $A_2 = 0.577$, $D_3 = 0$ and $D_4 = 2.115$

(i) Control limits for $\bar{X}$ chart :

$$CL \text{ (Central line)} = \bar{X} = 44.2$$

$$LCL = \bar{X} - A_2 \bar{R} = 44.2 - (0.577)(5.8) = 40.8534 = 40.85$$

$$UCL = \bar{X} + A_2 \bar{R} = 44.2 + (0.577)(5.8) = 47.5466 = 47.55$$



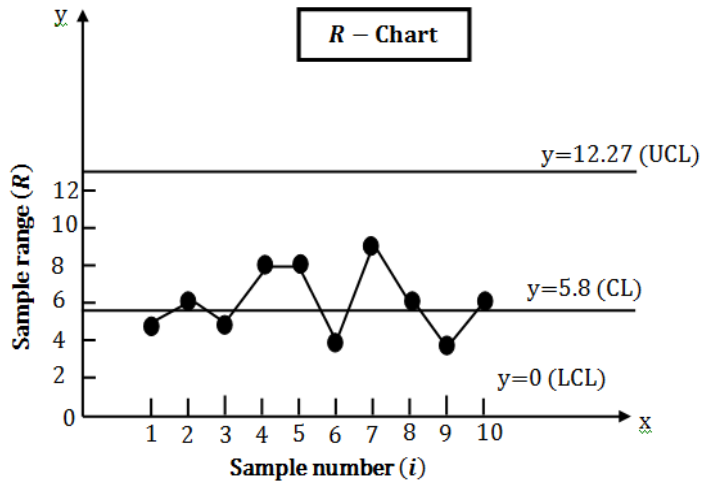
Conclusion:

Since 2nd, 3rd, 6th and 7th sample means fall outside the control limits the statistical process is out of control according to $\bar{X}$ - Chart.

(ii) Control limits for R - Chart:

$$CL \text{ (Central line)} = \bar{R} = 5.8$$

$$LCL = D_3; \bar{R} = 0; UCL = D_4 \bar{R} = (2.115)(5.8) = 12.267 \approx 12.27$$



Conclusion:

Since all the sample mean fall within the control limits the statistical process is under control according to $\bar{X}$ – Chart.

Inference:

From both $\bar{X}$ and R – chart, we see that a point in $\bar{X}$ – Chart lies outside control limits while all points in R – chart lie within control limits. Through the range variation is under control, we conclude that the process is out of statistical control.

Note (i): If the process is to be under control, then all sample points in both $\bar{X}$ and R – chart must be within control limits.

Note (ii): Eliminating the sample no.8 which goes outside control limits, we can get new control limits to set up testing of quality.

Example 2:

The following are the sample means and ranges for ten samples, each of size 5. Construct the control chart for mean and range and comment on the nature of control.

Sample No.:	1	2	3	4	5	6	7	8	9	10
Mean :	12.8	13.1	13.5	12.9	13.2	14.1	12.1	15.5	13.9	14.2
Range :	2.1	3.1	3.9	2.1	1.9	3.0	2.5	2.8	2.5	2.0

Solution:

$$\bar{\bar{X}} = \frac{12.8 + 13.1 + 13.5 + \dots + 14.2}{10}$$

$$= \frac{135.5}{10} = 13.53$$

$$\bar{R} = \frac{\sum R_i}{N} = \frac{2.1 + 3.1 + 3.9 + \dots 2.5 + 2.0}{10}$$

$$= \frac{25.9}{10} = 2.59$$

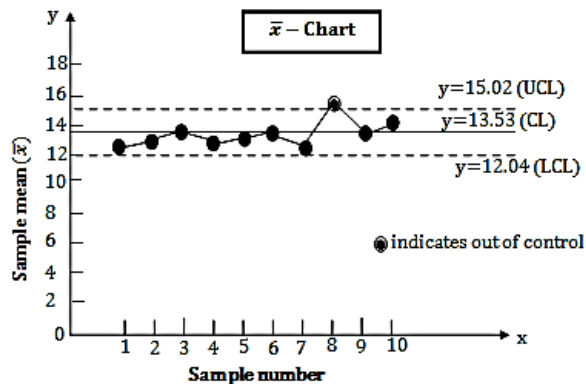
From the table of control charts constants, for sample size $n = 5$, we have $A_2 = 0.577$, $D_3 = 0$ and $D_4 = 2.115$

(i) Control limits for $\bar{X}$ chart :

$$CL = \bar{\bar{X}} = 13.53$$

$$\begin{aligned} LCL &= \bar{\bar{X}} - A_2 \bar{R} = 13.53 - (0.577)(2.59) \\ &= 12.03557 \approx 12.04 \end{aligned}$$

$$\begin{aligned} UCL &= \bar{\bar{X}} + A_2 \bar{R} = 13.53 + (0.577)(2.59) \\ &= 15.02443 \approx 15.02 \end{aligned}$$



Conclusion:

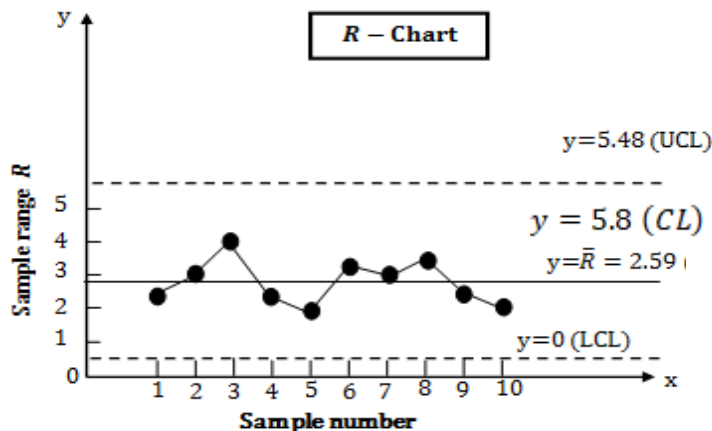
Since 8th sample means fall outside the control limits the statistical process is out of control according to $\bar{X}$ - Chart.

(ii) Control limits for R - Chart :

$$UCL = D_4 \bar{R} = 2.115 \times 2.59 \approx 5.48$$

$$LCL = D_3 \bar{R} = 0$$

$$UR = \bar{R} = 2.59$$



Conclusion:

Since all the sample mean fall within the control limits the statistical process is under control according to $\bar{X}$ – Chart.

Example 3:

The following table gives the sample means and ranges for 10 samples, each of size 6, in the production of certain component. Construct the control chart for mean and range and comment on the nature of control.

Sample No.:	1	2	3	4	5	6	7	8	9	10
Mean $\bar{X}$:	37.3	49.8	51.5	59.2	54.7	34.7	51.4	61.4	70.7	75.3
Range :	9.5	12.8	10.0	9.1	7.8	5.8	14.5	2.8	3.7	8.0

Solution:

$$\begin{aligned}
 \bar{\bar{X}} &= \frac{\sum \bar{x}}{N} \quad [\because N = 10] \\
 &= \frac{37.3 + 49.8 + 51.5 + 59.2 + 54.7 + 34.7 + 51.4 + 61.4 + 70.7 + 75.3}{10} \\
 &= \frac{546.0}{10} = 54.6 \\
 \bar{R} &= \frac{\sum R}{N} \\
 &= \frac{9.5 + 12.8 + 10.0 + 9.1 + 7.8 + 5.8 + 14.5 + 2.8 + 3.7 + 8.0}{10} \\
 &= \frac{84.0}{10} = 8.4
 \end{aligned}$$

From the table of control chart, for sample size of 6,

$$A_2 = 0.483, D_3 = 0 \text{ and } D_4 = 2.004$$

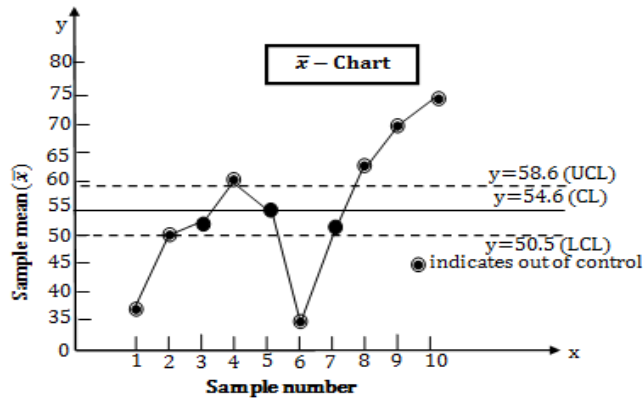
(i) Control limits for $\bar{X}$ chart :

$$UCL = \bar{X} + A_2 \bar{R}$$

$$= 54.6 + (0.483)(8.4) = 58.657$$

$$LCL = \bar{X} - A_2 \bar{R}$$

$$= 54.6 - (0.483)(8.4) = 50.543$$

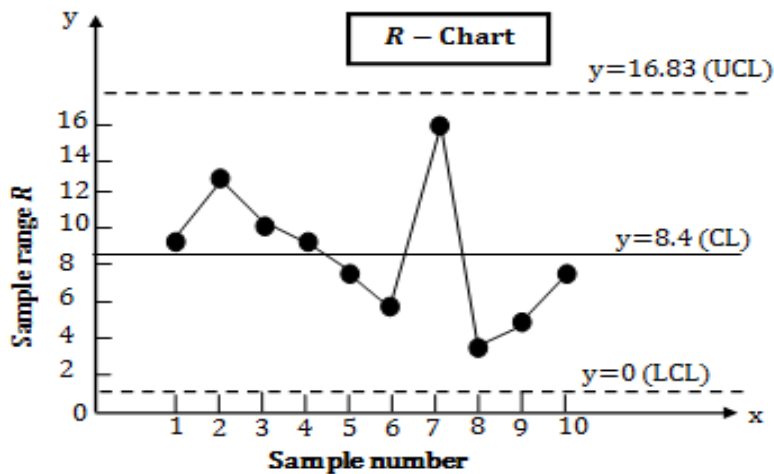


(ii) Control limits for R - Chart :

$$\bar{R} = 8.4$$

$$UCL = D_4 \bar{R} = 2.004 \times 8.4 = 16.834$$

$$LCL = D_4 \bar{R} = 0$$



Conclusion:

Since 1st, 2nd, 4th, 6th, 8th, 9th, 10th sample means fall outside the control limits the statistical process is out of control according to $\bar{X}$ - Chart.

Since all the sample mean fall within the control limits the statistical process is under control according to R - Chart.

Inference:

Through the sample points in $\bar{R}$ – Chart lie within control limits, some of the sample points in $\bar{X}$ – Chart lie outside the control limits. Hence, we conclude that the process is out of control; corrective measures are necessary.

Example 4:

The following data give the measurements of 10 samples each of size 5 in the production process taken in an interval of 2 hours. Calculate the sample means and ranges and drew the control charts for mean and range.

Sample Number	1	2	3	4	5	6	7	8	9	10
Observed Measurements X	49	50	50	48	47	52	49	55	53	54
	55	51	53	53	49	55	49	55	50	54
	54	53	48	51	50	47	49	50	54	52
	49	46	52	50	44	56	53	53	47	54
	53	50	47	53	45	50	45	57	51	56

Solution:

We shall find $\bar{X}$ and R for each sample.

Sample No.:	1	2	3	4	5	6	7	8	9	10
$\sum X$:	260	250	250	255	235	260	245	270	255	270
$\bar{X}$:	52	50	50	51	47	52	49	54	51	54
R:	6	7	6	5	6	9	8	7	7	4

$$\begin{aligned}\bar{\bar{X}} &= \frac{\sum \bar{X}}{N} \\ &= \frac{52 + 50 + 50 + 51 + 47 + 52 + 49 + 54 + 51 + 54}{10} = \frac{510}{10} = 51.0 \\ \bar{R} &= \frac{\sum R}{N} = \frac{6 + 7 + 6 + 5 + 6 + 9 + 8 + 7 + 7 + 4}{10} \\ &= \frac{65}{10} = 6.5\end{aligned}$$

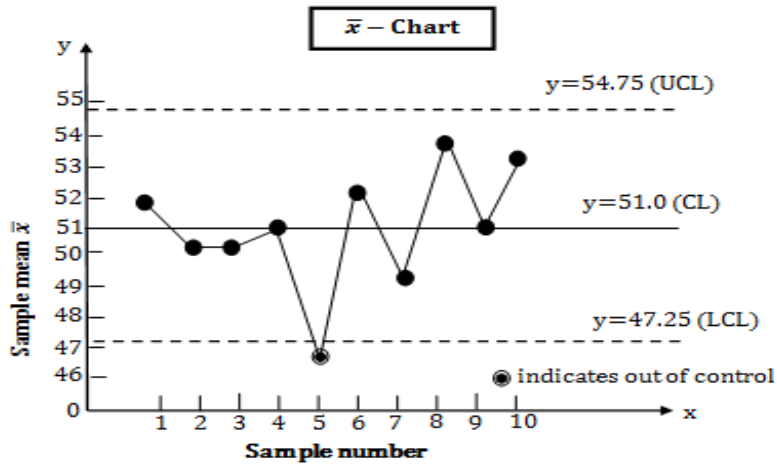
From the table for sample size of $n = 5$, $A_2 = 0.577$, $D_3 = 0$ and $D_4 = 2.115$

(i) Control limits for $\bar{X}$ chart :

$$\begin{aligned}\text{UCL} &= \bar{\bar{X}} + A_2 \bar{R} \\ &= 51.0 + (0.577) 6.5 = 54.7505\end{aligned}$$

$$\begin{aligned}\text{LCL} &= \bar{\bar{X}} - A_2 \bar{R} \\ &= 51.0 - (0.577)(6.5) = 47.2495\end{aligned}$$

$$LCL = \bar{X} = 51.0$$

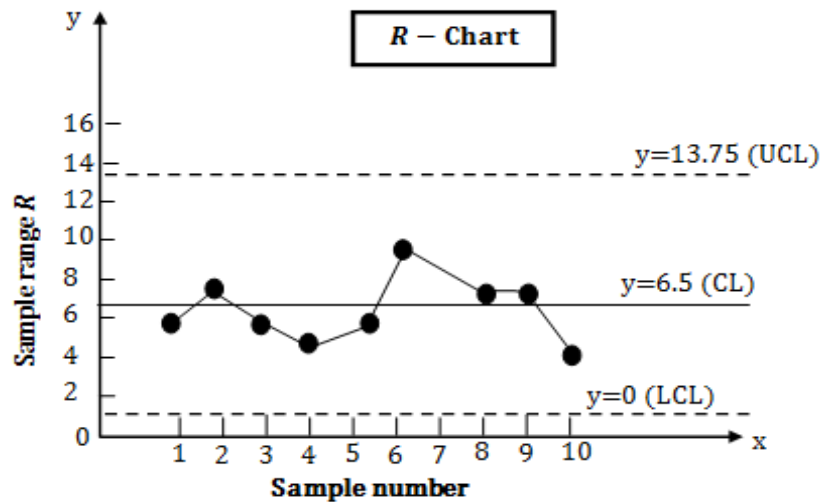


(ii) Control limits for R - Chart :

$$UCL = D_4 \bar{R} = (2.115)(6.5) = 13.7475$$

$$LCL = D_3 \bar{R} = 0$$

$$CL = \bar{R} = 6.5$$



Conclusion:

Since 5th sample mean full outside the control limits the statistical process is out of control according to $\bar{X}$ - Chart.

Since all the sample mean fall within the control limits the statistical process is under control according to R - Chart.

Example 5:

Control on measurements of pitch diameter of thread in air–craft fittings is checked with 5 samples each containing 5 items at equal intervals of time. The measurements are given below. Construct $\bar{X}$ and R charts and state your inference from the charts.

Sample No.	Measurements				
1	46	45	44	43	42
2	41	41	44	42	40
3	40	40	42	40	42
4	42	43	43	42	45
5	43	44	47	47	45

Solution:

For each sample, calculate $\bar{X}$ and R and tabulate

Sample No.:	$\sum x$	$\bar{X}$	R
1	220	44.0	4
2	208	41.6	4
3	204	40.8	2
4	215	43.0	3
5	226	45.2	4

$$\bar{\bar{X}} = \frac{\sum \bar{X}}{N} = \frac{44 + 41.6 + 40.8 + 43.0 + 45.2}{5} = 42.92$$

$$\bar{R} = \frac{\sum R}{N} = \frac{17}{5} = 3.4$$

From table, for sample size of 5 items,

$$A_2 = 0.577, D_3 = 0, D_4 = 2.115$$

(i) **Control limits for $\bar{X}$ chart :**

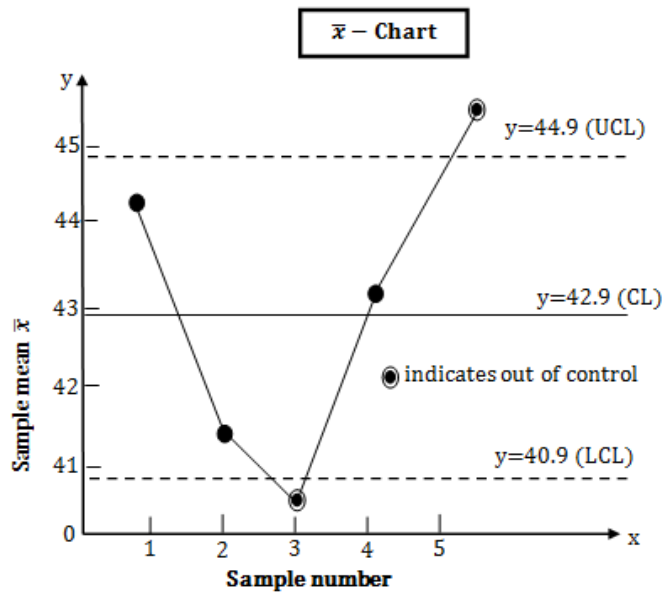
$$UCL = \bar{\bar{X}} + A_2 \bar{R}$$

$$= 42.92 + (0.577)(3.4) = 44.88$$

$$LCL = \bar{\bar{X}} - A_2 \bar{R}$$

$$= 42.92 - (0.577)(3.4) = 40.96$$

$$CL = \bar{\bar{X}} = 42.92$$



Conclusion:

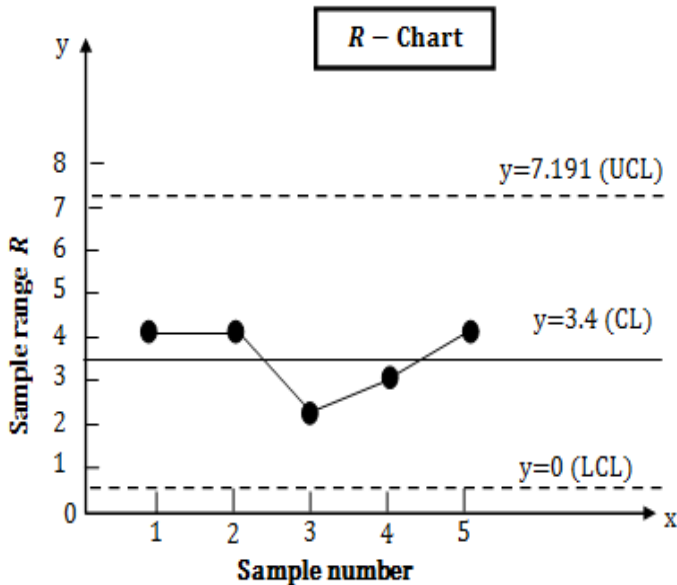
$\bar{X}_5 = 45.2 > UCL = 44.88$. All sample points do not lie between control limits.

Hence, the process is out of control.

(ii) Control limits for R - Chart :

$$UCL = D_4 \bar{R} = (2.115)(3.4) = 7.191$$

$$LCL = D_3 \bar{R} = 0 \quad CL = \bar{R} = 3.4$$



Conclusion: All sample points lie between control limits. Hence, the variability is under control. But process is out of control due to $\bar{X}$ - chart.

CONTROL CHART FOR SAMPLE STANDARD DEVIATION

OR S – CHART:

The standard deviation is an ideal measure of dispersion, a combination of control charts for the sample mean $\bar{X}$ and the sample S.D.

s is more appropriate than a combination of $\bar{X}$ and R charts for controlling process average and process variability. We know that s , the S.D of a sample of size n , is a random variable that is normally distributed with mean σ and S.D.

$\frac{s}{\sqrt{2n}}$, where σ is the S.D. of the population from which the sample is drawn.

$$\therefore P\left\{\sigma - \frac{3\sigma}{\sqrt{2n}} \leq s \leq \sigma + \frac{3\sigma}{\sqrt{2n}}\right\} = 0.9973$$

$$\therefore \text{The lower and upper control limits are } \sigma - \frac{3\sigma}{\sqrt{2n}} \text{ and } \sigma + \frac{3\sigma}{\sqrt{2n}}$$

Since σ is not known, it is estimated approximated by

$\bar{s} = \frac{1}{N}(s_1 + s_2 + \dots + s_N)$, where s_i is the S.D. of the i th sample and N is the number of sample considered.

$$\therefore \text{LCL for } s = \left(1 - \frac{3}{\sqrt{2n}}\right)\bar{s} \approx B_3\bar{s} \text{ and}$$

$$\text{UCL for } s = \left(1 + \frac{3}{\sqrt{2n}}\right)\bar{s} \approx B_4\bar{s}$$

The values of B_3 and B_4 can be read for various values of sample size n from the table of control chart constants.

If $\bar{x}$ values and s values only are given, then CL for $\bar{x} = \bar{\bar{x}}$,

$$\text{LCL for } \bar{X} = \bar{\bar{X}} - A_i \sqrt{\frac{n-1}{n}} \bar{s} \text{ and}$$

$$\text{UCL for } \bar{X} = \bar{\bar{X}} + A_i \sqrt{\frac{n-1}{n}} \bar{s}, \text{ when } n \leq 25.$$

Example 1:

The following data give coded measurements of 10 samples each of size 5, drawn from a production process at intervals of 1 hour. Calculate the sample means and S.D's and draw the draw control charts for $\bar{X}$ and s .

Sample No.	1	2	3	4	5	6	7	8	9	10
Coded measurements (X)	9	10	10	8	7	12	9	15	10	16
	15	11	13	13	9	15	9	15	13	14
	14	13	8	11	10	7	9	10	14	12

	9	6	12	10	4	16	13	13	7	14
	13	10	7	13	5	10	5	17	11	14

Solution:

Sample No.:	1	2	3	4	5	6	7	8	9	10
$\sum X$:	60	50	50	55	35	60	45	70	55	70
$\bar{X}$:	12	10	10	11	7	12	9	14	11	14
$\sum (X - \bar{X})^2$:	32	26	26	18	26	54	32	28	30	8
s :	2.5	2.3	2.3	1.9	2.3	3.3	2.5	2.4	2.4	1.3

$$\bar{X} = \frac{1}{N} \sum \bar{X}_i = \frac{1}{10} \times (12 + 10 + 10 + \dots + 14) = \frac{110}{10} = 11$$

$$\bar{s} = \frac{1}{N} \sum s_i = \frac{1}{10} \times (2.5 + 2.3 + 2.3 + \dots + 1.3) = \frac{23.2}{10} = 2.32$$

From the table of control chart constants, for sample size $n = 5$, we have

$$A_1 = 1.596; B_3 = 0 \text{ and } B_4 = 2.089$$

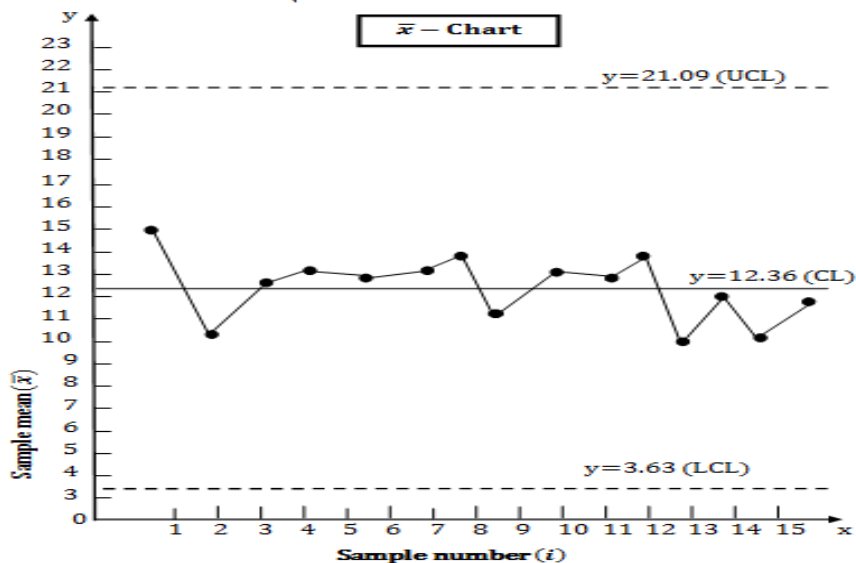
Control limits for $\bar{X}$ chart:

$$CL = \bar{X} = 11; \quad LCL = \bar{X} - A_0 \sqrt{\frac{n}{n-1}} \bar{s}$$

$$= 11 - 1.596 \sqrt{\frac{5}{4}} \times 2.32 = 6.86$$

$$UCL = \bar{X} + A_1 \sqrt{\frac{n}{n-1}} \bar{s}$$

$$= 11 + 1.596 \sqrt{\frac{5}{4}} \times 2.32 = 15.14$$

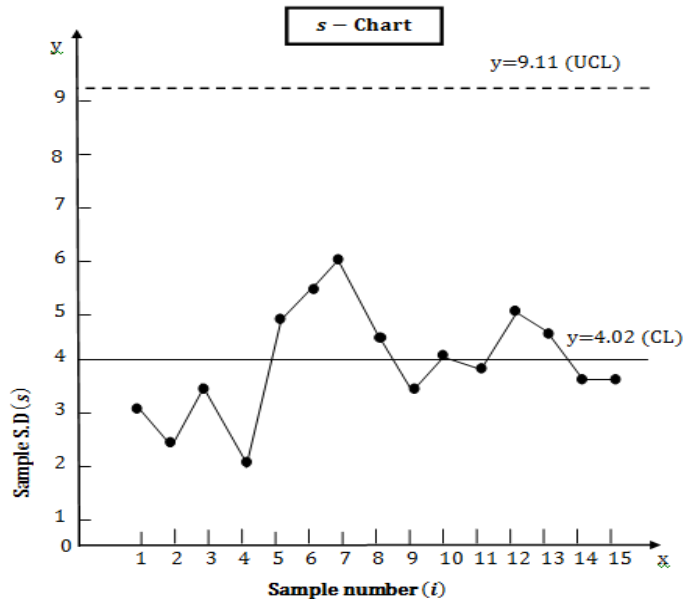


Control limits for s chart:

$$CL = \bar{s} = 2.32$$

$$LCL = B_3 \bar{s} = 0$$

$$UCL = B_4 \bar{s} = (2.089)(2.32) = 4.85$$



Conclusion:

The sample mean $\bar{X}$ values lie between 6.86 and 15.14 and the given $S.D(s)$ values between 0 and 4.85.

Hence, the process is under with respect to average and variability.

Control charts for attributes (p, c and np charts):

To control the quality of certain products whose attribute are available, the following control charts are used:

- (i) np - chart of number of defectives
- (ii) p - chart for proportion of defectives
- (iii) c - chart for the number of defects in a unit.

(i) np - chart

If the proportion of defectives (successes) in the population of items produced is p . The number of defectives in a sample of size n is X , then X follows a binomial distribution with mean np and S.D. is $\sqrt{npq}$. When n is sufficiently large and when np and S.D. is $\sqrt{npq}$.

$$\text{Hence } P\{np - 3\sqrt{npq} \leq X \leq np + 3\sqrt{npq}\} = 0.9973$$

∴ The control limits for x , the number of defectives, are $np \pm 3\sqrt{npq}$.

The population proportion of defects p will not be known, it is estimated approximately as $\bar{p} = \frac{1}{N}(p_1 + p_2 \dots + p_N)$ or equivalently $n\bar{p} = \frac{1}{N}(np_1 + np_2 + \dots + np_N)$, where all the samples are of the same size n , the number of sample is N and the number of defectives in the i th sample is np_i .

Hence, the control limits for the number of sample defectives are $n\bar{p} \pm 3\sqrt{n\bar{p}(1 - \bar{p})}$, where each sample is size n . Here the target value is $n\bar{p}$.

To draw the np – chart, the sample number is represented on the x – axis and the number of defectives is represented on the y – axis.

The line $y = n\bar{p}$, $y = n\bar{p} - 3\sqrt{n\bar{p}(1 - \bar{p})}$ and $y = n\bar{p} + 3\sqrt{n\bar{p}(1 - \bar{p})}$, which represent respectively the central line, LCL line and UCL line are drawn. We plot the points $(1, np_1), (2, np_2), \dots (N, np_N)$ on the graph sheet and adjacent points are joined by line segments. The state of control of the process is decided as before.

(ii) p – chart

Under the same assumption as those for np – chart, since X follows a normal distribution with mean np and S.D. $\sqrt{npq}$, the proportion of defectives, viz. $\frac{X}{n}$ follows a normal distribution with mean p and S. D. $\sqrt{\frac{pq}{n}}$.

$$\text{Hence } P\left\{p - 3\sqrt{\frac{pq}{n}} \leq \frac{X}{n} \leq p + 3\sqrt{\frac{pq}{n}}\right\} = 0.9973$$

∴ The control limits for $\frac{X}{n}$, the proportion of defectives, are

$$p \pm 3\sqrt{\frac{pq}{n}}$$

As in the precious case, p is estimated as

$\bar{p} = \frac{1}{N}(p_1 + p_2 \dots + p_N)$, where the proportion of defectives in the i th sample is p_i .

Hence, the control limits for the number of fraction proportion of sample defectives are

$$\bar{p} \pm 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}} \dots (1)$$

The formula (1) holds good when all the samples are of the same size n .

Note: np – chart and p – chart are used when $\bar{p} \geq 0.05$ or $n\bar{p} \geq 4$.

(iii) $c - \text{chart}$

It is required to control (minimize) the number of defects per unit, $c - \text{chart}$ is used. ' c ' represents the number of defects in a unit. For construction of $c - \text{chart}$, a record of the number of defects in each of the N articles inspected should be known since the probability of occurrence of a defect in a unit is very small, the number X of defects in a unit follows a Poisson distribution with parameter λ , viz., with mean λ and S.D. $\sqrt{\lambda}$. In the limit, X follows a normal distribution with mean λ and S.D. λ .

$$\text{Hence } P\{\lambda - 3\sqrt{\lambda} \leq X \leq \lambda + 3\sqrt{\lambda}\} = 0.9973$$

$\therefore$ The control limits for X , the number of defects in a unit are $\lambda \pm 3\sqrt{\lambda}$.

As the value of λ will not be known, it is estimated approximately by $\bar{c}$, where $\bar{c} = \frac{1}{N}(c_1 + c_2 + \dots + c_N)$, which c_i is the number of defects in the i th unit.

Hence, the control limits for the number of defects c in a unit are $\bar{c} \pm 3\sqrt{\bar{c}}$.

Note:

$c - \text{chart}$ is used when $\bar{c} \geq 4$ or when $\bar{c}$ is small compared with the maximum number of defects given in the data.

Example 1:

Construct a control chart for the number of defects from the following data which give the number defects in 15 pieces cloth of equal length when inspected in a textile mill and find the nature of the process, Number of defects : 3, 4, 2, 7, 9, 6, 5, 4, 8, 10, 5, 8, 7, 7, 5.

Solution:

Let c denote the number of defects in each piece.

$$\bar{c} = \frac{\sum c}{N} = \frac{90}{15} = 6$$

$$UCL = \bar{c} + 3\sqrt{\bar{c}} = 6 + 3\sqrt{6} = 13.35$$

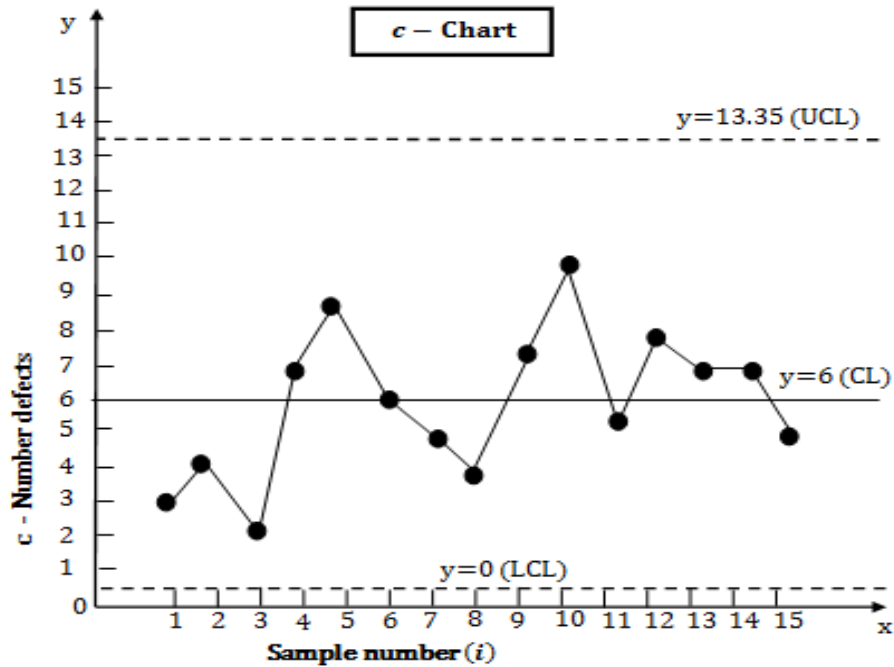
$$LCL = \bar{c} - 3\sqrt{\bar{c}} = 6 - 3\sqrt{6} = -ve$$

We take $LCL = 0$ [Since it become negative]

$$CL = \bar{c} = 6$$

Scanning the given values of c , we find all the values of c lie between $LCL = 0$ and $UCL = 13.35$.

Hence, the process is under control.



Example 2:

A plant produces paper for newsprint and tolls of paper are inspected for defects. The results inspections of 20 rolls of papers are given below: Draw the **c - chart** and comment on the state of control.

Roll No. (i):	1	2	3	4	5	6	7	8	9	10
No. of defects (c):	19	10	8	12	15	22	7	13	18	13
(i):	11	12	13	14	15	16	17	18	19	20
(c):	16	14	8	7	6	4	5	6	8	9

Solution:

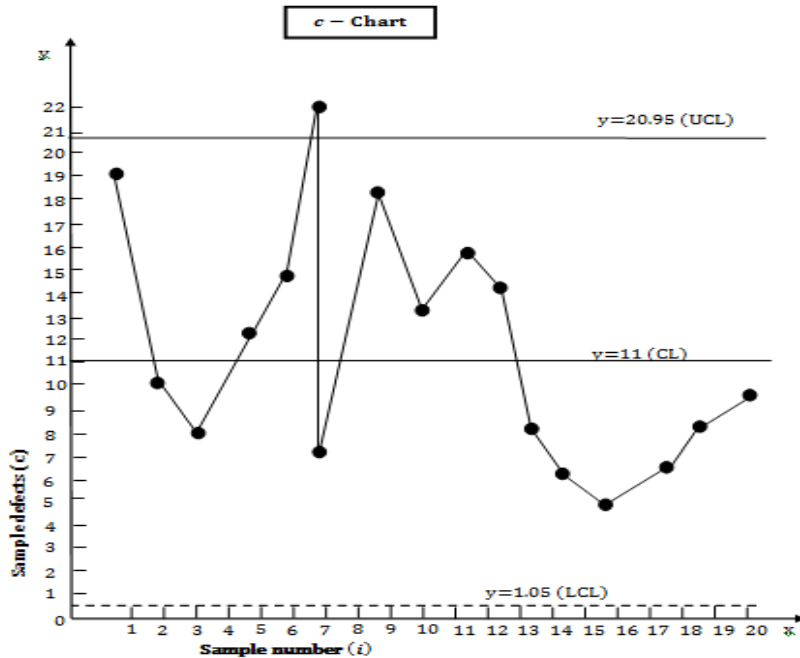
Let c denote the number of defects in each piece.

$$\bar{c} = \frac{1}{N} \sum c_i = \frac{1}{20} \times 220 = 11$$

$$CL = \bar{c} = 11$$

$$LCL = \bar{c} - 3\sqrt{\bar{c}} = 11 - 3\sqrt{11} = 11 - 3\sqrt{11} = 1.05$$

$$UCL = \bar{c} + 3\sqrt{\bar{c}} = 11 + 3\sqrt{11} = 20.95$$



Since one point falls outside the control lines, the process is out of control.

Example 3:

Construct a control chart for defectives for the following data:

Sample No.:	1	2	3	4	5	6	7	8	9	10
No. inspected:	90	65	85	70	80	80	70	95	90	75
No. of defectives:	9	7	3	2	9	5	3	9	6	7

Solution:

We note that the size of the sample varies from sample to sample. We can construct **p - chart**, provided $0.75 \bar{n} < n_i < 1.25 \bar{n}$, for all i

$$\begin{aligned} \text{Here } \bar{n} &= \frac{1}{N} \sum n_i = \frac{1}{10} (90 + 65 + \dots + 90 + 75) \\ &= \frac{1}{10} (800) = 80 \end{aligned}$$

$$\text{Hence } 0.75\bar{n} = 60 \text{ and } 1.25\bar{n} = 100$$

The values of n_i be between 60 and 100. Hence **p - chart**, can be drawn by the method given below.

$$\text{Now } \bar{p} = \frac{\text{Total no. of defectives}}{\text{Total no. of items inspected}} = \frac{60}{800} = 0.075$$

Hence for the p - chart to be constructed,

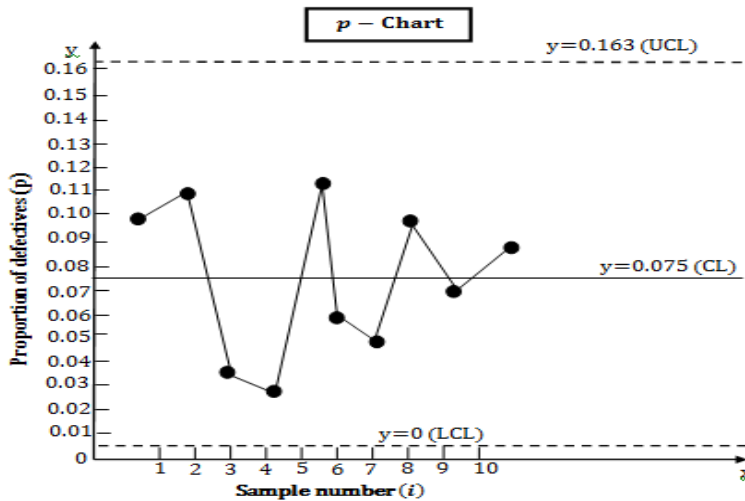
$$CL = \bar{p} = 0.075$$

$$LCL = \bar{p} - 3 \sqrt{\frac{\bar{p}(1 - \bar{p})}{\bar{n}}} = 0.075 - 3 \sqrt{\frac{0.075 \times 0.925}{80}} = -0.013$$

Since LCL cannot be negative, it is taken as 0.

$$UCL = \bar{p} + 3 \sqrt{\frac{\bar{p}(1 - \bar{p})}{\bar{n}}} = 0.075 + 3 \sqrt{\frac{0.075 \times 0.925}{80}} = 0.163$$

The values of p_i for the various samples are 0.100, 0.108, 0.035, 0.029, 0.113, 0.063, 0.043, 0.095, 0.067, 0.093.



Since all the sample points lie within the control lines, the process is under control.

Example 4:

The data given below are the number of defectives in 10 samples of 100 items each. Construct a p - chart and as np - chart and comment on the results:

Sample No.:	1	2	3	4	5	6	7	8	9	10
No. of defectives:	6	16	7	3	8	12	7	11	11	4

Solution: Sample size is constant for all samples, $n = 100$.

Total no. of defectives

$$= 6 + 16 + 7 + 3 + 8 + 12 + 7 + 11 + 11 + 4 = 85$$

Total no. of inspected = $10 \times 100 = 1000$

Average fraction defective = $\bar{p} = \frac{\text{No. of defectives in the sample}}{\text{No. of items inspected}}$

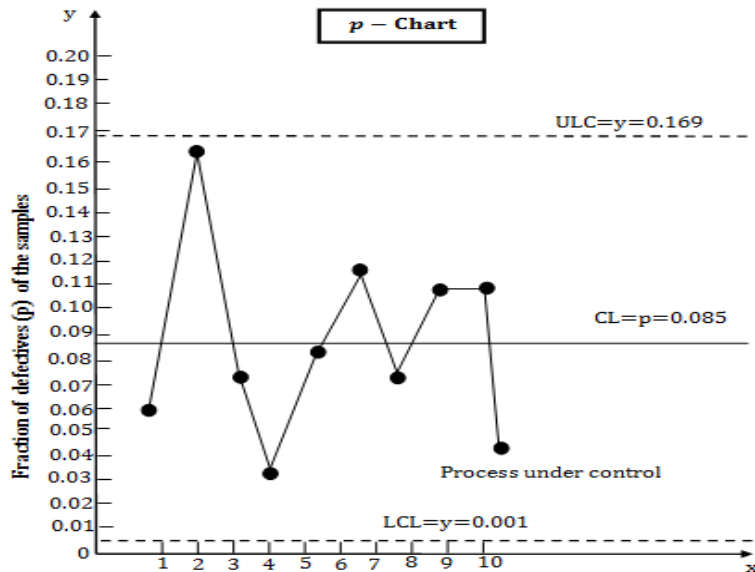
$$= \frac{85}{100} = 0.085$$

For p - chart:

$$UCL = \bar{p} + 3 \sqrt{\frac{\bar{p}(1 - \bar{p})}{\bar{n}}} = 0.085 + 3 \sqrt{\frac{(0.085 \times 0.0915)}{100}} = 0.1687$$

$$LCL = \bar{p} - 3 \sqrt{\frac{\bar{p}(1 - \bar{p})}{\bar{n}}} = 0.085 - 3 \sqrt{\frac{(0.085 \times 0.0915)}{100}} = 0.0013$$

CL corresponds to $\bar{p} = 0.085$



MM

Fraction defectives for sample are 0.06, 0.16, 0.07, 0.03, 0.08, 0.12, 0.07, 0.11, 0.11, 0.04.

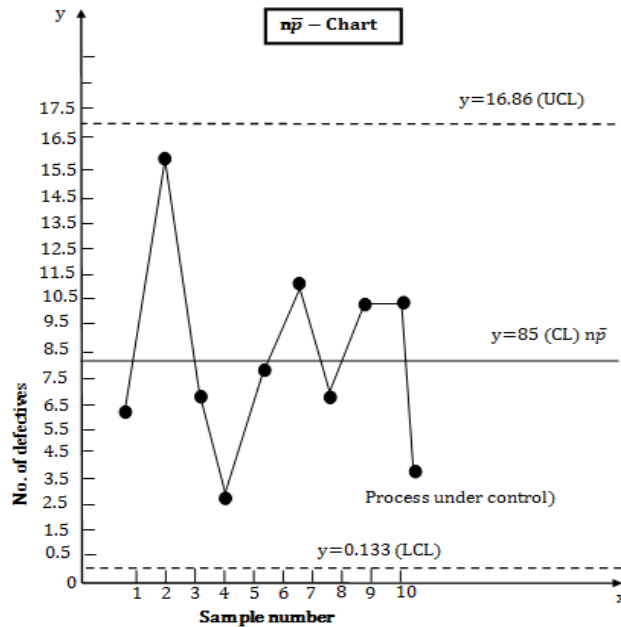
Conclusion: All the values are less than $UCL = 0.1687$ and greater than $LCL = 0.0013$. In the control chart, all sample points lie within the control limits. Hence, the process is under statistical control.

For np - chart:

$$UCL = n\bar{p} + 3\sqrt{n\bar{p}(1 - \bar{p})}$$

$$= n \left[\bar{p} + 3 \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}} \right]$$

$$= 100 \times 0.1687 = 16.87$$



$$n\bar{p} = 100 \times 0.085 = 8.5$$

$$UCL = n\bar{p} - 3\sqrt{n\bar{p}(1 - \bar{p})}$$

$$= n \left[\bar{p} - 3 \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}} \right]$$

$$= 100(0.0013) = 0.13$$

Conclusion:

All the values of number of defectives in the table lie between 16.87 and 0.13. Hence, the process is under control even in np - chart.

TOLERANCE LIMITS:

We have seen confidence limits are used to estimate a parameter of a population. But tolerance limits are used to indicate the limits within which a certain proportion of the population lie. For example, from a normal population with known mean μ and variance σ^2 , the limits which ever the middle 95% of the population observations is $\mu - 1.96\sigma$, $\mu + 1.96\sigma$. This interval ($\mu - 1.96\sigma$, $\mu + 1.96\sigma$) is called tolerance interval.

But when μ and σ^2 are not known in a normal population, they are estimated by sample mean $\bar{x}$ and variance s^2 . Then the **tolerance limits** are given by $\bar{x} \pm ks$, where k is determined such that one can assert with $100(1-\alpha)\%$ confidence that the proportion p of the population lie between $\bar{x} - ks$ and $\bar{x} + ks$.

Thus of large sample of size n , k is 1.96 for $p = 0.95$

ACCEPTANCE SAMPLING:

Two commonly used quality control techniques are **control charts** and **acceptance sampling**.

We have already seen control charts of various types. They are useful only for the regulation of the manufacturing process. Acceptance sampling is another important aspect of quality control.

For example, Whether to accept or not to accept a shipment of product on inspection by a customer on the basis of sampling is known as **acceptance sampling**. When a sample of the shipment is inspected, if the number of defective items is not more than a specified number, then the lot will be accepted.

This specified number is denoted by c and is called the **acceptance number**. This procedure is equivalent to a test of the null hypothesis that the proportion of defective p in a lot equals some specified value P_0 against an alternative value P_1 where $P_1 > P_0$. The value P_0 is called the **acceptable quality level (AQL)** and P_1 is called the **lot tolerance percent defectives (LTPD)**.

The statistical techniques used in acceptance sampling will be familiar hypothesis testing ideas. In acceptance sampling there is always a possibility of committing two types of errors.

- 1) Type I error or producer's risk. That is a sampling plan leading to the rejection of a lot which is good.
- 2) Type II error or consumer's risk. That is a sampling plan leading to acceptance of a lot which is not good.

Types of acceptance sampling plans.

1. Single sampling plan

When decision is taken to accept or reject a lot on the basis of only one sample, the acceptance plan is known as a single sampling plan. In this plan, the sample size n from a lot containing N items and the acceptance number c to be used and its choice is based on a specified AQL and (or) LTPD in association with given producers risk and (or) consumer's risk.

The sampling plan may be specified as

$$N = 250; \quad n = 25; c = 1$$

This means "take a random sample of 25 from a lot containing 250 items. If the sample contains more than 1 defective, reject the lot, otherwise accept the lot."

2. Double sampling plan

Double sampling involves the possibility of putting off the decision on a lot until a second sample is taken. A lot is accepted readily if the first sample is good enough. If the first sample is neither good nor bad, the second sample will be taken and the decision is based on the first and second samples combined. In a double sampling plan five things are specified: n_1 , c_1 , n_2 , n_1+n_2 and c_2

n_1 = size of the first sample from a lot.

c_1 = acceptance number of the first sample (i.e., the maximum number of defectives that will permit acceptance of the lot on the basis of first sample)

n_2 = size of the second sample from a lot.

$n_1 + n_2$ = size of the combined sample of the first and second samples.

c_2 = acceptance number for the two samples combined.

The double sampling plan may be specified as

$N = 500$

$n_1 = 25$ $c_1 = 1$

$n_2 = 50$ $c_2 = 3$

This plan involves selection of first a random sample of 25 items from the lot of 500 items. If the sample contains less than or equal to 1 defective, the lot is accepted. If more than 1 defective is found, the lot is rejected.

A second sample of 50 items is taken from the lot and the lot is accepted if the combined sample of 75 items contains less than or equal to 3 defectives and reject the lot otherwise.

3. Multiple or sequential sampling plan

When a sampling plan consists of more than two sample stages, it is called a **multiple sampling plan**.

A sampling procedure is said to be sequential if, after each stage of observation of sample, if the lot is rejected then take another sample and continue this process.

Operating characteristic curve (OC-curve)

A sampling plan is best described by its operating characteristic curve or OC curve. The OC curve gives the probability of acceptance for each value that can be assumed by the lot proportion defective P . The OC curve of an acceptance sampling plan shows the ability of the plan to distinguish between good and bad lots.

By an appropriate choice of the sample sizes and the acceptance numbers it is possible to match the OC curve of a double or multiple-sampling plan closely to that of an equivalent single – sampling plan. Thus the degree of protection given by a double or multiple sampling plan can be made essentially the same as that given by single-sampling plan.

It is usual in acceptance sampling the OC curve is constructed by using binomial distribution for the probability of accepting a lot containing the proportion defective P .

If P is very small, say $P < 0.10$ (or np is less than 5) and the lot is at least 10 times the size of the sample, we use Poisson distribution for constructing the OC curve.

Example: 5

For a sampling plan $N = 1000$, $n = 600$, $c = 1$, determine the probability of acceptance of the following lots.

(i) 0.6% defective

(ii) 0.7% defective

(iii) 2% defective

(iv) 3% defective

(v) 5% defective

(vi) 9% defective

Also draw the OC curve.

Solution:

The lot has 0.6% defective, the samples from it will have an average of 0.6% defective. So, in a sample of size 60, the average number of defectives will be $60 \times \frac{0.6}{100} = 0.36$ under the sampling plan with $c = 1$, the sample contains 0 or 1 defectives, we shall find the cumulative probability of drawing a sampling of 6 containing 0 or 1 defective using Poisson approximation.

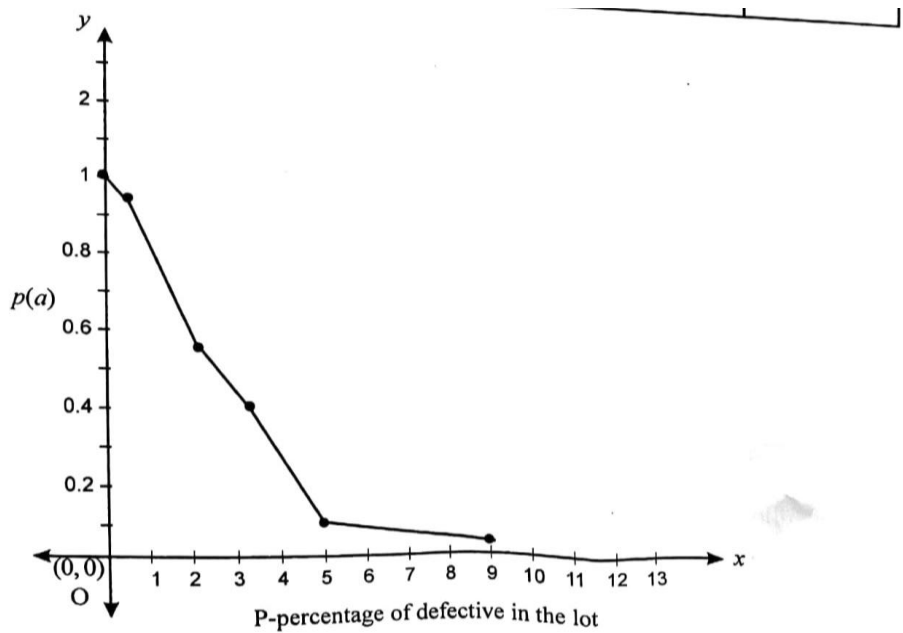
OC curve for single – sampling

$P(a)$: Probability of acceptance

The value $0.9489 \cong 0.95$ represents the prob. of drawing a sample of size 60 with 0 or 1 defective from a lot known to be 0.6% defective. In words, such a sample will enable acceptance of 95% of lots containing 0.5 percent defective or in a lot of 1000.

960 will be accepted and only 40 will be rejected.

S.No	% defective in the lot	Average no. of defective in sample	$P(0) = e^{-\lambda}$	$P(1) = e^{-\lambda} \cdot \lambda$	$P(0)+P(1)=P(a)$
1	0.6	$60 \times \frac{0.6}{100} = 0.36$			
2	0.7	$60 \times \frac{0.7}{100} = 0.42$	0.6977	0.2512	0.9489
3	2	$60 \times \frac{2}{100} = 1.2$	0.6570	0.2759	0.9329
4	3	$60 \times \frac{3}{100} = 1.8$	0.3012	0.3614	0.6626
5	5	$60 \times \frac{5}{100} = 3.0$	0.1653	0.2975	0.4628
6	9	$60 \times \frac{9}{100} = 5.4$	0.0498	0.1494	0.1992
			0.0045	0.0243	0.0288



Taking percentage defective on x-axis and probability of acceptance $p(a)$ on y-axis, the curve obtained is known as operating characteristic curve (OC curve) of the sampling plan.

Probability of rejection is $1-p(a)$ corresponding to any lot.